NEET 2017

OFFICIAL QUESTION PAPERS WITH DETAILED SOLUTION

All questions with detailed solutions of Physics, Chemistry and Biology

CAREERS 360

Physics

Q. 1 Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk up on the moving escalator will be:

Option 1:

$\frac{t_1 + t_2}{2}$

Option 2:

$\frac{t_1 t_2}{t_2 - t_1}$

Option 3:

$\frac{t_1 t_2}{t_2 + t_1}$

Option 4:

$t_1 - t_2$

Correct Answer:

$\frac{t_1 t_2}{t_2 + t_1}$

Solution:

$V_1 =$ Velocity of Preeti

$V_2 \rightarrow$ Velocity of escalator

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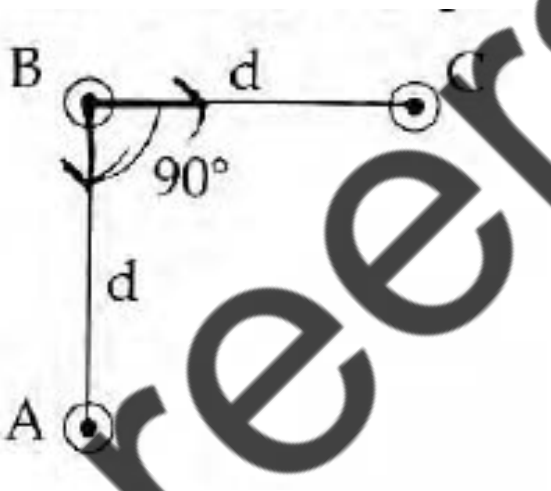
$l \rightarrow$ distance

$$t = \frac{l}{v_1 + v_2} = \frac{l}{\frac{l}{t_1} + \frac{l}{t_2}}$$

$$= \frac{t_1 t_2}{t_1 + t_2}$$

Hence, the answer is the option (3).

Q. 2 An arrangement of three parallel straight wires placed perpendicular to plane of paper carrying same current 'I' along the same direction is shown in Fig. Magnitude of force per unit length on the middle wire ' B' is given by :



Option 1:

$\frac{\mu_o i^2}{2\pi d}$

Option 2:

$\frac{2\mu_o i^2}{\pi d}$

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Option 3:

$$\frac{\sqrt{2}\mu_o i^2}{\pi d}$$

Option 4:

$$\frac{\mu_o i^2}{\sqrt{2}\pi d}$$

Correct Answer:

$$\frac{\mu_o i^2}{\sqrt{2}\pi d}$$

Solution:

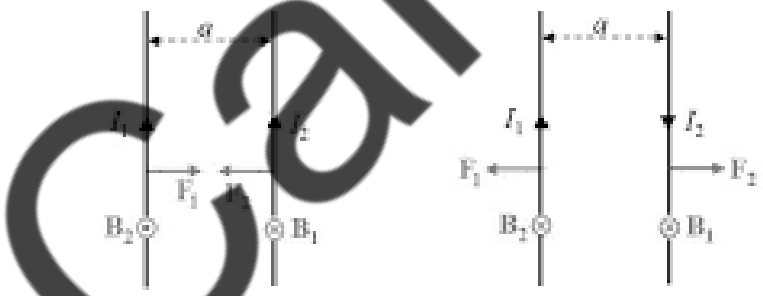
As we learnt in

Force between two parallel current carrying conductors -

$$F = \frac{\mu}{4\pi}\frac{2I_1 I_2}{a}l$$

$$\frac{F}{l} = \frac{\mu o}{4\pi}\frac{2I_1 I_2}{a}$$

- wherein



I_1 and I_2 current carrying two parallel wires

a-seperation between two wires

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Force between wires A and B = Force between B and C

$$F_{BC} = F_{AB} = \frac{\mu_0 I^2 l}{2\pi d}$$

$\vec{F_{AB}} \perp \vec{F_{BC'}}$ net force on wire B

$$F_{net} = \sqrt{2} F_{BC} = \frac{\sqrt{2}\mu_0 I^2 l}{2\pi d} \Rightarrow \frac{F_{net}}{l} = \frac{\mu_0 I^2}{\sqrt{2}\pi d}$$

Q. 3 A particle executes linear simple harmonic motion with an amplitude of 3 cm. When the particle is at 2 cm from the mean position, the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is:

Option 1:

$\frac{\sqrt{5}}{\pi}$

Option 2:

$\frac{\sqrt{5}}{2\pi}$

Option 3:

$\frac{4\pi}{\sqrt{5}}$

Option 4:

$\frac{2\pi}{\sqrt{3}}$

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Correct Answer:

$\frac{4\pi}{\sqrt{5}}$

Solution:

$v = \omega\sqrt{A^2 - x^2}$

$a = x\omega^2$

$v = a$

$\omega\sqrt{A^2 - x^2} = X\omega^2$

$\sqrt{(3)^2 - (2)^2} = 2\left(\frac{2\pi}{T}\right)$

$\sqrt{5} = \frac{4\pi}{T}$

$T = \frac{4\pi}{\sqrt{5}}$

Hence correct option is (3).

Q. 4 A spherical black body with a radius of 12 cm radiates 450 watt power at 500 K. If the radius were halved and the temperature doubled, the power radiated in watt would be :

Option 1:
225

Option 2:
450

Option 3:
1000

Option 4:
1800

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Correct Answer:
1800

Solution:
As we learnt in

Radiant Power (P) -

$$P = \frac{Q}{t} = A\epsilon\sigma\theta^4$$

-

Power of radiation is

$P = \sigma AT^4$ (For blackbody)

$$\frac{P_2}{P_1} = \left(\frac{A_2}{A_1}\right) \cdot \left(\frac{T_2}{T_1}\right)^4 = \left(\frac{r_2}{r_1}\right)^2 \cdot \left(\frac{T_2}{T_1}\right)^4$$

According to question

$$\frac{r_2}{r_1} = \frac{1}{2} \quad and \quad \frac{T_2}{T_1} = 2$$

Then $\frac{P_2}{P_1} = \left(\frac{1}{2}\right)^2 .2^4 = 2^2$

or $P_2 = 4P_1 = 4 \times 450$ W = 1800 W

Correct option is 4.

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Q. 5 A long solenoid of diameter 0.1 m has 2 x 10^4 turns per meter. At the centre of the solenoid, a coil of 100 turns and a radius of 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0 A from 4 A in 0.05 s. If the resistance of the coil is $10\pi^2\Omega$, the total charge flowing through the coil during this time is:

Option 1:
$32\pi\mu C$

Option 2:
$16\mu C$

Option 3:
$32\mu C$

Option 4:
$16\pi\mu C$

Correct Answer:
$32\mu C$

Solution:
The field at the centre of the solenoid $B = \mu_0 N_l i$

where Nl-number of turns per unit length
Flux through the inner loop =
$100 \times \pi 0.01^2 B = 100 \times \pi 0.01^2 \times \mu_0 N_l i$

emf induced = rate of change of flux = $\frac{\triangle\phi}{\triangle t} = 100 \times \pi 0.01^2 \times \mu_0 N_l \frac{\triangle i}{\triangle t}$

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Current through loop =

$$i = \frac{emf}{R} = \frac{\triangle\phi}{\triangle t} = 100 \times \pi 0.01^2 \times \mu_0 N_l \frac{\triangle i}{\triangle t}\frac{1}{R}$$

Total charge through the coil during this time

$$i\triangle t = \frac{\triangle\phi}{\triangle t} = 100 \times \pi 0.01^2 \times \mu_0 N_l \triangle i \frac{1}{R} = 3.2 \times 10^{-5} C$$

Hence, the answer is the option (3).

Q. 6 Two rods *A* and *B* of different materials are welded together as shown in figure. Their thermal conductivities are K_1 and K_2. The thermal conductivity of the composite rod will be :

Option 1:

$\frac{K_1 + K_2}{2}$

Option 2:

$\frac{3(K_1 + K_2)}{2}$

Option 3:

$K_1 + K_2$

Option 4:

2(K1 + K2)

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Correct Answer:

$$\frac{K_1 + K_2}{2}$$

Solution:

$$R_1 = \frac{d}{k_1 A}$$

$$R_2 = dK_2 A$$

$$Req = \frac{R_1 R_2}{R_1 + R_2}$$

$$\frac{d}{keq(2A)} = \frac{(\frac{d}{k_1 A})(\frac{d}{k_2 A})}{\frac{d}{A}(\frac{1}{k_1} + \frac{1}{k_2})}$$

$$\frac{1}{2keq} = \frac{\frac{1}{k_1 k_2}}{\frac{k_1 + k_2}{k_1 k_2}} = \frac{1}{k_1 + k_2}$$

$$keq = \frac{k_1 + k_2}{2}$$

Q. 7 A Carnot engine having an efficiency of $\frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is :

Option 1:

1 J

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Option 2:
90 J

Option 3:
99 J

Option 4:
100 J

Correct Answer:
90 J

Solution:
As we learnt in

Coefficient of performance (beta) -

$$\beta = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

- wherein

$T_1 > T_2$

For a perfect refrigerator $\beta \rightarrow \infty$

Efficiency of a carnot cycle -

$$\eta = \frac{W}{Q_1 - Q_2} = 1 - \frac{T_2}{T_1}$$

$T_1 \, and \, T_2$ are in kelvin

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- wherein

$T_1 =$ Source temperature

$T_2 =$ Sink Temperature

$(T_1 > T_2)$

For a carnot cycle

$$\beta = \frac{1}{\eta} - 1$$

β = Coefficient of performance

η = Efficiency of carnot engine

$$\eta = \frac{1}{10} \qquad => \beta = 9$$

$$=> \beta = \frac{Q_2}{W}$$

$$Q_2 = \beta.W = 9 \times 10J = 90J$$

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Q. 8 A spring of force constant k is cut into lengths of ratio 1 : 2 : 3. They are connected in series and the new force constant is k'. Then they are connected in parallel and force constant is k'' . Then k' : k'' is:

Option 1:
1:6

Option 2:
1:9

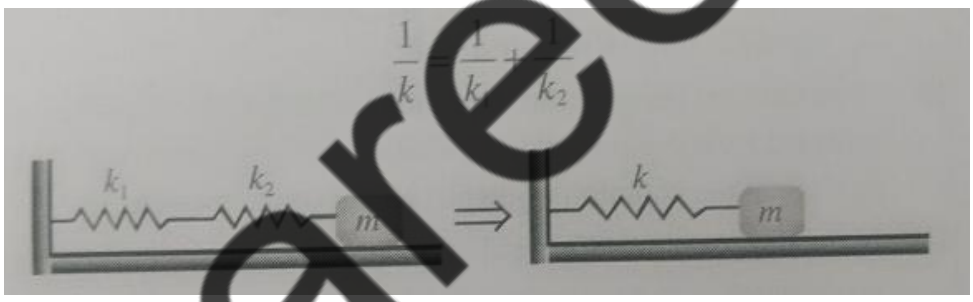
Option 3:
1:11

Option 4:
1:14

Correct Answer:
1:11

Solution:
As we learnt in

Series combination of spring -



- wherein

$$\frac{1}{K_{eq}}=\frac{1}{K_1}+\frac{1}{K_2}$$

$K_1 and\ K_2$ are spring constants of spring 1 & 2 respectively.

and

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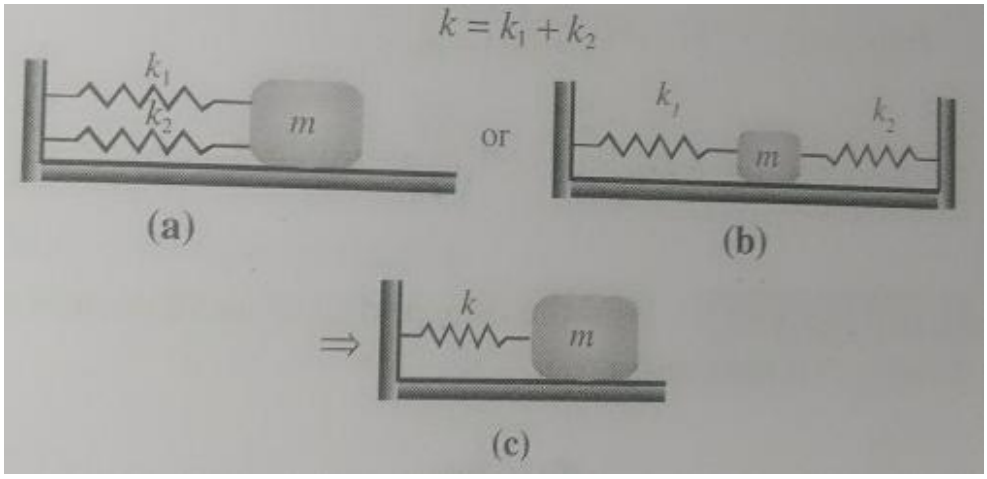
Parallel combination of spring -



- wherein

$$K_{eq} = K_1 + K_2$$

$K_1 and\ K_2$ are spring constants of spring 1 & 2 respectively.

Three strings of lengths $\frac{l}{6}$, $\frac{2l}{6}$, $and\ \frac{3l}{6}$ with force constant

$$K_1 = \frac{Kl}{\frac{l}{6}} = 6K$$

$$K_2 = \frac{Kl}{\frac{2l}{6}} = 3K$$

$$K_3 = \frac{Kl}{\frac{3l}{6}} = 2K$$

Connected in series

$$\frac{l}{K'} = \frac{1}{6K} + \frac{1}{3K} + \frac{1}{2K} = \frac{1+2+3}{6K} = \frac{1}{K}$$

$$\Rightarrow K' = K$$

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When connected in paralllel

K" = 6K + 3K + 2K = 11K

$$\frac{K'}{K''} = \frac{K}{11K} = \frac{1}{11}$$

Q. 9 The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water at a temperature T (Kelvin) and mass m, is :

Option 1:
$\frac{\mathrm{h}}{\sqrt{\mathrm{mkT}}}$

Option 2:
$\frac{\mathrm{h}}{\sqrt{3\mathrm{mkT}}}$

Option 3:
$\frac{2\mathrm{h}}{\sqrt{3\mathrm{mkT}}}$

Option 4:
$\frac{2\mathrm{h}}{\sqrt{\mathrm{mkT}}}$

Correct Answer:
$\frac{\mathrm{h}}{\sqrt{3\mathrm{mkT}}}$

Solution:
De - Broglie wavelength with charged particle -

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$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mE}}$$

$$\lambda = \frac{h}{\sqrt{2mqv}}$$

- wherein

$$E \rightarrow kinetic\ energy\ of particle$$

$$q \rightarrow charged\ particle$$

The kinetic energy of neutrons is

$$K.E = \frac{3}{2}KT \text{(Due to thermal motion)}$$

$$=> \lambda = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2m.\left(\frac{3}{2}KT\right)}}$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{3mkT}}$$

Hence, the answer is the option (2).

Q. 10 Radioactive material 'A' has decay constant '8λ' and material 'B' has decay constant 'λ'. Initially they have same number of nuclei. After what time, the ratio of number of nuclei of material 'B' to that 'A' will be $\frac{1}{e}$?

Option 1:

$\frac{1}{\lambda}$

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Option 2:

$\frac{1}{7\lambda}$

Option 3:

$\frac{1}{8\lambda}$

Option 4:

$\frac{1}{9\lambda}$

Correct Answer:

$\frac{1}{7\lambda}$

Solution:

As we learnt in

Number of nuclei after disintegration -

$N = N_0 e^{-\lambda t}$ or $A = A_0 e^{-\lambda t}$

- wherein

Number of nucleor activity at a time is exponentional function

At any time t, number of nuclei

N= $N_0.e^{-\lambda t}$

For A, $N_A = N_0\, e^{-8\lambda t}$

For B, $N_B = N_0\, e^{-\lambda t}$

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$$\frac{N_B}{N_A} = \frac{1}{e} = e^{-7\lambda t}$$

$$\Rightarrow 7\lambda t = 1 \; or \; t = \frac{1}{7\lambda}$$

Q. 11 Young's double slit experiment is first performed in air and then in a medium other than air. It is found that 8th bright fringe in the medium lies where 5th dark fringe lies in air. The refractive index of the medium is nearly:

Option 1:
1.25

Option 2:
1.59

Option 3:
1.69

Option 4:
1.78

Correct Answer:
1.78

Solution:
Fringe Width in a medium -

$$\beta' = \frac{\beta}{\mu}$$

wherein

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Since wavelength in a medium become $\lambda = \frac{\lambda_o}{\mu}$

Fringe width in air =$\frac{\lambda_0 D}{d}$

Fringe width in any other medium =$(\frac{\lambda_0}{\mu}).\frac{D}{d}$

According to question

$$8\left(\frac{\lambda_0}{\mu}.\frac{D}{d}\right) = (\frac{9}{2})\frac{\lambda_0 D}{d}$$

or $\mu = \frac{16}{9} = 1.78$

Hence, the answer is the option (4).

Q. 12 A potentiometer is an accurate and versatile device to make electrical measurements of E.M.F. because the method involves :

Option 1:
cells

Option 2:
potential gradients

Option 3:
a condition of no current flow through the galvanometer

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Option 4:

a combination of cells, galvanometer and resistances

Correct Answer:

a condition of no current flow through the galvanometer

Solution:

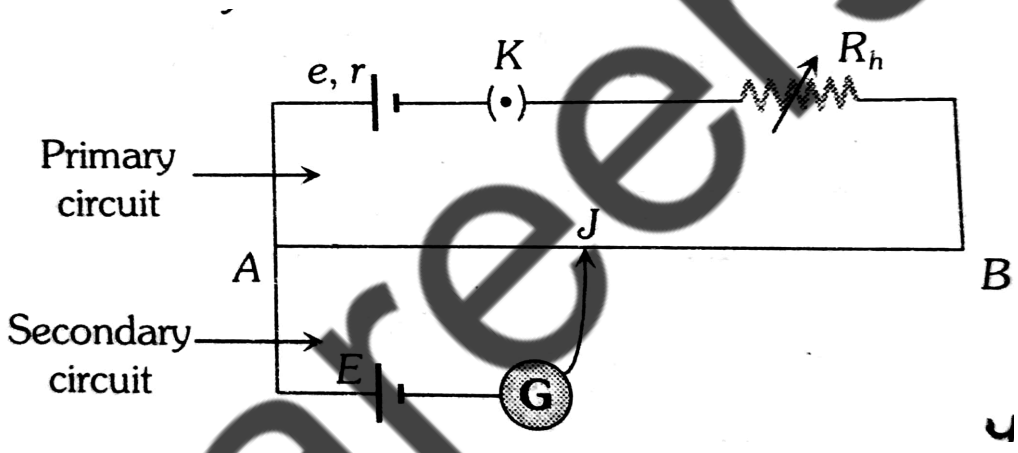
As we learnt in

Potentiometer -

It is a device used to measure e.m.f of a given cell and to compare e.m.f's of cells

- wherein

It is also used to measure internal resistance of given cell



Because the method involves a condition of no current flow through the galvanometer the device can be used to measure P.D internal resistance of a cell and compare emf's of two sources.

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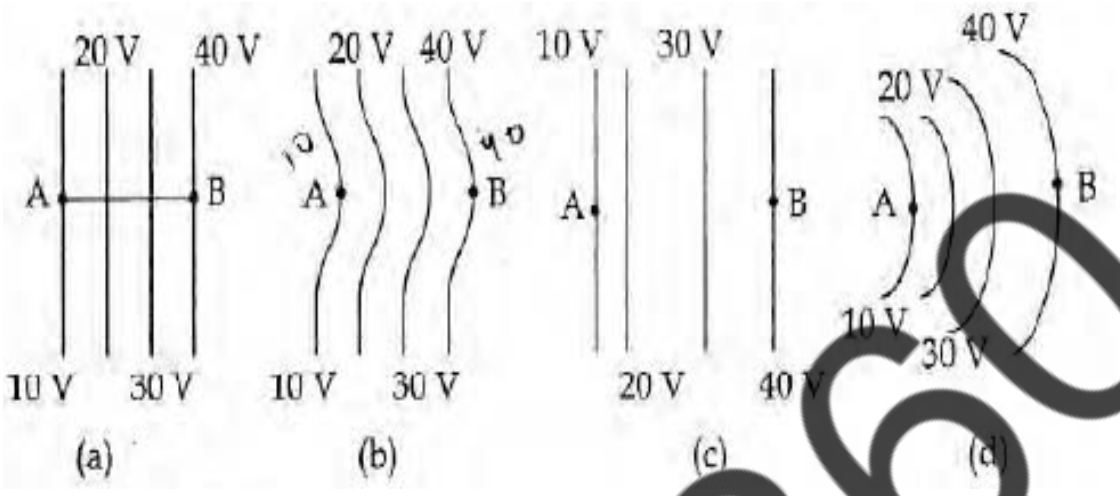
Q. 13 The diagrams below show regions of equipotentials.



A positive charge is moved from A to B in each diagram.

Option 1:
Maximum work is required to move *q* in figure (c).

Option 2:
In all the four cases the work done is the same.

Option 3:
Minimum work is required to move q in figure (a).

Option 4:
Maximum work is required to move q in figure (b).

Correct Answer:
In all the four cases the work done is the same.

Solution:
As we learnt in

Work done in moving a charge -

Always be zero in equipotential surface.

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-

work done is given as $w = q.\Delta v$

In all the four cases the potential difference from A to B is same

$\therefore$ In all the four cases the work done is same.

Q. 14 Two cars moving in opposite directions approach each other with speed of 22 m/s and 16.5 m/s respectively. The driver of the first car blows a horn having a frequency 400 Hz. The frequency heard by the driver of the second car is [velocity of sound 340 m/s]:

Option 1:
350 Hz

Option 2:
361 Hz

Option 3:
411 Hz

Option 4:
448 Hz

Correct Answer:
448 Hz

Solution:
As we learnt in

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Frequency of sound when source and observer are moving towards each other -

$$\nu' = \nu_0 . \frac{C + V_0}{C - V_s}$$

- wherein

$C =$ Speed of sound

$V_0 =$ Speed of observer

$V_s =$ speed of source

$\nu_0 =$ Original frequency

$\nu' =$ apparent frequency

$$f_{app} = f_o \left[\frac{v + v_o}{v - v_s}\right] = 400 \left[\frac{340 + 16.5}{340 - 22}\right]$$

$$\therefore f_{app} = 448 Hz$$

Q. 15 Which one of the following represents forward bias diode?

Option 1:

Option 2:

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Option 3:

Option 4:

Correct Answer:

Solution:

As we learnt in

P -N junction as diode -

It is a one way device. It offers a low resistance when forward biased and high resistance when reverse biased.

- wherein

R = 0, Forward

R $\rightarrow \infty$ Reverse

For forward biased P type should be at higher potential from the figure only 1 is in forward biased mode.

Correct option is 1.

Q. 16 A thin prism having refracting angle 10° is made of glass of refractive index 1.42. This prism is combined with another thin prism of glass of refractive index 1.7. This combination produces dispersion without deviation. The refracting angle of second prism should be :

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Option 1:
4°

Option 2:
6°

Option 3:
8°

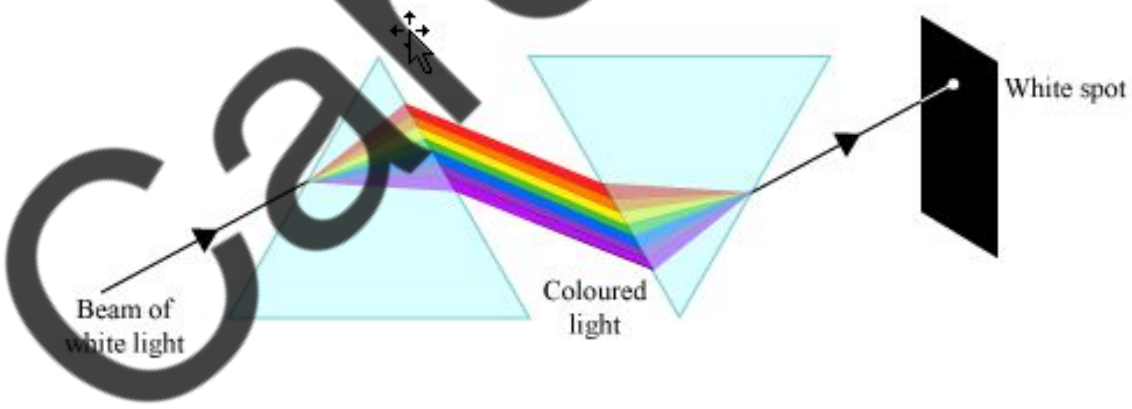
Option 4:
10°

Correct Answer:
6°

Solution:
As we learnt in

Condition for deviation without dispersion -

$$(\mu_v - \mu_r)\, A + (\mu'_v - \mu'_r)\, A' = 0$$



- wherein

μ_v = Refractive index of violet (prism 1)

μ_r = Refractive index of red (prism 1)

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μ_v' = Refractive index of violet (prism 2)

$\mu_r' =$ Refractive index of red (prism 2)

For dispersion without deviation $|\delta_1| = |\delta_2|$

$$(\mu_{1-1})A_1 = (\mu_2 - 1).A_2$$

$$or\ (1.42 - 1) \times 10 = (1.7 - 1)(A_2)$$

$$or\ A_2 = \frac{0.42 \times 10}{0.7} = 6^o$$

Q. 17 One end of string of length l is connected to a particle of mass 'm' and the other end is connected to a small peg on a smooth horizontal table. If the particle moves in circle with speed *v*, the net force on the particle (directed towards centre) will be (T represents the tension in the string)

Option 1:
T

Option 2:
$T + \frac{mv^2}{l}$

Option 3:
$T - \frac{mv^2}{l}$

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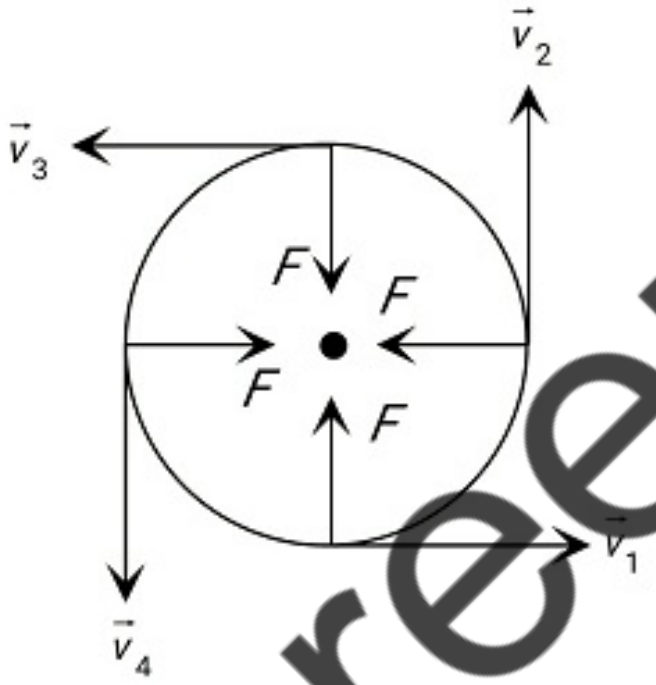
Option 4:
zero

Correct Answer:
T

Solution:
As we learnt in

Centripetal Force -



$$F = 4m\pi^2 n^2 r$$

$$F = \frac{4m\pi^2 n^2 r}{T^2}$$

F = Centripetal force

$\omega =$ Angular velocity

n = frequency

- wherein

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Force acts on the body along the radius and towards the centre.

The net force is the tension on the string = Centripetal force

$= \frac{mv^2}{l}$

Q. 18 The *x* and *y* coordinates of the particle at any time are *x* = 5t - $2t^2$ and *y* = 10t respectively, where *x* and *y* are in meters and t in seconds. The acceleration of the particle at t = 2s is

Option 1:
0

Option 2:
5m /s^2

Option 3:
-4m /s^2

Option 4:
-8m /s^2

Correct Answer:
-4m /s^2

Solution:
As we learnt in

Introduction to Differentiation -

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$$\frac{d}{dx}\left(x^n\right) = nx^{n-1}$$

- wherein

$$\frac{d}{dx}\left(x^5\right) = (n = 5)$$

$$\Rightarrow nx^{n-1}$$

$$\Rightarrow 5x^{5-1}$$

$$\Rightarrow 5x^4$$

x = 5t - $2t^2$ y = 10t

$$\frac{dx}{dt} = 5 - 4t \qquad \frac{dy}{dt} = 10$$

vx=5-4t vy=10

ax=-4 ay=0

Acceleration of particle at $(t = 2) = -4m/s^2$

Correct option is 3.

Q. 19 Suppose the charge of a proton and an electron differ slightly. One of them is -e, the other is (e + $\triangle$e). If the net of electrostatic force and gravitational force between two hydrogen atoms placed at a distance d (much greater than atomic size) apart is zero, then $\triangle$e is of the order of [Given mass of hydrogen m_h = 1.67 x 10^{-27} kg]

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Option 1:
10^{-20} C

Option 2:
10^{-23} C

Option 3:
10^{-37} C

Option 4:
10^{-47} C

Correct Answer:
10^{-37} C

Solution:
As we learnt in

Coulombic force -

$$F \propto Q_1 Q_2 = F \propto \frac{Q_1 Q_2}{r^2} = F = \frac{K Q_1 Q_2}{r^2}$$

- wherein

K - proportionality Constant

Q_1 and Q_2 are two Point charge

Charge on one hydrogen atoms

$$q_e + q_p = -e + (e + \Delta e) = \Delta e$$

$$\therefore F_e = \frac{1}{4\pi\varepsilon_0}\frac{(\Delta e)^2}{d^2}$$

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gravitational force between two hydrogen

$$F_g = \frac{Gm_n m_n}{d^2}$$

net force on the system is zero

$$\therefore F_e = F_g \Rightarrow \frac{(\Delta e)^2}{4\pi\varepsilon_0 d^2} = \frac{Gm_n^2}{d^2}$$

$$(\Delta e)^2 = 4\pi\varepsilon_0 G m_n^2$$

$$\Delta e = 10^{-37} C$$

Q. 20 Which of the following statements are correct?

(a) Centre of mass of a body always coincides with the centre of gravity of the body.

(b) Centre of mass of a body is the point at which the total gravitational torque on the body is zero.

(c) A couple on a body produce both translational and rotational motion in a body.

(d) Mechanical advantage greater than one means that small effort can be used to lift a large load.

Option 1:
(b) and (d)

Option 2:
(a) and (b)

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Option 3:
(b) and (c)

Option 4:
(c) and (d)

Correct Answer:
(b) and (d)

Solution:
Centre of gravity -

The Centre of gravity of a body is a point, through which the resultant of all the forces experienced by various particles of the body due to attraction of earth, passes irrespective of the orientation of the body.

1) The centre of mass of a body always coincides with the centre of gravity of the body is wrong

2) M.A=load / effort

Hence, the answer is the option (1).

Q. 21 The ratio of wavelengths of the last line of Balmer series and the last line of Lyman series is:

Option 1:
2

Option 2:
1

Option 3:
4

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Option 4:
0.5

Correct Answer:
4

Solution:
As we learned in

The energy emitted due to the transition of electron- -

$$\Delta E = Rhcz^2\left(\frac{1}{{n_f}^2} - \frac{1}{{n_i}^2}\right)$$

$$\frac{1}{\lambda} = Rz^2\left(\frac{-1}{{n_i}^2} + \frac{1}{{n_f}^2}\right)$$

- wherein

$$R = Rhydberg\ constant$$

$$n_i = initial\ state$$

$$n_f = final\ state$$

The last line of the Lyman series means the transition from

$n = \infty \rightarrow n = 1$

$$\frac{1}{\lambda_{lymen}} = Rz^2\left(1 - \frac{1}{\infty^2}\right) \quad or \quad \lambda_{lymen} = \frac{1}{Rz^2}$$

The last line of the Balmer series means the transition from

$n = \infty\ to\ n = 2$

$$\frac{1}{\lambda_{Balmer}} = Rz^2(\frac{1}{4} - \frac{1}{\infty}) \quad or \quad \lambda_{Balmer} = \frac{4}{Rz^2}$$

$$\therefore \frac{\lambda_{Balmer}}{\lambda_{lymer}} = 4$$

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Hence, the answer is the option (3).

Q. 22 Consider a drop of rainwater having a mass of 1 g falling from a height of 1 km. It hits the ground with a speed of 50 m/s. Take the 'g' constant with a value of 10 m/s^2. The work done by the (i) gravitational force and the (ii) resistive force of air is:

Option 1:
(i) -10 J (ii) - 8.25 J

Option 2:
(i) 1.25 J (ii) - 8.25 J

Option 3:
(i) 100 J (ii) 8.75 J

Option 4:
(i) 10 J (ii) -8.75 J

Correct Answer:
(i) 10 J (ii) -8.75 J

Solution:
If only conservative forces act on a system, total mechanical energy remains constant -

$$K + U = E\,(constant)$$

$$\Delta K + \Delta U = 0$$

$$\Delta K = -\Delta U$$

From work energy theorem

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$W_g = W_a = \Delta k$

$mgh + W_a = \frac{1}{2}mv^2$

$10^{-3} \times 10 \times 10^3 + W_a = \frac{1}{2} \times 10^{-3} \times 50^2$

$W_a = -8.75J$ (work due to air resistance)

work done by gravity=$mgh = 10^{-3} \times 10 \times 10^3 = 10J$

Hence, the answer is the option (4).

Q. 23 The two nearest harmonics of a tube closed at one end and open at the other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system?

Option 1:
10 Hz

Option 2:
20 Hz

Option 3:
30 Hz

Option 4:
40 Hz

Correct Answer:
20 Hz

Solution:

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The frequency in open organ pipe -

$$\nu = n \cdot \frac{V}{2l}$$

The nearest harmonics of an organ pipe closed at one and differ by twice its fundamental frequency

$$\therefore 260 - 220 = 2v$$

$$v = 20Hz$$

Hence, the answer is the option (2).

Q. 24 Two Polaroids P_1 and P_2 are placed with their axes perpendicular to each other. Unpolarised light I_0 is incident on P_1. A third polaroid P_3 is kept in between P_1 and P_2 such that its axis makes an angle of 45° with that of P_1. The intensity of transmitted light through P_2 is:

Option 1:
$\frac{I_0}{2}$

Option 2:
$\frac{I_0}{4}$

Option 3:
$\frac{I_0}{8}$

Option 4:
$\frac{I_0}{16}$

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Correct Answer:

$\frac{I_0}{8}$

Solution:

Malus Law -

$$I = I_0 \cdot \cos^2\theta$$

Intensity of light after crossing $P_1 = \frac{I_0}{2}$

Intensity of light after crossing $P_3 = \frac{I_0}{2}.cos^2 45^\circ = \frac{I_0}{4}$

Intensity of light after crossing $P_2 = \frac{I_0}{4}.cos^2 45^\circ$

$$I = \frac{I_0}{8}$$

Hence, the answer is the option (3).

Q. 25 The acceleration due to gravity at a height 1 km above the earth is the same as at a depth d below the surface of earth. Then :

Option 1:

$d = \frac{1}{2}\text{km}$

Option 2:

$d = 1\,\text{km}$

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Option 3:

$$\mathrm{d} = \frac{3}{2}\mathrm{km}$$

Option 4:

$$\mathrm{d} = 2\,\mathrm{km}$$

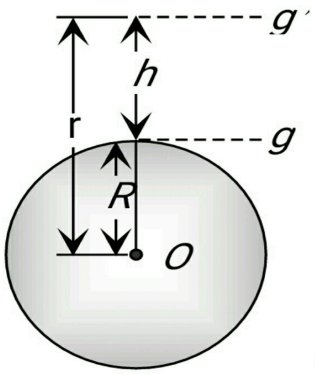
Correct Answer:

$$\mathrm{d} = 2\,\mathrm{km}$$

Solution:

As we learnt in

Variation in 'g' with height -



$$g' = g\left(\frac{R}{R+h}\right)^2$$

$g' \rightarrow$ gravity at height h from surface of earth.

$R \rightarrow$ Radius of earth

$h \rightarrow$ height above surface

- wherein

$$g' \alpha \frac{1}{r^2}$$

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$r = R + h$

Acceleration due to gravity above earth $g' = g_o\left(1 - \frac{2h}{R}\right)$. Since h<<R

Acceleration due to gravity below earth surface $g_2 = g_o\left(1 - \frac{d}{R}\right)$

$\because g_1 = g_2$

or $g_o\left(1 - \frac{2h}{R}\right) = g_o\left(1 - \frac{d}{R}\right)$

or d = 2h

d = 2 Km

Q. 26 The bulk modulus of a spherical object is 'B'. If it is subjected to uniform pressure 'p', the fractional decrease in radius is :

Option 1:

$\frac{\mathrm{p}}{\mathrm{B}}$

Option 2:

$\frac{\mathrm{B}}{3\mathrm{p}}$

Option 3:

$\frac{3\mathrm{p}}{\mathrm{B}}$

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Option 4:

$\frac{\mathrm{p}}{3\mathrm{B}}$

Correct Answer:

$\frac{\mathrm{p}}{3\mathrm{B}}$

Solution:

Bulk Modulus -

Ratio of normal stress to volumetric strain.

$$K = \frac{f/A}{-\Delta v/v} = \frac{-Fv}{A\Delta v}$$

$$K = \frac{-Pv}{\Delta v}$$

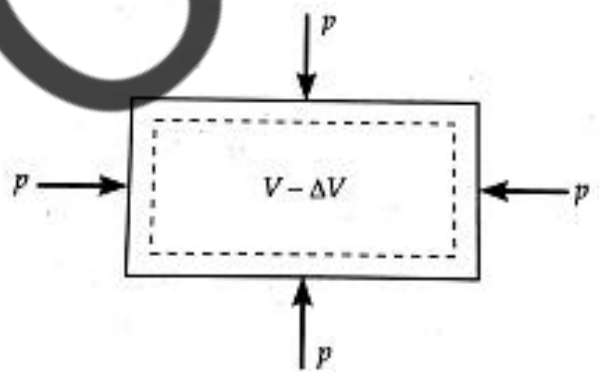
v = Original volume

Δv = Change in volume

P = Increase in pressure

-ve(sign) shows volume (Δv) decrease.

- wherein



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Buck modulus is

$$B = -V\frac{\Delta P}{\Delta V}$$

or $\frac{\Delta V}{V} = \frac{-\Delta P}{B} = -\frac{P}{B} - - - - - (1)$

$$V = \frac{4\pi}{3}r^3 \; or \; \frac{dv}{v} = 3\frac{dr}{r}$$

$$= 3\left(\frac{\Delta r}{r}\right) = -\frac{P}{B}$$

or

$$= \frac{-\Delta r}{r} = +\frac{P}{3B}$$

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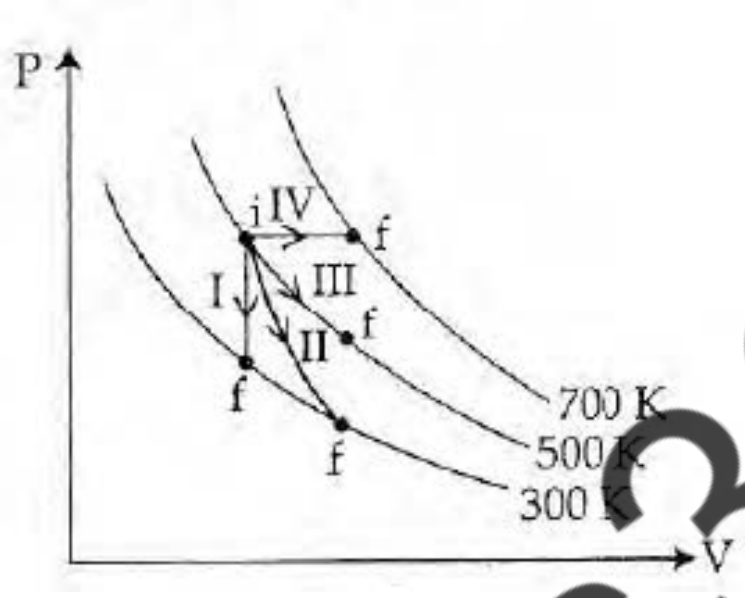
Q. 27 Thermodynamic processes are indicated in the following diagram.



Match the following

Column-1	**Column-2**
P. Process I	a. Adiabatic
Q. Process II	b. Isobaric
R. Process III	c. Isochoric
S. Process IV	d. Isothermal

Option 1:
P $\rightarrow$ a, Q $\rightarrow$ c, R $\rightarrow$ d, S $\rightarrow$ b

Option 2:
P $\rightarrow$ c, Q $\rightarrow$ a, R $\rightarrow$ d, S $\rightarrow$ b

Option 3:
P $\rightarrow$ c, Q $\rightarrow$ d, R $\rightarrow$ b, S $\rightarrow$ a

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Option 4:
P ⟶ d, Q ⟶ b, R ⟶ a, S ⟶ c

Correct Answer:
P ⟶ c, Q ⟶ a, R ⟶ d, S ⟶ b

Solution:

Path (I) → constant volume → isochoric
Path (II) → $PV^{\gamma} = k$ → adiabatic
Path (III) → constant temperature → Isothermal
Path (IV) → constant pressure → isobaric

Q. 28 In an electromagnetic wave in free space the root mean square value of the electric field is E_{rms} = 6 V/m. The peak value of the magnetic field is

Option 1:
1.41 x 10^{-8} T

Option 2:
2.83 x 10^{-8} T

Option 3:
0.70 x 10^{-8} T

Option 4:
4.23 x 10^{-8} T

Correct Answer:
2.83 x 10^{-8} T

Solution:

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$$\frac{E_{rms}}{B_{rms}} = c$$

$$B_{rms} = \frac{E_{rms}}{c} = \frac{6}{3 \times 10^8}$$

$$B_{\text{rms}} = 2 \times 10^{-8}$$

$$B_{rms} = \frac{B_0}{\sqrt{2}}$$

$$B_0 = \sqrt{2} \times B_{rms}$$

$$= \sqrt{2} \times 2 \times 10^{-8}$$

$$= 2.83 \times 10^{-8}\ \text{T}$$

Hence, the answer is the equation (2).

Q. 29 A rope is wound around a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N?

Option 1:
25 m/s^2

Option 2:
0.25 rad/s^2

Option 3:
25 rad/s^2

Option 4:
5 m/s^2

Correct Answer:
25 rad/s^2

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Solution:

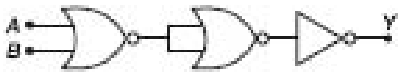
$$T = I\alpha$$

$$RF = mr^2\alpha$$

$$\alpha = \frac{F}{mR} = \frac{30}{3 \times \frac{40}{100}} = 25 rad/s^2$$

Hence, the answer is the option (3).

Q. 30 The given electrical network is equivalent to:



Option 1:
AND gate

Option 2:
OR gate

Option 3:
NOR gate

Option 4:
NOT gate

Correct Answer:
NOR gate

Solution:
As we learnt in

NOR Gate -

NOT + OR Gate

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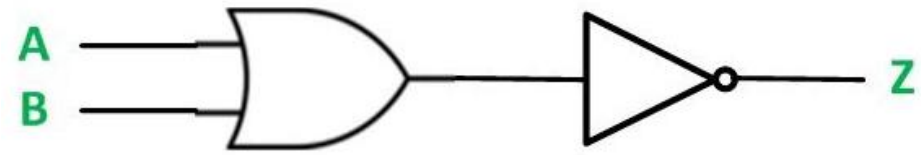
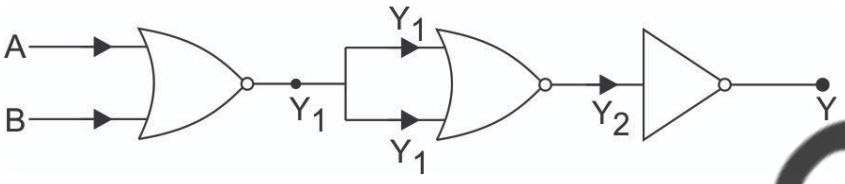


- wherein

$$Y = \overline{A + B}$$

A and B are input

Y is out put



$$Y_1 = \overline{A + B}$$

$$Y_2 = \overline{Y_1} = \phi.\overline{\overline{A + B}} = A + B$$

$$Y = \overline{Y}_2 = \overline{A + B}$$

It represent NOR gate

Q. 31 If θ_1 and θ_2 be the apparent angles of dip observed in two vertical planes at right angles to each other, then the true angle of dip θ is given by :

Option 1:

$\cot^2\theta = \cot^2\theta_1 + \cot^2\theta_2$

Option 2:

$\tan^2\theta = \tan^2\theta_1 + \tan^2\theta_2$

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Option 3:
$\cot^2\theta = \cot^2\theta_1 - \cot^2\theta_2$

Option 4:
$\tan^2\theta = \tan^2\theta_1 - \tan^2\theta_2$

Correct Answer:
$\cot^2\theta = \cot^2\theta_1 + \cot^2\theta_2$

Solution:
If θ_1 and θ_2 are apparent angles of dip. Let α be the angle that one of the planes makes with the magnetic meridian.

$$\tan\theta_1 = \frac{v}{H\cos\alpha} \text{ i.e., } \cos\alpha = \frac{v}{H\tan\theta_1} .....(1)$$

$$\tan\theta_2 = \frac{v}{H\sin\alpha}, \text{ i.e., } \sin\alpha = \frac{v}{H\tan\theta_2} .....(2)$$

Squaring and adding (1) and (2), we get

$$\cos^2\alpha + \sin^2\alpha = \left(\frac{v}{H}\right)^2\left(\frac{1}{\tan^2\theta_1} + \frac{1}{\tan^2\theta_2}\right)$$

$$\text{i.e., } 1 = \frac{v^2}{H^2}\left[\cot^2\theta_1 + \cot^2\theta_2\right]$$

$$\text{or } \frac{H^2}{v^2} = \cot^2\theta_1 + \cot^2\theta_2$$

$$\text{i.e., } \cot^2\theta = \cot^2\theta_1 + \cot^2\theta_2$$

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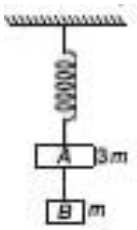
Q. 32 Two blocks A and B of masses 3m and m respectively are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in figure. The magnitudes of acceleration of A and B immediately after the string is cut, are respectively:



Option 1:

$g, \frac{g}{3}$

Option 2:

$\frac{g}{3}, g$

Option 3:

g, g

Option 4:

$\frac{g}{3}, \frac{g}{3}$

Correct Answer:

$\frac{g}{3}, g$

Solution:

As we learnt in

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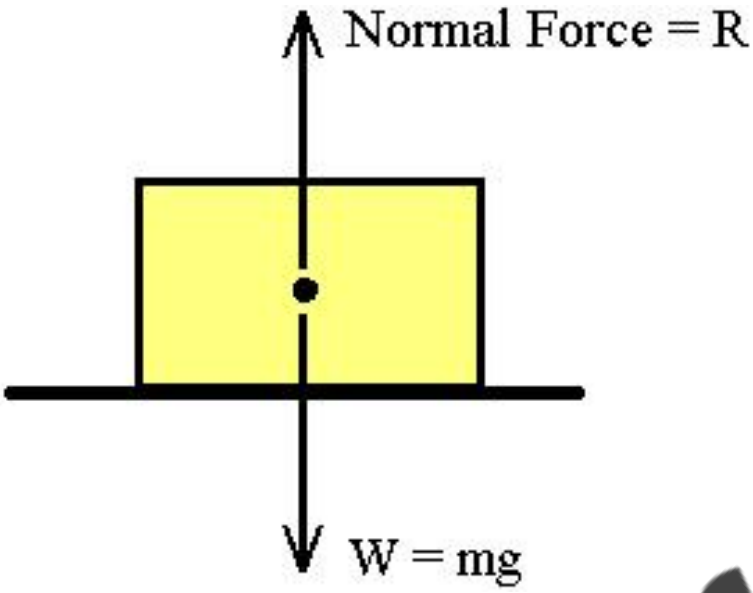
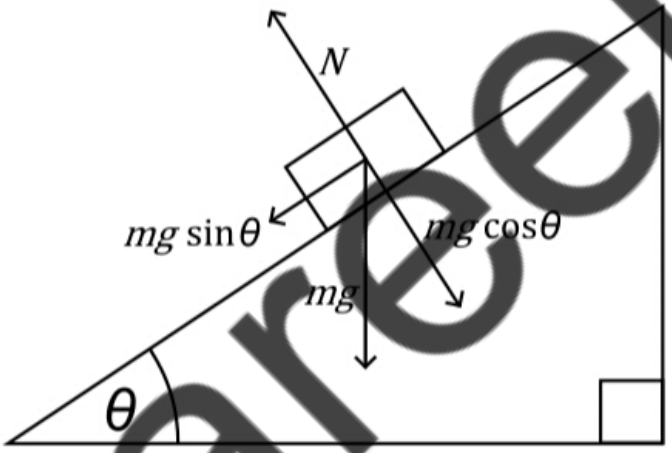
Reaction or Normal Force -

A force which is perpendicular to the surface in contact.

- wherein





Before cut Kx = 4mg

After cut T=0

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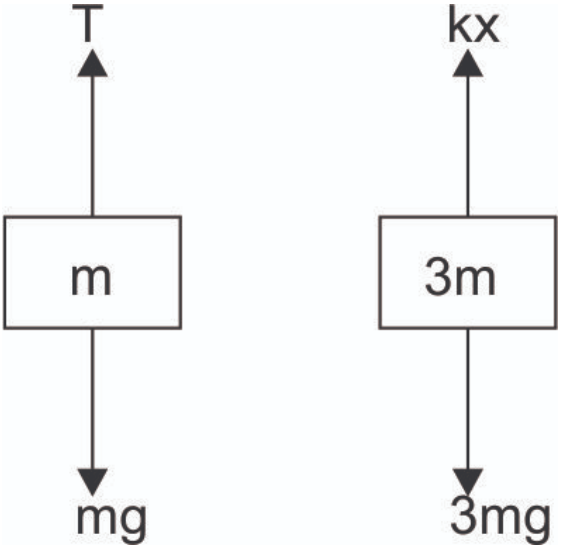

$$a = \frac{kx - 3mg}{3m}$$

$$a = \frac{4mg - 3mg}{3m}$$

$$a = \frac{g}{3}$$

Q. 33 Two discs of same moment of inertia rotating about their regular axis passing through centre and perpendicular to the plane of disc with angular velocities ω_1 and ω_2. They are brought into contact face to face coinciding the axis of rotation. The expression for loss of energy during this process is:

Option 1:

$$\frac{1}{2}\mathrm{I}\left(\omega_1 + \omega_2\right)^2$$

Option 2:

$$\frac{1}{4}\mathrm{I}\left(\omega_1 - \omega_2\right)^2$$

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Option 3:

$\mathrm{I}\left(\omega_1-\omega_2\right)^2$

Option 4:

$\frac{\mathrm{I}}{8}\left(\omega_1-\omega_2\right)^2$

Correct Answer:

$\frac{1}{4}\mathrm{I}\left(\omega_1-\omega_2\right)^2$

Solution:

As we learnt in

Resolving power $\propto \frac{1}{\lambda}$

$\therefore \frac{R_1}{R_2}=\frac{\lambda_2}{\lambda_1}=3:2$

Law of conservation of angular moment -

$\vec{\tau}=\frac{d\vec{L}}{dt}$

- wherein

If net torque is zero

i.e. $\frac{d\vec{L}}{dt}=0$

$\vec{L}=constant$

angular momentum is conserved only when external torque is zero .

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$$Iw_1 + Iw_2 = 2Iw \Rightarrow w = \frac{w_1 + w_2}{2}$$

$$(K.E)_i = \frac{1}{2}Iw_1^2 + \frac{1}{2}w_2^2$$

$$(K.E)_f = \frac{1}{2} \times 2Iw^2 = I\left(\frac{w_{1+w_2}}{2}\right)^2$$

Loss in K.E = $\frac{1}{4}I\left(w_1 - w_2\right)^2$

Q. 34 A 250-Turn rectangular coil of length 2.1 cm and width 1.25 cm carries a current of 85 μA and subjected to a magnetic field of strength 0.85 T. Work done for rotating the coil by 180° against the torque is:

Option 1:
9.1μ J

Option 2:
4.55μ J

Option 3:
2.3μ J

Option 4:
1.15μ J

Correct Answer:
9.1μ J

Solution:

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$W = MB(\cos\theta_1 - \cos\theta_2)$

When it is rotated by angle 180° then

$W = 2MB$

$W = 2(NIA)B$

$= 2 \times 250 \times 85 \times 10^{-6}[1.25 \times 2.1 \times 10^{-4}] \times 85 \times 10^{-2}$

9.1μJ

Q. 35 A capacitor is charged by a battery. The battery is removed and another identical uncharged capacitor is connected in parallel. The total electrostatic energy of resulting system:

Option 1:
increases by a factor of 4

Option 2:
decreases by a factor of 2

Option 3:
remains the same

Option 4:
increases by a factor of 2

Correct Answer:
decreases by a factor of 2

Solution:
As we learnt in

Energy Stored -

$$U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}$$

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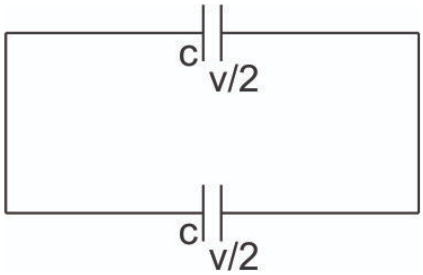
-

When the capacitor is charged by a battery of potential V, then energy stored in the capaciter

$$U_i = \frac{1}{2}CV^2$$

When the battery is removed and another identical unchanged capacitor is connected in parallel



Common potential $V' = \frac{CV}{C+C} = \frac{V}{2}$

$$U_f = \frac{1}{2}(2C)(\frac{V}{2})^2 = \frac{1}{4}CV^2$$

$$U_f = \frac{U_i}{2}$$

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Q. 36 Two astronauts are floating in gravitational free space after having lost contact with their spaceship. The two will:

Option 1:
keep floating at the same distance between them.

Option 2:
move towards each other.

Option 3:
move away from each other.

Option 4:
will become stationary.

Correct Answer:
keep floating at the same distance between them.

Solution:
As we learnt in

Vector form of formula -

$$\vec{F}_{12} = \frac{-Gm_1m_2}{r^2}(\hat{r}_{21})$$

$$\vec{F}_{12} = \frac{-Gm_1m_2}{r^3}(\vec{r}_{21})$$

$\hat{r}_{21} \rightarrow$ Positive vector

- wherein

$\hat{r}_{12} = -\hat{r}_{21}$

So, $\vec{F}_{12} = -\vec{F}_{21}$

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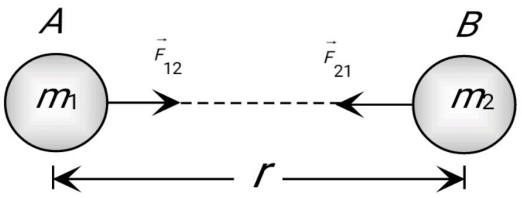
2



The distance between the two astronauts will remain same as in space there is no significant force acting on them . the gravitational force between them is negligibly small

---

Q. 37 A beam of light from a source L is incident normally on a plane mirror fixed at a certain distance *x* from the source. The beam is reflected back as a spot on a scale placed just above the source L. When the mirror is rotated through a small angle θ, the spot of the light is found to move through a distance *y* on the scale. The angle θ is given by :

Option 1:

$\frac{y}{2x}$

Option 2:

$\frac{y}{x}$

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Option 3:

$\frac{x}{2y}$

Option 4:

$\frac{x}{y}$

Correct Answer:

$\frac{y}{2x}$

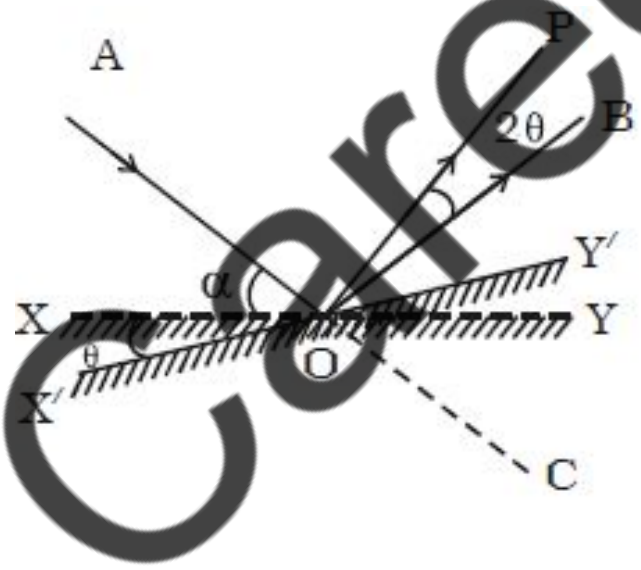
Solution:

As we learnt in

Rotaion of Mirror -

Keeping the incidence ray fixed, When mirror is rotated by an angle θ, the reflected ray is rotated by an angle 2θ.

- wherein



When mirror is rotated by an angle Θ then reflected ray will be rotated by an angle 2Θ

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$$\therefore 2\Theta = \frac{y}{x}$$

$$\therefore \Theta = \frac{y}{2x}$$

Q. 38 In a common emitter transistor amplifier the audio signal voltage across the collector is 3 V. The resistance of collector is 3 $K\Omega$. If current gain is 100 and the base resistance is 2 $K\Omega$, the voltage and power gain of the amplifier is :

Option 1:
200 and 1000

Option 2:
15 and 200

Option 3:
150 and 15000

Option 4:
20 and 2000

Correct Answer:
150 and 15000

Solution:
As we learnt in

Amplifier -

Transistor is used as an amplifier

- wherein

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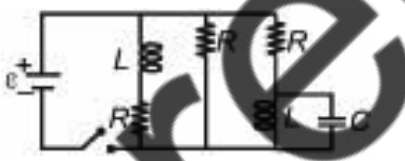
in such circuit output is amplified as compared to input .

$$\begin{array}{l|l} V_o = 3V & \beta = 100 \\ R_L = 3K\Omega & R_{in} = 2K\Omega \end{array}$$

Voltage gain $=\beta.\frac{R_L}{R_{in}} = \frac{3k\Omega}{2k\Omega} \times 100 = 150$

Power gain= $\beta^2 \frac{R_L}{R_{in}} = 10^4 \times \frac{3}{2} = 15000$

Q. 39 Figure shows a circuit contains three identical resistors with resistance R = 9.0 Ω each, two identical inductors with inductance L = 2.0 mH each, and an ideal battery with emf ε = 18 V. The current 'i' through the battery just after the switch closed is, ...



Option 1:
2 mA

Option 2:
0.2 A

Option 3:
2 A

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Option 4:
0 ampere

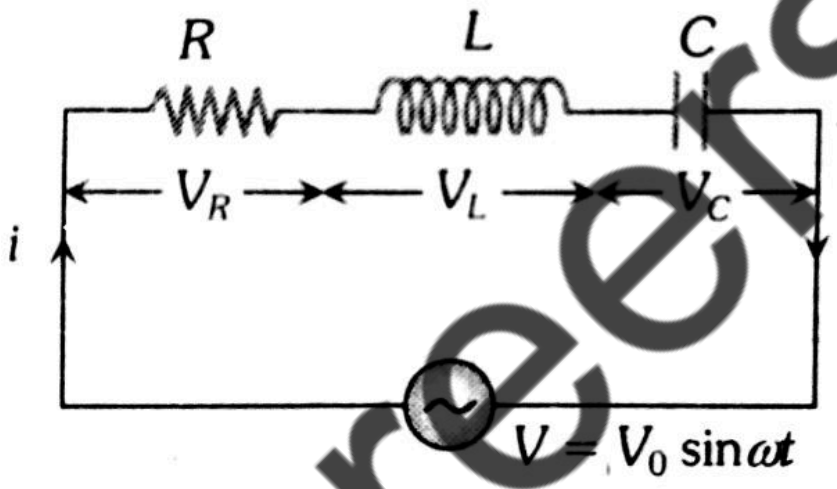
Correct Answer:
2 A

Solution:
As we learnt in

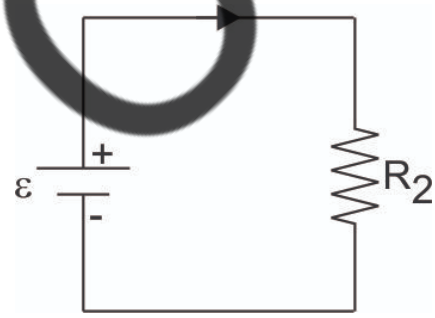
RLC circuit Voltage -

$$V_R = iR \quad V_L = i\,X_L \quad V_C = i\,X_C$$

- wherein



At t=0, no current flows through R_1 and R_3



Current through battery just after the switch closed is

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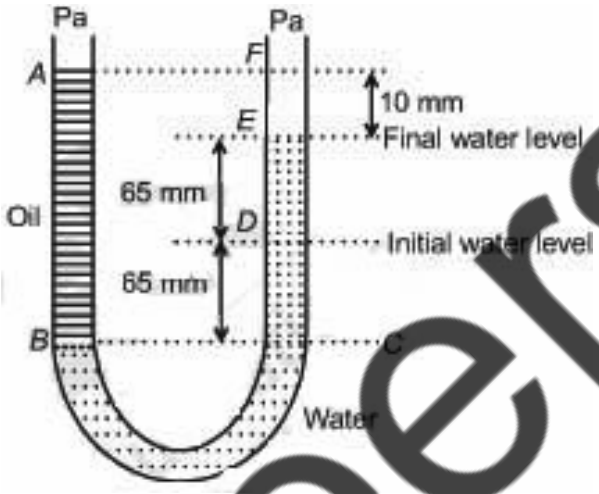
$$i = \frac{\varepsilon}{9} = \frac{18}{9} = 2A$$

Q. 40 A U tube with both ends open to the atmosphere, is partially filled with water. Oil, which is immiscible with water, is poured into one side until it stands at a distance of 10 mm above the water level on the other side. Meanwhile the water rises by 65 mm from its original level (see diagram). The density of the oil is



Option 1:
650 kg m^{-3}

Option 2:
425 kg m^{-3}

Option 3:
800 kg m^{-3}

Option 4:
928 kg m^{-3}

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Correct Answer:
928 kg m^{-3}

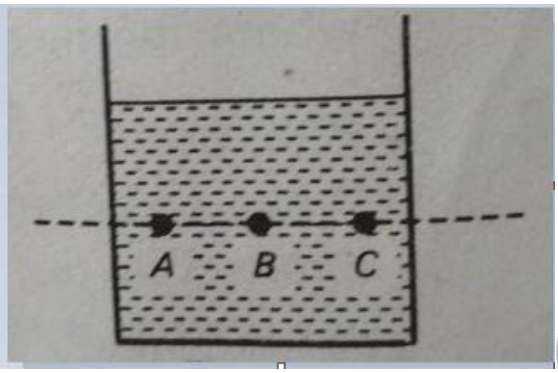
Solution:
As we learnt in

Pressure is uniform on a horizontal line -

In horizontal line or in horizontal plane in stationary liquid

$$P_A = P_B = P_C$$

- wherein



Gauge pressure -

Pressure diffrence between hydrostatic and atmospheric pressure

- wherein

$$P - P_0 = \rho g h$$

$$P - P_0 = gauge\ pressure$$

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Pressure of two points lie in the same horizontal level should be same and $P = \rho g h$

Both ends of tube is open, so the pressure on both the free surface must be equal.

$$P_1 = P_2$$

$$\Rightarrow h_{oil}\ \rho_{oil}\ g = h_{water} \cdot \rho_{water}\ g$$

$$\Rightarrow \rho_{oil} = \frac{h_{water} \cdot \rho_{water}}{h_{oil}}$$

$$\rho_{oil} = \frac{(65+65) \times 1000}{(65+65+10)} = 928\ kgm^{-3}$$

Correct option is 4.

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Q. 41 The photoelectric threshold wavelength of silver is 3250 x 10^{-10} m. The velocity of the electron ejected from a silver surface by ultraviolet light of wavelength 2536 x 10^{-10} m is :

(Given h = 4.14 x 10^{-15} eVs and c = 3 x 10^{8} ms^{-1})

Option 1:

$\approx 6 \times 10^{5} \mathrm{ms}^{-1}$

Option 2:

$\approx 6 \times 10^{6} \mathrm{ms}^{-1}$

Option 3:

$\approx 61 \times 10^{3} \mathrm{ms}^{-1}$

Option 4:

$\approx 0.3 \times 10^{6} \mathrm{ms}^{-1}$

Correct Answer:

$\approx 6 \times 10^{5} \mathrm{ms}^{-1}$

Solution:

As we learnt in

Conservation of energy -

$$h\nu = \phi_0 + \frac{1}{2}mv_{max}^2$$

$$h\nu = h\nu_0 + \frac{1}{2}mv_{max}^2$$

$$h\left(\nu - \nu_0\right) = \frac{1}{2}mv_{max}^2$$

$where, h-Plank's constant\ \nu-Frequency\ \nu_0-threshold\ frequency\ \phi_0-work function$

- wherein

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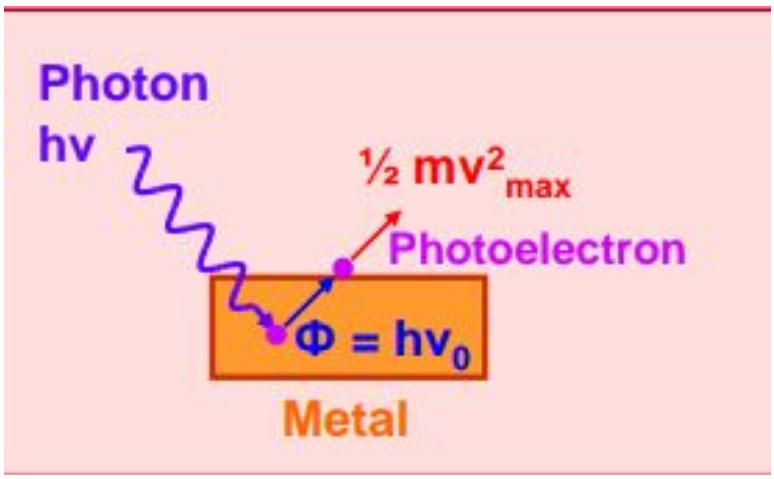

maximum kinetic energy = $\frac{hc}{\lambda} - \phi$

$$K.E. = \frac{hc}{\lambda} = \frac{hc}{\lambda_o} = hc(\frac{1}{\lambda} - \frac{1}{\lambda_o}) = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2hc}{m}.\left(\frac{1}{\lambda} - \frac{1}{\lambda_o}\right)}$$

$h = 6.67 \times 10^{-34} J - s$
$c = 3 \times 10^{18} m/s$
$\lambda = 2536 \times 10^{-10} m$
$\lambda_0 = 3250 \times 10^{-10}$
$m = 9.1 \times 10^{-31} kg$

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Q. 42 A physical quantity of the dimensions of length that can be formed out of *c*, G and $\frac{e^2}{4\pi\epsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and e is charge]:

Option 1:

$$\therefore L = \frac{1}{C^2}\left[G\frac{e^2}{4\pi\epsilon_0}\right]^{1/2}$$

Option 2:

$$L = \frac{1}{C}\left[G\frac{e^2}{4\pi\epsilon_0}\right]^{1/2}$$

Option 3:

$$L = \frac{1}{C^2}\left[\frac{e^2}{4G\pi\epsilon_0}\right]$$

Option 4:

$$\therefore L = \frac{1}{C}\left[G\frac{e^2}{4\pi\epsilon_0}\right]$$

Correct Answer:

$$\therefore L = \frac{1}{C^2}\left[G\frac{e^2}{4\pi\epsilon_0}\right]^{1/2}$$

Solution:

Coulombic force -

$$F \propto Q_1Q_2 = F \propto \frac{Q_1Q_2}{r^2} = F = \frac{KQ_1Q_2}{r^2}$$

- wherein

K - proportionality Constant

Q_1 and Q_2 are two-point charge

$$[L] = [C]^a[G]^b\left[\frac{e^2}{4\pi\epsilon_0}\right]^c$$

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$$[L] = [lT^{-1}]^a [M^{-1}L^3T^{-2}]^b [ML^3T^{-2}]^c$$

$$[L] = L^{a+3b+3c}\ M^{-b+c}\ T^{-a-2b-2c}$$

$$a + 3b + 3c = 1$$

$$-b + c = 0$$

$$a + 2b + 2c = 0$$

after solving we get

$$a = -2, b = \frac{1}{2}, c = \frac{1}{2}$$

$$\therefore L = \frac{1}{C^2}\left[G\frac{e^2}{4\pi\epsilon_0}\right]^{1/2}$$

Hence, the answer is the option (1).

Q. 43 The ratio of resolving powers of an optical microscope for two wavelengths $\lambda_1 = 4000\ \text{Å}$ and $\lambda_2 = 6000\ \text{Å}$ is :

Option 1:
8 : 27

Option 2:
9 : 4

Option 3:
3 : 2

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Option 4:
16 : 81

Correct Answer:
3 : 2

Solution:

$$\text{Resolving power} \propto \frac{1}{\lambda}$$

$$\frac{R_1}{R_2} = \frac{\lambda_2}{\lambda_1}$$

$$= \frac{6000\ \text{Å}}{4000\ \text{Å}}$$

$$= \frac{3}{2}$$

Q. 44 A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is

Option 1:
4 RT

Option 2:
15 RT

Option 3:
9 RT

Option 4:
11 RT

Correct Answer:
11 RT

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Solution:
Total internal energy -

$$U = U_K + U_P$$

- wherein

Change in internal energy

$$\Delta U = n\frac{f}{2}R\Delta T$$

f is the degree of freedom

The internal energy of a system is

$$U = U_1 + U_2$$

$$= n_1 C_{V1} T + n_2 C_{V2} T$$

$$n_1 = 2, C_{V1} = \frac{fR}{2} = \frac{5R}{2}$$

$$n_2 = 4, C_{V2} = \frac{3R}{2}$$

$$\therefore U = \left(2 \times \frac{5R}{2} + 4 \times \frac{3R}{2}\right) T$$

or $U = 11RT$

Hence, the answer is the option (4).

Q. 45 The resistance of a wire is 'R' ohm. If it is melted and stretched to 'n' times its original length, its new resistance will be :

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Option 1:
nR

Option 2:
$\frac{\mathrm{R}}{\mathrm{n}}$

Option 3:
n^2R

Option 4:
$\frac{\mathrm{R}}{\mathrm{n}^2}$

Correct Answer:
n^2R

Solution:
As we learnt in

Resistance formula -

$$R = \rho\frac{l}{A} = \frac{m}{ne^2\tau}\cdot\frac{l}{A}$$

$$R = \rho\frac{1}{A}$$

$$R = \frac{\rho l^2}{V} \Rightarrow R \propto l^2$$

According to the question, $l_2 = nl_1$

$$\frac{R_2}{R_1} = \frac{n^2{l_1}^2}{{l_1}^2} \Rightarrow \frac{R_2}{R_1} = n^2$$

$$\Rightarrow R_2 = n^2R_1$$

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$\because R_1 = R \; and \; R_2 = R'$

$\therefore R' = n^2 R$

Hence, the answer is the option (3).

Chemistry

Q. 1 A first order reaction has a specific reaction rate of $10^{-2}\ \text{sec}^{-1}$. How much time will it take for 20 g of the reactant to reduce to 5 g?

Option 1:
238.6 sec

Option 2:
138.6 sec

Option 3:
346.5 sec

Option 4:
693.0 sec

Correct Answer:
138.6 sec

Solution:
As we learned in

First Order Reaction -

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$$\ln\left[\frac{A_o}{A}\right] = kt$$

where,

$A_o = \text{initial concentration}$
$A = \text{final concentration at time t}$

A first order reaction has a specific reaction rate of $10^{-2}\ \text{sec}^{-1}$.

So, $k = 10^{-2}\ \text{sec}^{-1}$

How much time will it take for 20 g of the reactant to reduce to 5 g?

So, $A_o = 20\ \text{g and } A = 5\ \text{g}$

Hence,

$$\ln\left[\frac{20}{5}\right] = 0.01 \times t$$

$$t = 100 \times \ln 4$$

$$t = 100 \times 2.303 \times \log 4$$

$$t = 138.6\ \text{sec}$$

So, option 2 is correct.

Q. 2 A gas is allowed to expand in a well-insulated container against a constant external pressure of 2.5 atm from an initial volume of 2.50 L to a final volume of 4.50 L. The change in internal energy ΔU of the gas in joules will be :

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Option 1:
1136.25 J

Option 2:
-500 J

Option 3:
-505 J

Option 4:
+505 J

Correct Answer:
-505 J

Solution:

$P = 2.5\ atm$

$v_i = 2.50L$

$v_f = 4.50L$

$\Delta U = ?$

Word

work is done,

$W = -P\Delta v$

$= -2.5\,(4.50 - 2.50)$

$= -5L\ atm$

$= -5 \times 101.3J$

$= -506.5J$

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Now, for adiabatic process, f=0

$$\therefore \Delta U = q + w$$

$$= -506.5J$$

$$= -505J$$

Hence, the answer is the option (3).

Q. 3 Which one is the **correct** order of acidity?

Option 1:
$CH_2 = CH_2 > CH_3 - CH = CH_2 > CH_3 - C \equiv CH > CH \equiv CH$

Option 2:
$CH \equiv CH > CH_3 - C \equiv CH > CH_2 = CH_2 > CH_3 - CH_3$

Option 3:
$CH \equiv CH > CH_2 = CH_2 > CH_3 - C \equiv CH > CH_3 - CH_3$

Option 4:
$CH_3 - CH_3 > CH_2 = CH_2 > CH_3 - C \equiv CH > CH \equiv CH$

Correct Answer:
$CH \equiv CH > CH_3 - C \equiv CH > CH_2 = CH_2 > CH_3 - CH_3$

Solution:
$$CH \equiv CH > CH_2 = CH_2 > CH_3 - CH_3$$

Acidity order

$sp > sp^2 > sp^3$

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More the percentage of s-character more will be electronegativity.

Hence, option number (2) is correct.

Q. 4 Which of the following is a sink for CO?

Option 1:
Haemoglobin

Option 2:
Micro organisms present in the soil

Option 3:
Oceans

Option 4:
Plants

Correct Answer:
Micro organisms present in the soil

Solution:
Carbon monoxide -

Produced due to incomplete combustion of carbon and due to metallurgical operations. Mainly released in air by automobile exhaust.

- wherein

$$2C_{(s)} + O_2 \rightarrow 2CO(g)$$

Microorganisms present in the soil is a sink for CO.

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Q. 5 If molality of the dilute solution is doubled, the value of molal depression constant (K_f) will be :

Option 1:
Doubled

Option 2:
Halved

Option 3:
Tripled

Option 4:
Unchanged

Correct Answer:
Unchanged

Solution:
Mathematical Expression of Depression in Freezing Point -

$$\Delta T_f = K_f m$$

- wherein

m = molarity of solvent

K_f = molal depress const

Units = K-kg-mol^{-1}

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K_f, molal depression constant is independent of the molality of the solution because it is the inherent property of the solvent.

Hence, the answer is the option (4).

Q. 6 With respect to the conformers of ethane, which of the following statements is true?

Option 1:
Bond angle changes but bond length remain same

Option 2:
Both bond angle and bond length change

Option 3:
Both bond angles and bond length remains same

Option 4:
Bond angle remains same but bond length changes

Correct Answer:
Both bond angles and bond length remains same

Q. 7 Pick out the correct statement with respect to $[Mn(CN)_6]^{3-}$:

Option 1:
It is sp^3d^2 hybridised and octahedral

Option 2:
It is sp^3d^2 hybridised and tetrahedral

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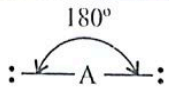
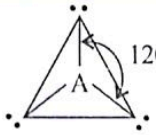
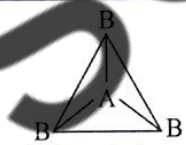
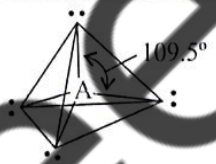
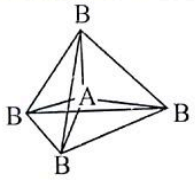
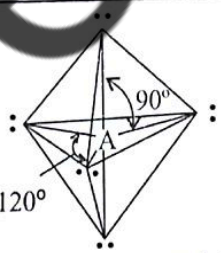
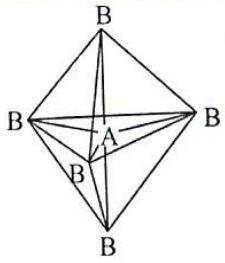
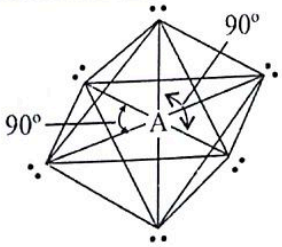
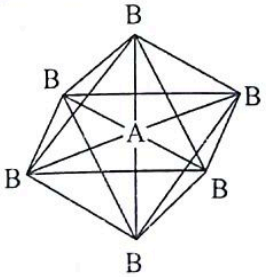
Option 3:
It is d^2sp^3 hybridised and octahedral

Option 4:
It is dsp^2 hybridised and square planar

Correct Answer:
It is d^2sp^3 hybridised and octahedral

Solution:
Shape of molecules -

S. No.	Number of electron pairs	Arrangement of electron pairs	Molecular geometry	Examples
1.	2	 Linear	Linear	$BeCl_2$, $HgCl_2$
2.	3	 Trigonal planar	 Trigonal planar	BF_3
3.	4	 Tetrahedral	 Tetrahedral	CH_4, $NH_4^{\oplus}$
4.	5	 Trigonal bipyramidal	 Trigonal bipyramidal	PCl_5
5.	6	 Octahedral	 Octahedral	SF_6

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-

$[Mn(CN)_6]^{3-} is\ d^2sp^3$ hybridised and it is octahedral.

Q. 8 Which of the following pairs of compounds is isoelectronic and isostructural?

Option 1:
$BeCl_2$, XeF_2

Option 2:
TeI_2, XeF_2

Option 3:
IBr_2^-, XeF_2

Option 4:
IF_3, XeF_2

Correct Answer:
IBr_2^-, XeF_2

Solution:

Species	Number of electron	Structure
$BeCl_2$	38	Linear
XeF_2	72	Linear
TeI_2	158	Bent
XeF_2	72	Linear

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IBr_2^- XeF_2	124 72	Linear Linear
IF_3 XeF_2	80 72	T-shaped Linear

In this question, in place of isoelectronic there should be same number of valence electron.

Hence, the answer is the option (3).

Q. 9 Which one of the following statements is not correct?

Option 1:
Catalyst does not initiate any reaction.

Option 2:
The value of the equilibrium constant is changed in the presence of a catalyst in the reaction at equilibrium.

Option 3:
Enzymes catalyze mainly biochemical reactions.

Option 4:
Coenzymes increase the catalytic activity of the enzyme

Correct Answer:
The value of the equilibrium constant is changed in the presence of a catalyst in the reaction at equilibrium.

Solution:
Catalyst - A catalyst is a substance that increases the rate of a chemical reaction without being consumed in the reaction.

Or

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Catalysts only change the rate of a reaction and do not bother the equilibrium constant in any way.

The catalyst reduces the time required to establish equilibrium but does not increase the yield of the product.

Hence, option 2 is an incorrect statement and the right option.

Q. 10 The concentration of the Ag+ ions in a saturated solution of $Ag_2C_2O_4$ is $2.2x10^{-4}molL^{-1}$. Solubility product of $Ag_2C_2O_4$ is :

Option 1:
2.42×10^{-8}

Option 2:
2.66×10^{-12}

Option 3:
4.5×10^{-11}

Option 4:
5.3×10^{-12}

Correct Answer:
5.3×10^{-12}

Solution:

$$Ag_2C_2O_4 \rightleftharpoons 2Ag^+ + C_2O_4^{2-}$$

$$[Ag^+] = 2[C_2O_4^{2-}] = 2.2 \times 10^{-4}$$

$$[C_2O_4^{2-}] = 1.1 \times 10^{-4}$$

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$$K_{sp} = [Ag^+]^2[C_2O_4^{2-}]$$

$$= \left(2.2 \times 10^{-4}\right)^2 \left(1.1 \times 10^{-4}\right)$$

$$= 5.324 \times 10^{-12}$$

Hence, the answer is the option (4).

Q. 11 The reason for the greater range of oxidation states in actinoids is attributed to:

Option 1:
the radioactive nature of actinoids

Option 2:
actinoid contraction

Option 3:
5f, 6d and 7s levels having comparable energies

Option 4:
4f and *5d* levels being close in energies

Correct Answer:
5f, 6d and 7s levels having comparable energies

Solution:
The reason for the greater range of oxidation states in actinoids is attributed to 5f, 6d and 7s levels having comparable energies. Due to this outermost electrons are excited to higher energy levels giving variable oxidation states.

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Hence, the answer is the option (3).

Q. 12 In the electrochemical cell : $Zn|ZnSO_4(0.01M)||CuSO_4(1.0M)|Cu$, the emf of this Daniel cell is E_1. When the concentration of $ZnSO_4$ is changed to 1.0 M and that of $CuSO_4$ changed to 0.01 M, the emf changes to E_2. From the followings, which one is the relationship between E_1 and E_2? (Given, $\frac{RT}{F} = 0.059$)

Option 1:
$E_1 = E_2$

Option 2:
$E_1 < E_2$

Option 3:
$E_1 > E_2$

Option 4:
$E_2 = 0 \neq E_1$

Correct Answer:
$E_1 > E_2$

Solution:
Cell reaction can be represented as

$$Zn + Cu^{2+} \rightarrow Cu + Zn^{2+}$$

So, the Nernst equation for the above reaction will be

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{2}\log\frac{[Zn^{2+}]}{[Cu^{2+}]}$$

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Now,

For case 1, $[\mathrm{Zn^{2+}}] = 0.01\ \mathrm{M}\ \text{and}\ [\mathrm{Cu^{2+}}] = 1.0\ \mathrm{M}$

So,

$$\mathrm{E_1} = \mathrm{E^{\circ}_{cell}} - \frac{0.0591}{2}\log\frac{0.01}{1.0}$$

$$\mathrm{E_1} = \mathrm{E^{\circ}_{cell}} - \frac{0.0591}{2}(-2)$$

$$\mathrm{E_1} = \mathrm{E^{\circ}_{cell}} + 0.0591$$

For case 2, $[\mathrm{Zn^{2+}}] = 1.0\ \mathrm{M}\ \text{and}\ [\mathrm{Cu^{2+}}] = 0.01\ \mathrm{M}$

So,

$$\mathrm{E_2} = \mathrm{E^{\circ}_{cell}} - \frac{0.0591}{2}\log\frac{1.0}{0.01}$$

$$\mathrm{E_2} = \mathrm{E^{\circ}_{cell}} - \frac{0.0591}{2}(+2)$$

$$\mathrm{E_2} = \mathrm{E^{\circ}_{cell}} - 0.0591$$

From the above two cases,

we get this relation:

$$\mathrm{E_1} > \mathrm{E_2}$$

Hence, the answer is the option (3).

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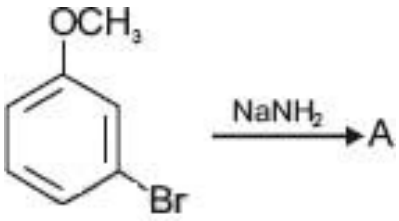
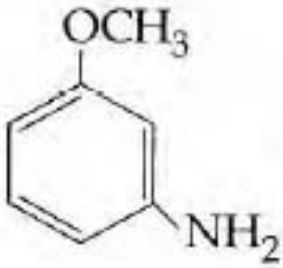
Q. 13 Identify A and predict the type of reaction

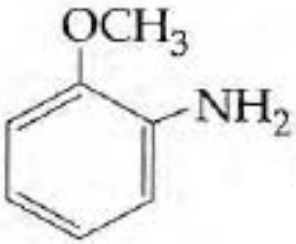


Option 1:



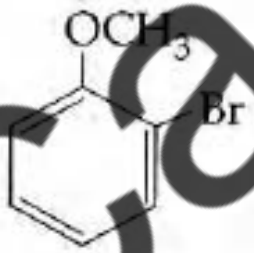
and substitution reaction

Option 2:



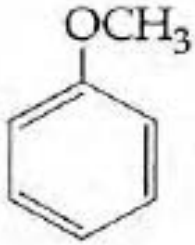
and elimination addition

Option 3:



and cine substitution reaction

Option 4:



and cine substitution reaction

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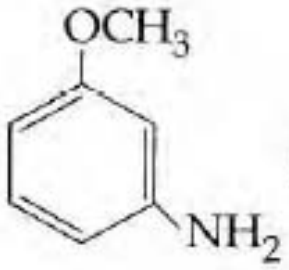
Correct Answer:



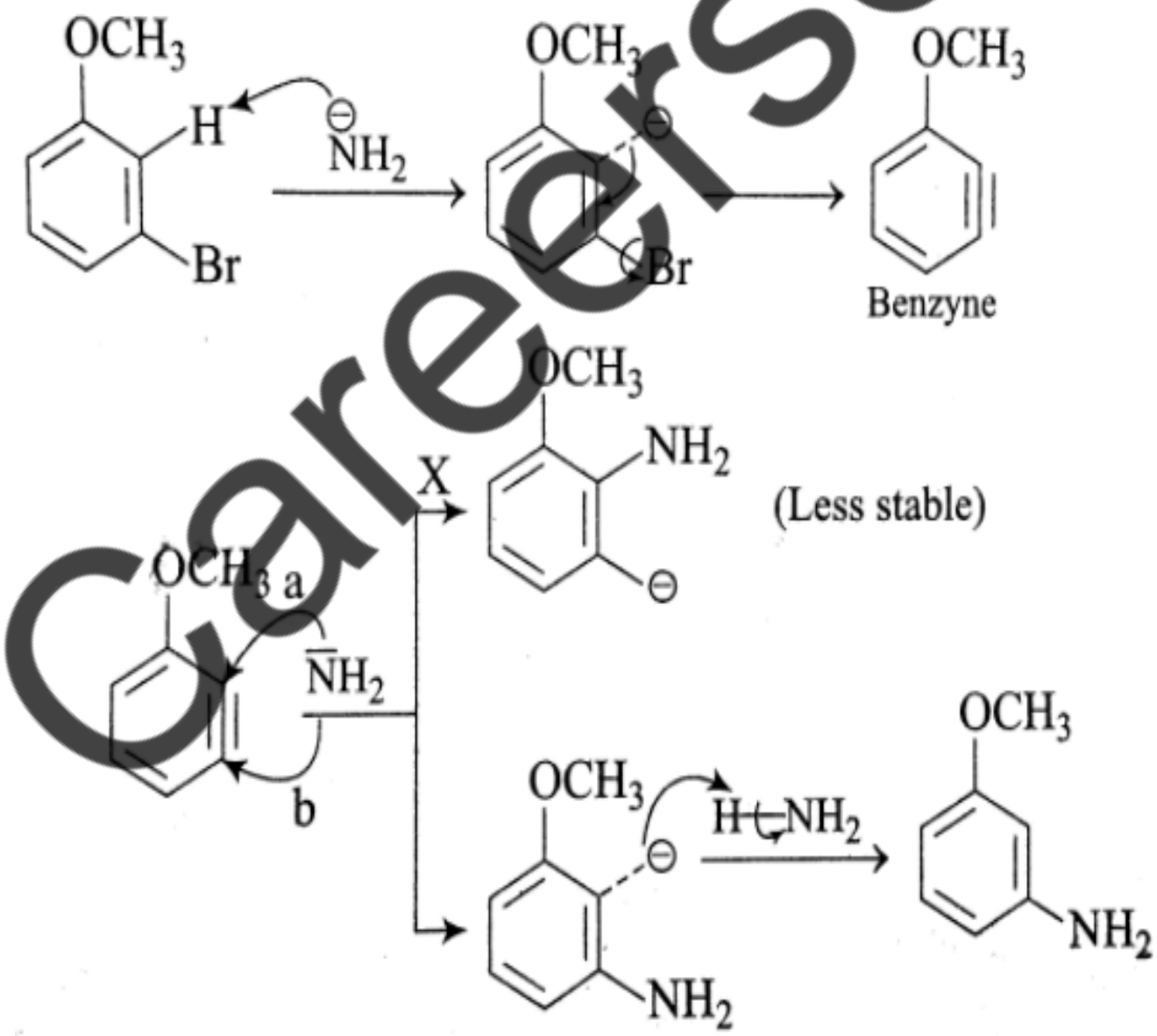
and substitution reaction

Solution:

This reaction is a substitution reaction

it will show cine substitution and ipso substitution.

Here, Cine will be Less stable(Ortha product.) and Ipso will be More stable (Meta product)



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hence correct option is 1

Q. 14 Which one is the wrong statement?

Option 1:

de-Broglie's wavelength is given by $\lambda = \frac{h}{mv}$, where m = mass of the particle, v = group velocity of the particle.

Option 2:

The uncertainty principle is

Option 3:

Half-filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement.

Option 4:

The energy of 2s orbital is less than the energy of 2p orbital in case of Hydrogen like atoms.

Correct Answer:

The energy of 2s orbital is less than the energy of 2p orbital in case of Hydrogen like atoms.

Solution:

As we learned in

The principal quantum number determines the size and to a large extent the energy of the orbital.

For a hydrogen-like atom, the energy of the 2s orbital is the **same** as the energy of the 2p orbital.

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Hence, the answer is the option (4).

Q. 15 The correct order of the stoichiometries of AgCl formed when $AgNO_3$ in excess is treated with the complexes: $CoCl_3.6NH_3$, $CoCl_3.5NH_3$, $CoCl_3.4NH_3$ respectively is :

Option 1:
1 AgCl, 3 AgCl, 2 AgCl

Option 2:
3 AgCl, 1 AgCl, 2 AgCl

Option 3:
3 AgCl, 2 AgCl, 1 AgCl

Option 4:
2 AgCl, 3 AgCl, 1 AgCl

Correct Answer:
3 AgCl, 2 AgCl, 1 AgCl

Solution:
The number of Cl atoms not in the coordinate complex are free to precipitate as AgCl.

Co forms a complex with 6 ligands with NH_3 molecules with priority over Cl here

So, $CoCl_3$. $6NH_3$ has $6NH_3$ as ligands and 3Cl will precipitate

Similarly $CoCl_3.5NH_3$, 2Cl will precipitate

$CoCl_3$.$4HN_3$ 1Cl will precipitate

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$$[Co(NH_3)_6]Cl_3 + 3AgNO_3 \rightarrow 3AgCl\downarrow + [Co(NH_3)_6](NO_3)_3$$

$$[Co(NH_3)_5Cl]Cl_2 + 2AgNO_3 \rightarrow 2AgCl\downarrow + [Co(NH_3)_5Cl](NO_3)_2$$

$$[Co(NH_3)_4Cl_2]Cl + AgNO_3 \rightarrow AgCl\downarrow + [Co(NH_3)_4Cl_2]NO_3$$

Hence, the option number (3) is correct.

Q. 16 Name the gas that can readily decolourise acidified $KMnO_4$ solution:

Option 1:
CO_2

Option 2:
SO_2

Option 3:
NO_2

Option 4:
P_2O_5

Correct Answer:
SO_2

Solution:
SO_2 easily gets oxidized in acidified $KMnO_4$ solution to sulfate ions.

in the case of, CO_2, NO_2 and P_2O_5 C,n and P are in maximum oxidation states.

Hence, the answer is the option (2).

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Q. 17 The correct statement regarding electrophile is :

Option 1:
Electrophile is a negatively charged species and can form a bond by accepting a pair of electrons from a nucleophile

Option 2:
Electrophile is a negatively charged species and can form a bond by accepting a pair of electrons from another electrophile

Option 3:
Electrophiles are generally neutral species and can form a bond by accepting a pair of electrons from a nucleophile

Option 4:
Electrophile can be either neutral or positively charged species and can form a bond by accepting a pair of electrons from a nucleophile

Correct Answer:
Electrophile can be either neutral or positively charged species and can form a bond by accepting a pair of electrons from a nucleophile

Solution:
Positively charged electrophiles -

Chemical species having capacity to attack e- rich portion of substance.

- wherein

$H^+, Cl^+, Br^+, NO^+, NO_2^+, R^+, R-\overset{\oplus}{C}=O$

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and

Neutral electrophile -

Those netural chemical species having capacity to attack e-rich portion of substrate.

- wherein

$BF_3, AlCl_3, FeCl_3, SO_3, :CH_2, :CCl_2$

Electrophiles are those having the capacity to accept an electron from nucleophiles.

Q. 18 The species, having bond angles of 120° is :

Option 1:
PH_3

Option 2:
ClF_3

Option 3:
NCl_3

Option 4:
BCl_3

Correct Answer:
BCl_3

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Solution:

Sp^2 Hybridisation -

When one s - and two p - orbitals of the same shell of an atom mix to form three new equivalent orbitals.The hybridised orbital is called sp^2 orbital.

- wherein

Shape is Trigonal planar

$BCl_3 \rightarrow Trigonal Planar\ Sp^2\ hybridised\ ,\ Bond\ angle = 120^\circ$

Q. 19 Which of the following statements is **not** correct?

Option 1:
Insulin maintains sugar level in the blood of a human body.

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Option 2:
Ovalbumin is a simple food reserve in egg-white.

Option 3:
Blood proteins thrombin and fibrinogen are involved in blood clotting.

Option 4:
Denaturation makes the proteins more active.

Correct Answer:
Denaturation makes the proteins more active.

Solution:
Denaturation of protein -

The change in the shape of the protein disrupts its function. Secondary and tertiary structures are destroyed but the primary structure remains intact.

- wherein

It occurs due to :

1. Temperature change
2. pH change
3. UV rays
4. X-rays, etc

Denaturation changes the shape of proteins and makes them inactive.

Hence, the answer is the option (4).

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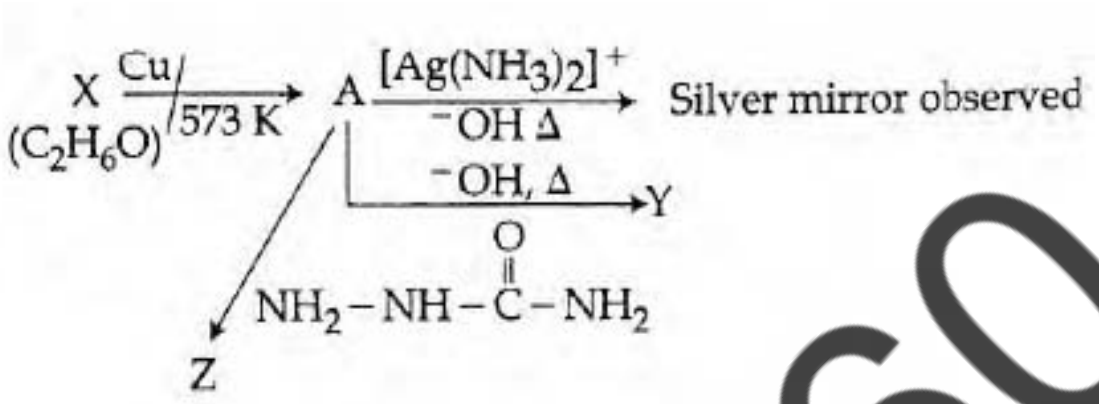
Q. 20 Consider the reactions:



Identify A, X, Y and Z

Option 1:
A-Methoxymethane, X-Ethanoic acid, Y-Acetate ion, Z-hydrazine.

Option 2:
A-Methoxymethane, X-Ethanol, Y-Ethanoic acid, Z-Semicarbazide.

Option 3:
A-Ethanal, X-Ethanol, Y-But-2-enal, Z-Semicarbazone.

Option 4:
A-Ethanol, X-Acetaldehyde, Y-Butanone, Z-Hydrazone.

Correct Answer:
A-Ethanal, X-Ethanol, Y-But-2-enal, Z-Semicarbazone.

Solution:
Addition of ammonia and its derivatives with aldehydes and ketones -

Product formation is favoured due to rapid dehydration of the intermediate to form > C = NG

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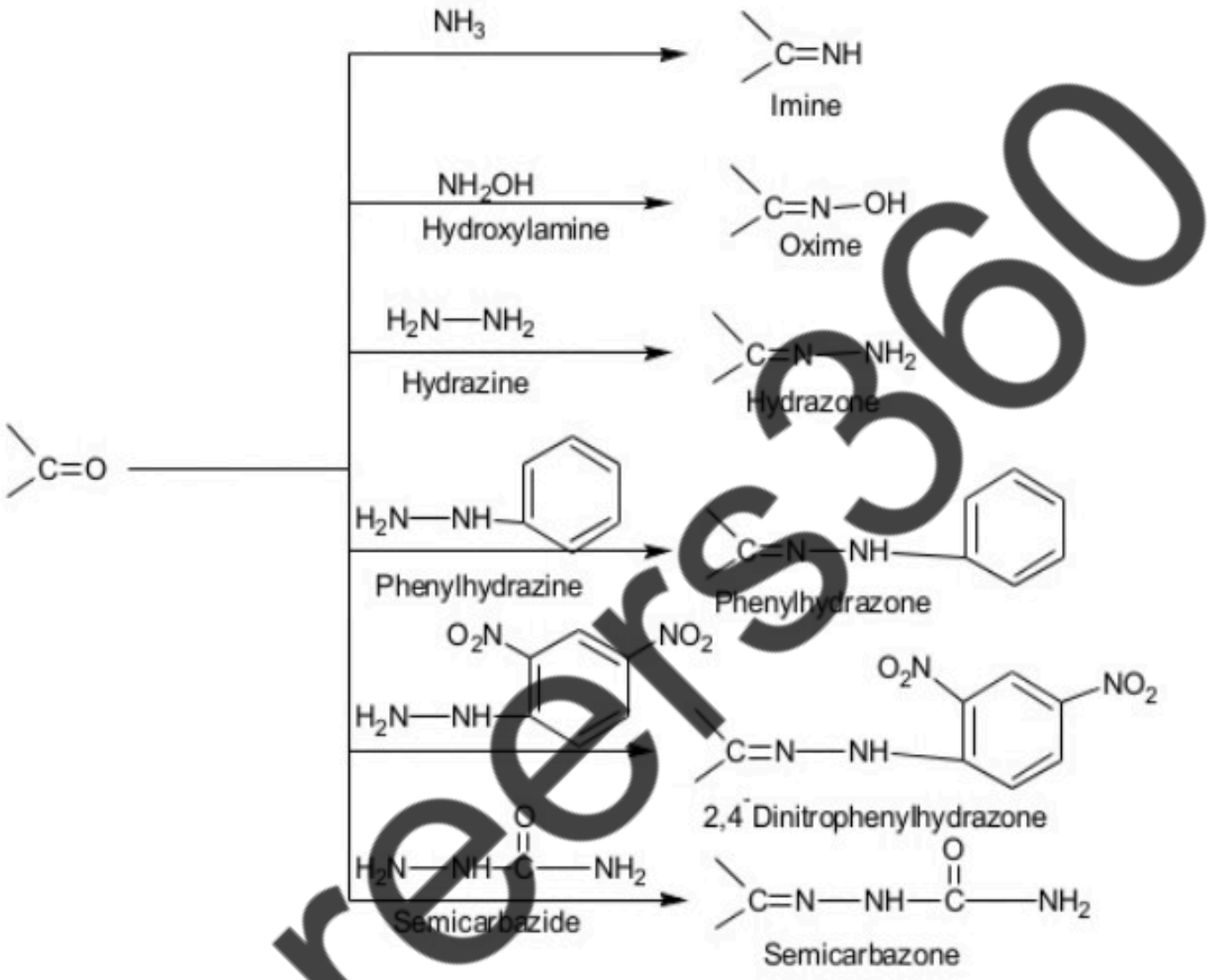
$G\ may\ be\ \ -OH,\ \ -NH_2,\ \ -NHC_6H_5,\ \ -NHCONH_2,\ \ etc$

- wherein



and

Aldol condensation -

Aldehydes and ketones possessing α - hydrogen atom in the presence of dilute alkali undergo self-condensation to form β - hydroxy aldehyde and

β - hydroxy ketone.

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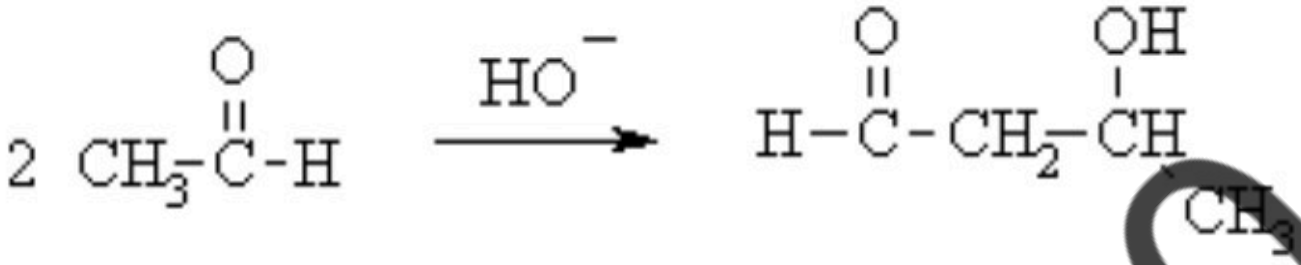
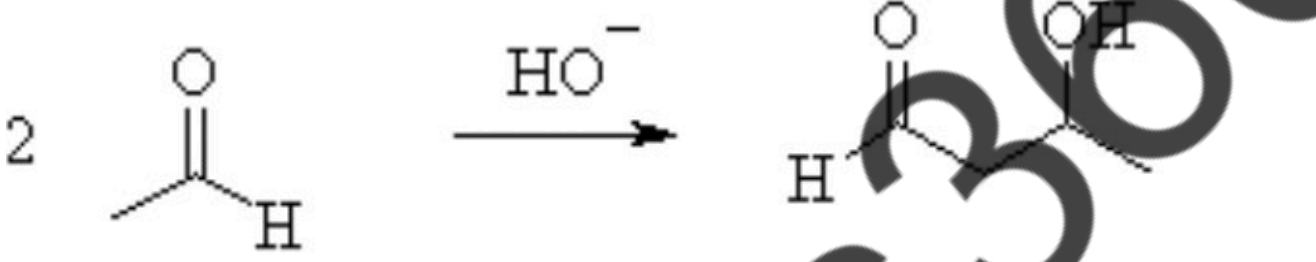
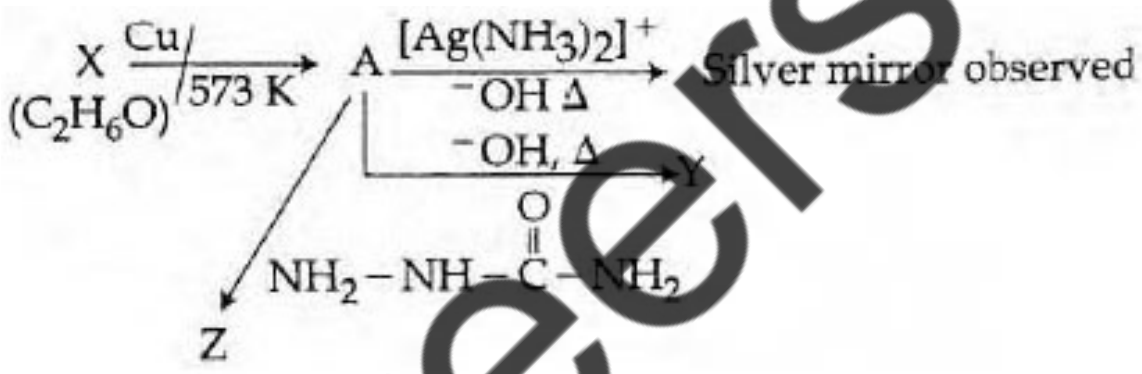
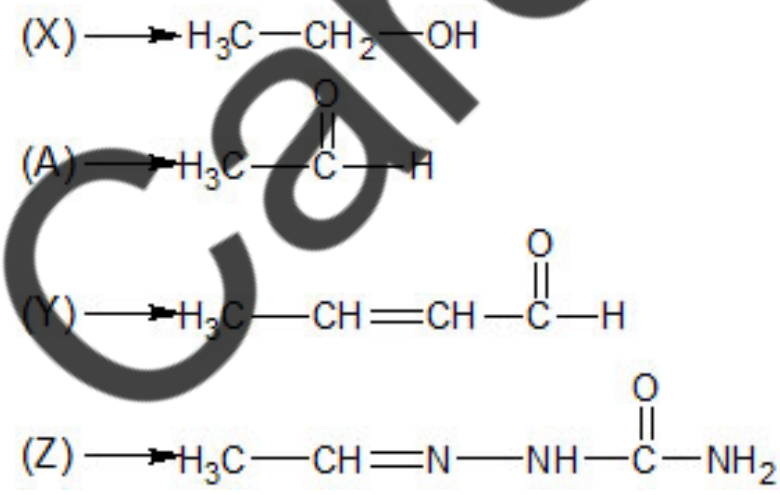
- wherein









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Q. 21 Mechanism of a hypothetical reaction $X_2 + Y_2 \rightarrow 2XY$ is given below :

(i) $X_2 \rightarrow X + X$ (fast)
(ii) $X + Y_2 \rightleftharpoons XY + Y$ (slow)

(iii) $X + Y \rightarrow XY$ (fast)

The overall order of the reaction will be :

Option 1:
1

Option 2:
2

Option 3:
0

Option 4:
1.5

Correct Answer:
1.5

Solution:
The given reaction

$$X_2 \rightarrow X + X(\text{fast})$$

$$X + Y_2 \rightleftharpoons XY + Y(\text{slow})$$

The slow step in the reaction is rate-determining

$$\therefore \text{Rate} = K[X][Y_2] \quad \text{-----(1)}$$

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The equilibrium constant for the last step

$$K' = \frac{[X]^2}{[X_2]}$$

$$\therefore [X] = \sqrt{K'[X_2]}$$

Substituting [X] in the first step will give

$$\text{rate} = K\sqrt{K'[X_2]}[Y_2]$$

$$\text{rate} = K''[X_2]^{1/2}[Y_2]$$

So.

$$\text{order of reaction} = 1 + \frac{1}{2} = \frac{3}{2} \text{order of reaction} = 1.5$$

Hence, the answer is the option (4).

Q. 22 Which of the following reactions is appropriate for converting acetamide to methanamine?

Option 1:
Carbylamine reaction

Option 2:
Hoffmann bromamide reaction

Option 3:
Stephens reaction

Option 4:
Gabriels phthalimide synthesis

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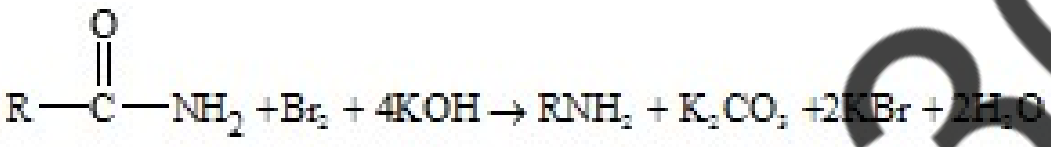
Correct Answer:
Hoffmann bromamide reaction

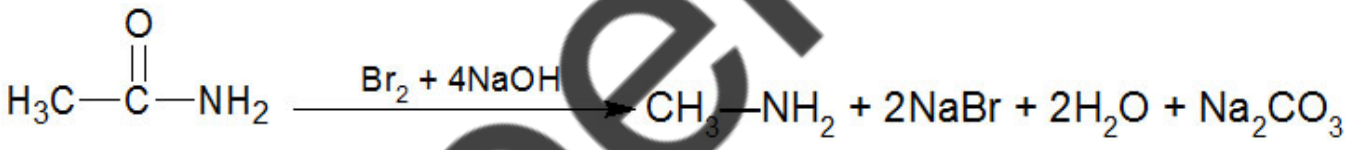
Solution:
Hofmann Bromamide Degradation -

Amines (only primary) can also be prepared by Hofmann degradation. In this method the amine will have one carbon atom less than the amide.

- wherein



Hoffmann hypobromamide reaction



Hence, the option number (2) is correct.

Q. 23 Correct increasing order for the wavelengths of absorption in the visible region for the complexes of Co^{3+} is :

Option 1:
$[Co(en)_3]^{3+}$, $[Co(NH_3)_6]^{3+}$, $[Co(H_2O)_6]^{3+}$

Option 2:
$[Co(H_2O)_6]^{3}+$, $[Co(en)_3]^{3+}$, $[Co(NH_3)_6]^{3+}$

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Option 3:
$[Co(H_2O)_6]^{3+}$, $[Co(NH_3)_6]^{3+}$, $[Co(en)_3]^{3+}$

Option 4:
$[Co(NH_3)_6]^{3+}$, $[Co(en)_3]^{3+}$, $[Co(H_2O)_6]^{3+}$

Correct Answer:
$[Co(en)_3]^{3+}$, $[Co(NH_3)_6]^{3+}$, $[Co(H_2O)_6]^{3+}$

Solution:
Spectrochemical series -

$$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < EDTA < NH_3 < en < CN^- < CO$$

On increasing strength of ligand wavelength of absorbed radiation decreases

Order of spectrochemical series $H_2O < NH_3 < en$

Order of absorbed wavelength

$$en < NH_3 < H_2O$$

Hence, the answer is the option (1).

Q. 24 Which is the incorrect statement?

Option 1:
$FeO_{0.98}$ has non stoichiometric metal deficiency defect.

Option 2:
Density decreases in case of crystals with Schottky's defect.

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Option 3:

NaCl(s) is insulator, silicon is semiconductor, silver is conductor, quartz is piezo electric crystal.

Option 4:

Frenkel defect is favoured in those ionic compounds in which sizes of cation and anions are almost equal.

Correct Answer:

Frenkel defect is favoured in those ionic compounds in which sizes of cation and anions are almost equal.

Solution:

As we learnt in

Frenkel defect is shown by ionic substances in which there is a large difference in the size of ions. Here, The smaller ion (usually cation) is dislocated from its normal site to an interstitial site. It creates a vacancy defect at its original site and an interstitial defect at its new location.

Fig. : Frenkel defects

Therefore, the correct option is (4).

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Q. 25 The equilibrium constants of the following are:

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 \quad K_1$$

$$N_2 + O_2 \rightleftharpoons 2NO \quad K_2$$

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O \quad K_3$$

The equilibrium constant (K) of the reaction:

$$2NH_3 + \frac{5}{2}O_2 \overset{K}{\rightleftharpoons} 2NO + 3H_2O$$

Option 1:
$\frac{K_1K_3^3}{K_2}$

Option 2:
$\frac{K_2K_3^3}{K_1}$

Option 3:
$\frac{K_2K_3}{K_1}$

Option 4:
$\frac{K_3K_2^3}{K_1}$

Correct Answer:
$\frac{K_2K_3^3}{K_1}$

Solution:

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The equilibrium constant for the reverse reaction is the inverse of the equilibrium constant for the reaction in the forward direction.

$$K_c' = \frac{1}{K_c}$$

K_c' is the equilibrium constant for the reverse direction.

$$N_2 + 3H_2 \rightleftharpoons 2NH_3, \quad K_1$$

$$2NH_3 \rightleftharpoons 3H_2 + N_2, \qquad \frac{1}{K_1} \text{ --------- (1)}$$

$$N_2 + O_2 \rightleftharpoons 2NO, \quad K_2 \quad \text{.....(2)}$$

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O, \quad K_3$$

$$3H_2 + \frac{3}{2}O_2 \rightarrow 3H_2O, \qquad K_3^3 \text{ ---------- (3)}$$

adding (1) (2) (3)

$$3H_2 + \frac{3}{2}O_2 \rightleftharpoons 3H_2O, \qquad K = \frac{K_2K_3^3}{K_1}$$

Hence, the answer is the option (2).

Q. 26 Extraction of gold and silver involves leaching with CN^- ion. Silver is later recovered by :

Option 1:
liquation

Option 2:
distillation

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Option 3:
zone refining

Option 4:
displacement with Zn

Correct Answer:
displacement with Zn

Solution:
Leaching - In this process, we use a suitable agent which reacts with ore to form a water-soluble complex while impurities are insoluble.
Reactions involved in leaching of silver -

$$Ag_2S + 4NaCN \overset{O_2}{\rightarrow} 2Na[Ag(CN)_2] + Na_2S$$

In the process of leaching, we use an agent which reacts with ore to form a water-soluble complex while impurities are insoluble.

$$Na[Ag(CN)_2] + Zn(dust) \rightarrow Na_2[Zn(CN)_4] + 2Ag \downarrow$$

On addition of Zn Dust to soluble complex results in the precipitation of silver.
The correct option is 4.

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Q. 27 The most suitable method of separation of 1:1 mixture of ortho and para-nitrophenols is:

Option 1:
Chromatography

Option 2:
Crystallisation

Option 3:
Steam distillation

Option 4:
Sublimation

Correct Answer:
Steam distillation

Solution:
Steam Distillation -

This method is used to separate steam volatile substances that are immiscible with water.

- wherein

e.g. O-nitrophenol and P-nitrophenol

Steam distillation is the most suitable method of separation of a 1: 1 mixture of ortho and para nitrophenols, as ortho nitrophenol has intramolecular H-bonds.

Hence, the answer is the option (3).

Q. 28 It is because of inability of ns^2 electrons of the valence shell to participate in bonding that:

Option 1:
Sn^{2+} is reducing while Pb^{4+} is oxidising

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Option 2:
Sn^{2+} is oxidising while Pb^{4+} is reducing

Option 3:
Sn^{2+} and Pb^{2+} are both oxidising and reducing

Option 4:
Sn^{4+} is reducing while Pb^{4+} is oxidising

Correct Answer:
Sn^{2+} is reducing while Pb^{4+} is oxidising

Solution:
Inert pair effect - The phenomenon in which the outer shell s electrons (ns^2) penetrate to $(n-1)d$ electrons and thus become closer to the nucleus and are more effectively pulled by the nucleus

This results in less availability of ns^2 electrons pair for bonding or ns^2 electron pair becomes inert.

The inert pair effect starts from period number four and has maximum consequences in period number six and it is shown by only groups 13, 14, and 15.

Stability of $Sn^{+4} > Sn^{2+}$ but the stability of $Pb^{2+} > Pb^{+4}$ which result Sn^{2+} is reducing while Pb^{+4} is oxidizing in nature.
Hence, the answer is the option (1).

Q. 29 An example of a sigma bonded organometallic compound is :

Option 1:
Ruthenocene

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Option 2:
Grignard's reagent

Option 3:
Ferrocene

Option 4:
Cobaltocene

Correct Answer:
Grignard's reagent

Q. 30 For a given reaction, ΔH = 35.5 kJ mol^{-1} and ΔS = 83.6 JK^{-1} mol^{-1}. The reaction is spontaneous at : (Assume that ΔH and ΔS do not vary with temperature)

Option 1:
T < 425 K

Option 2:
T > 425 K

Option 3:
all temperatures

Option 4:
T > 298 K

Correct Answer:
T > 425 K

Solution:
Since, $\Delta G = \Delta H - T\Delta S$

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for spontaneous reaction $\Delta G = -ve$

i.e. $\Delta H < T\Delta S$

Therefore, $\frac{\Delta H}{\Delta S} < T$

$$T > \frac{35.5 \times 10^3 J}{83.6 \quad JK^{-1}} > 425K$$

$$\therefore T > 425K$$

Hence, the answer is the option (2).

Q. 31 Ionic mobility of which of the following alkali metal ions is lowest when aqueous solution of their salts are put under an electric field?

Option 1:
Na

Option 2:
K

Option 3:
Rb

Option 4:
Li

Correct Answer:
Li

Solution:
Li^+ being the smallest has maximum charge density.

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Li^+ is the most heavily hydrated among all others.

Due to Li^+ being the most hydrated, So the size of the hydrated ion will largest among these.

We know,

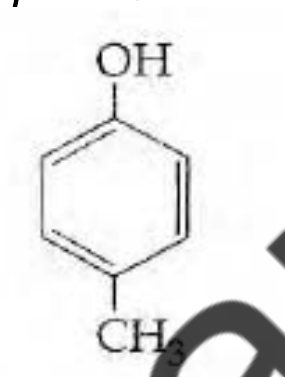
$$\text{Ionic mobility} \propto \frac{1}{\text{size of hydrated ion}}$$

Due to its larger size, mobility in the solution will be very low.

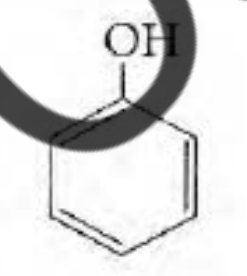
Hence, option number (4) is correct.

Q. 32 Which one is the most acidic compound?

Option 1:



Option 2:



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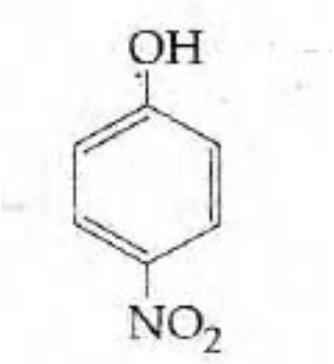
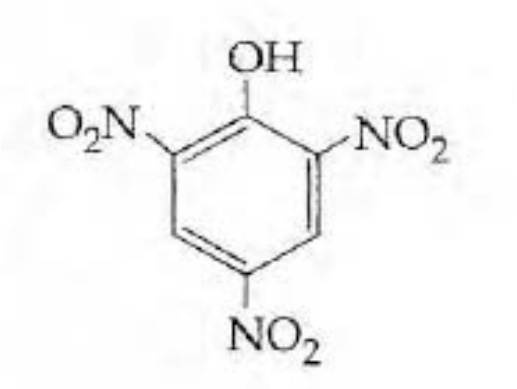
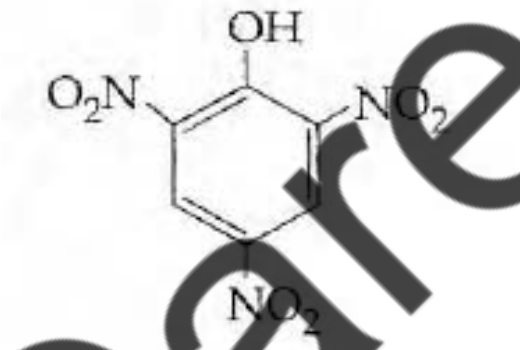
Option 3:



Option 4:



Correct Answer:



Solution:

Effects of substituents on acidity of phenols -

1. Electron withdrawing group like $-NO_2, -CN, -X, -CHO$

2. Electron donating group like $-R, -OR, -NH_2$

- wherein

1. Increase the acidic strength of phenol.

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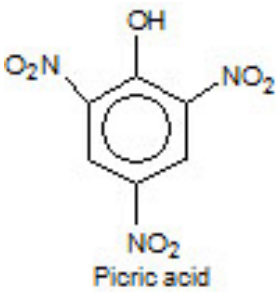
2. Decrease the acidic strength of phenol.



Picric acid

It has 3-strong withdrawing group which increases acidity hence correct answer is (4)

Q. 33 Mixture of chloroxylenol and terpineol acts as :

Option 1:
analgesic

Option 2:
antiseptic

Option 3:
antipyretic

Option 4:
antibiotic

Correct Answer:
antiseptic

Solution:
A mixture of chloroxylenol and terpineol acts as an antiseptic (Dettol).

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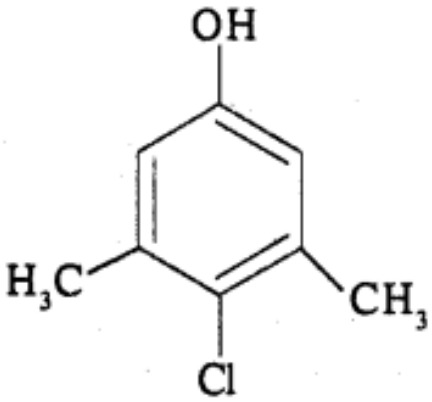
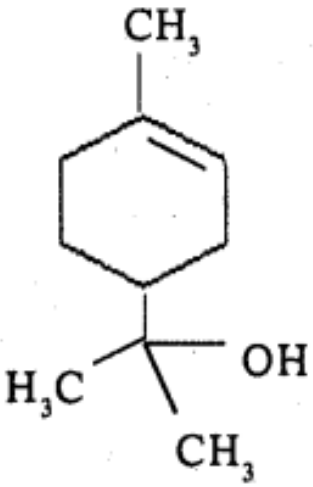


chloroxylenol

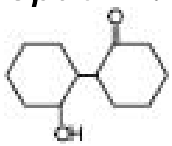


α -terpineol

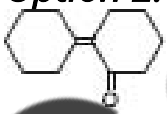
Hence, option number (2) is correct.

Q. 34 Of the following, which is the product formed when cyclohexanone undergoes aldol condensation followed by heating?

Option 1:

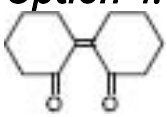


Option 2:



Option 3:

Option 4:



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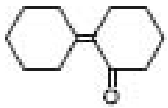
Correct Answer:



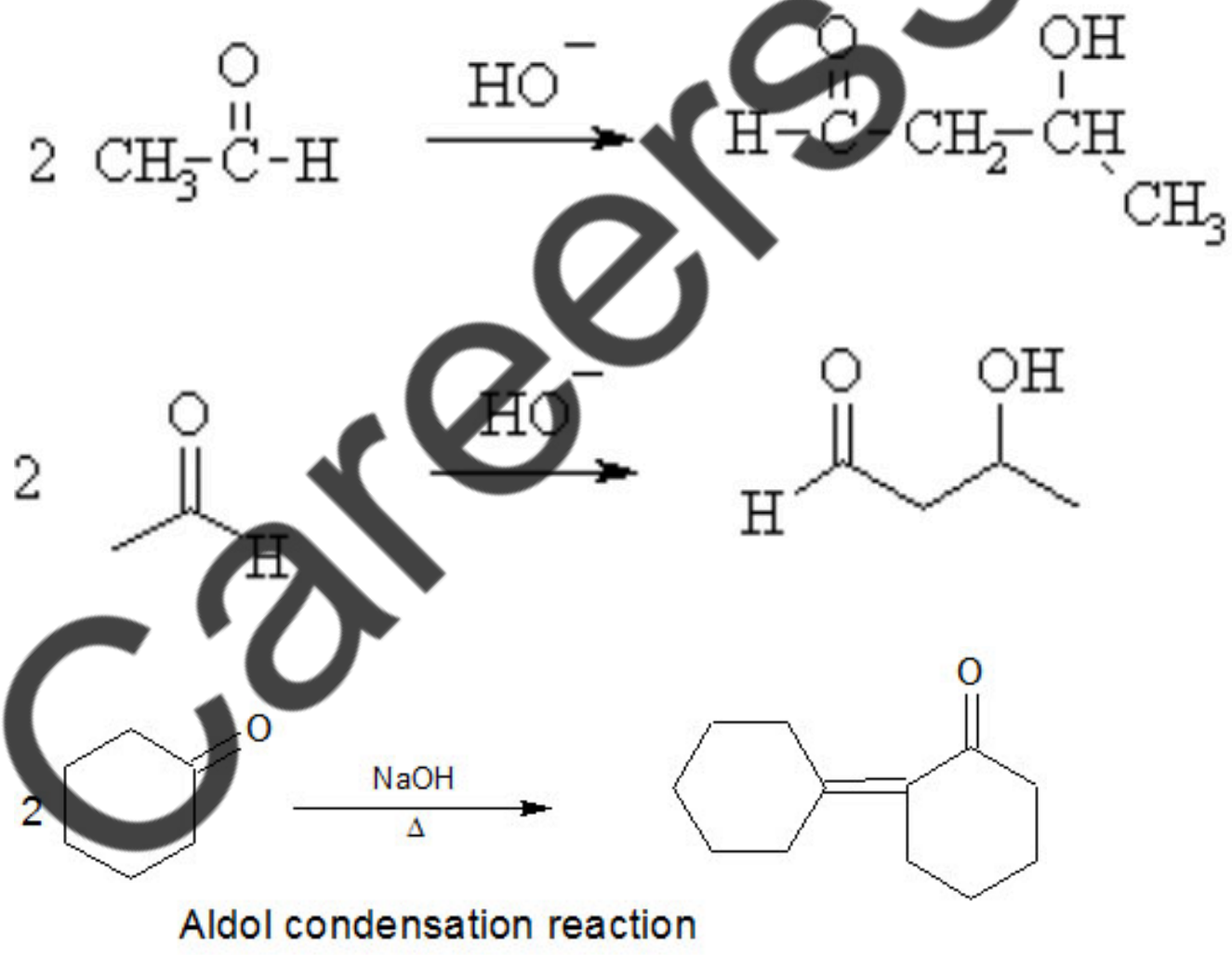
Solution:

Aldol condensation -

Aldehydes and ketones possessing α - hydrogen atom in the presence of dilute alkali undergo self-condensation to form β - hydroxy aldehyde and

β - hydroxy ketone.

- wherein



Aldol condensation reaction

Hence, the option number (2) is correct.

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Q. 35 The element Z = 114 has been discovered recently. It will belong to which of the following family/group and electronic configuration?

Option 1:
Halogen family, [Rn] $5f^{14}6d^{10}7s^{2}7p^{5}$

Option 2:
Carbon family, [Rn] $5f^{14}6d^{10}7s^{2}7p^{2}$

Option 3:
Oxygen family, [Rn] $5f^{14}6d^{10}7s^{2}7p^{4}$

Option 4:
Nitrogen family, [Rn] $5f^{14}6d^{10}7s^{2}7p^{6}$

Correct Answer:
Carbon family, [Rn] $5f^{14}6d^{10}7s^{2}7p^{2}$

Solution:
The electronic configuration of the element with z = 114 (flerovium) is
$[\mathrm{Rn}]5\mathrm{f}^{14}6\mathrm{d}^{10}7\mathrm{s}^{2}7\mathrm{p}^{2}$

Hence it belongs to the carbon family which has the same outer electronic configuration.

Hence, the answer is the option (2).

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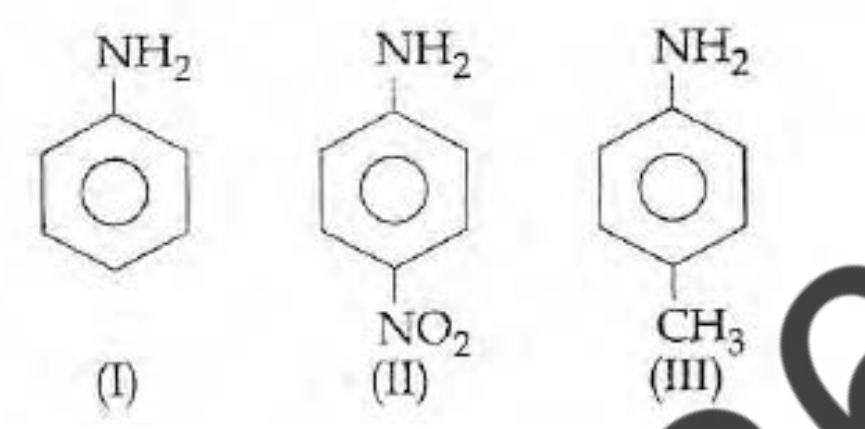
Q. 36 The correct increasing order of basic strength for the following compounds is :



Option 1:
II < III < I

Option 2:
III < I < II

Option 3:
III < II < I

Option 4:
II < I < III

Correct Answer:
II < I < III

Solution:
Basicity of Aromatic Amines -

The unshared pair of electrons at the nitrogen atom is in resonance with the benzene ring and hence not fully available for donation as in the case of aniline.

-

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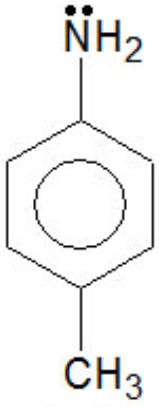
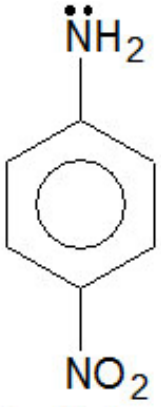
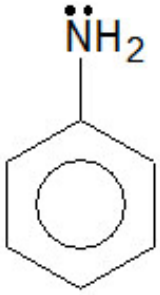
(III) 

e^- donating group increases basicity

(II) 

e^- withdrawing group decreases basicity

(I) 

Hence, the option (4) is correct.

Q. 37 Which of the following is dependent on temperature?

Option 1:
Molality

Option 2:
Molarity

Option 3:
Mole fraction

Option 4:
Weight percentage

Correct Answer:
Molarity

Solution:

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As we learned in

$$\text{Molarity} = \frac{\text{Moles of solute}}{\text{Vol.of solution(L)}}$$

Volume is dependent on temperature thereby making molarity dependent on temperature.

Hence, the answer is the option (2).

Q. 38 The heating of phenyl-methyl ethers with HI produces.

Option 1:
ethyl chlorides

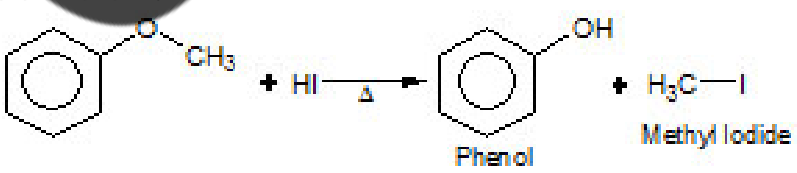
Option 2:
iodobenzene

Option 3:
phenol

Option 4:
benzene

Correct Answer:
phenol

Solution:



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Q. 39 Predict the correct intermediate and product in the following recation.

$$H_3C-C\equiv CH \xrightarrow[HgSO_4]{H_2O,H_2SO_4} \underset{(A)}{Intermediate} \rightarrow \underset{(B)}{product}$$

Option 1:

$A: H_3C-\underset{\underset{OH}{|}}{C}=CH_2 : B: H_3C-\underset{\underset{SO_4}{|}}{C}=CH_2$

Option 2:

$A: H_3C-\underset{\underset{O}{||}}{C}=CH_3 B: H_3C-C\equiv CH$

Option 3:

$A: H_3C-\underset{\underset{OH}{|}}{C}=CH_2 B: CH_3-\underset{\underset{O}{||}}{C}-CH_3$

Option 4:

$A: H_3C-\underset{\underset{SO_4}{|}}{C}=CH_2 B: H_3C-\underset{\underset{O}{|}}{C}-CH_3$

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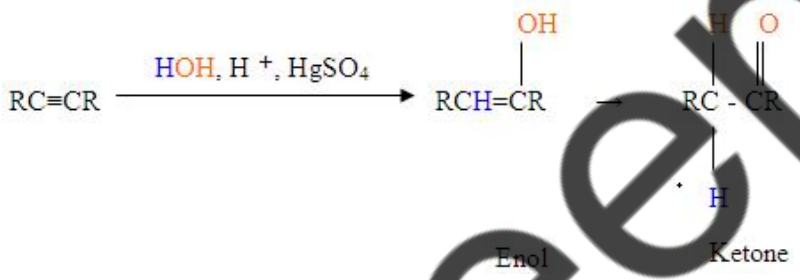
Correct Answer:

$$A : H_3C - \underset{\underset{OH}{|}}{C} = CH_2B : CH_3 - \underset{\underset{O}{\|}}{C} - CH_3$$

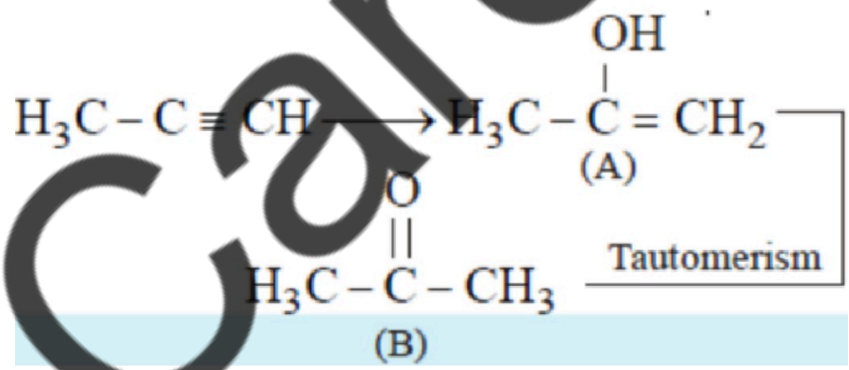
Solution:
Addition of water on alkyne -

By the addition of water on alkyne in presence of $HgSO_4$ and H_2SO_4 alkyne give aldehyde or ketone.

- wherein



Hydration of alkynes give ketones.



Hence, the option number (3) is correct.

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Q. 40 In which pair of ions both the species contain $S-S$ bond?

Option 1:
$S_2O_7^{2-}, S_2O_3^{2-}$

Option 2:
$S_4O_6^{2-}, S_2O_3^{2-}$

Option 3:
$S_2O_7^{2-}, S_2O_8^{2-}$

Option 4:
$S_4O_6^{2-}, S_2O_7^{2-}$

Correct Answer:
$S_4O_6^{2-}, S_2O_3^{2-}$

Solution:
Trick - If the calculated oxidation state is less than that of the maximum oxidation state of the element then atom-atom linkages are present.

The maximum oxidation state of S is +6.

The oxidation state of S in the case of

$S_2O_7^{2-}$ is +6

$S_2O_3^{2-}$ is +2

$S_4O_6^{2-}$ is +2.5

$S_2O_8^{2-}$ is +7 (peroxy linkage)

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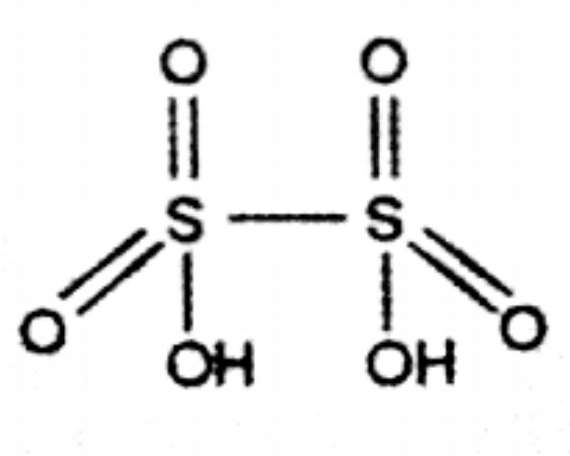
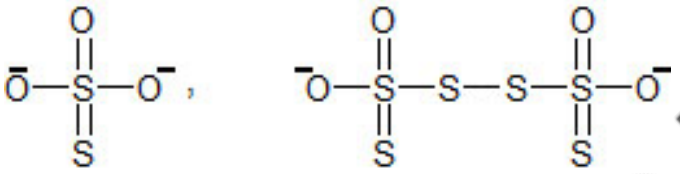
Hence the $S-S$ linkages are present in $S_2O_3^{2-}, S_4O_6^{2-}$

Dithionic acid -

$H_2S_2O_6$-Free acid is moderately stable, but its salt is quite stable.





The correct answer is option 4.

Q. 41 $HgCl_2$ and I_2 both when dissolved in water containing I^- ions the pair of species formed is :

Option 1:
HgI_2, I_3^-

Option 2:
HgI_2, I^-

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Option 3:
HgI_4^-, I_3^-

Option 4:
Hg_2I_2, I^-

Correct Answer:
HgI_4^-, I_3^-

Solution:
In addition of $HgCl_2 \ and \ I_2$ to a solution containing I^- ions.

$$HgCl_2 + 4I^- \rightarrow [HgI_4]^{2-} + 2Cl^-$$

$$I_2 + I^- \rightarrow I_3^-$$

Hence, the answer is the option (3).

Q. 42 A $20 \ litre$ container at $400K$ contains $CO_2(g)$ at pressure $0.4atm$ and an excess of SrO (neglect the volume of solid SrO). The volume of the containers is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value, will be :

(Given that :
$\mathrm{SrCO_3(s) \rightleftharpoons SrO(s) + CO2(g),\ K_p = 1.6\,atm}$)

Option 1:
$5 \ litre$

Option 2:
$10 \ litre$

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Option 3:

$4\ litre$

Option 4:

$2\ litre$

Correct Answer:

$5\ litre$

Solution:

$SrCO_3 \rightleftharpoons SrO + CO_2(g), K_p = 1.6\ atm$

$K_p = p_{CO_2}$

Before compression,

$$n_{CO_2} = \frac{pV}{RT} = \frac{0.4 \times 20}{R \times 400} = \frac{8}{R400} = \frac{1}{50R}$$

After Compression

For the maximum Volume of the container, let's assume that

$n = constant,\ p = 1.6atm$

$$V = \frac{nRT}{p} = \frac{1}{50R} \times \frac{R \times 400}{1.6} = 5L$$

Hence, the answer is the option (1).

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Q. 43 Match the interhalogen compounds of column-I with the geometry in column II and assign the correct code.

	Column-I		**Column-II**
1.	XX'	(i)	T-shape
2.	XX_3'	(ii)	Pentagonal bipyramidal
3.	XX_5'	(iii)	Linear
4.	XX_7'	(iv)	Square-pyramidal
		(v)	Tetrahedral

Option 1:
Code:

1	2	3	4
(iii)	(i)	(iv)	(ii)

Option 2:
Code:

1	2	3	4
(v)	(iv)	(iii)	(ii)

Option 3:
Code:

1	2	3	4
(iv)	(iii)	(ii)	(i)

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Option 4:

Code:

1	2	3	4
(iii)	(iv)	(i)	(ii)

Correct Answer:

Code:

1	2	3	4
(iii)	(i)	(iv)	(ii)

Solution:

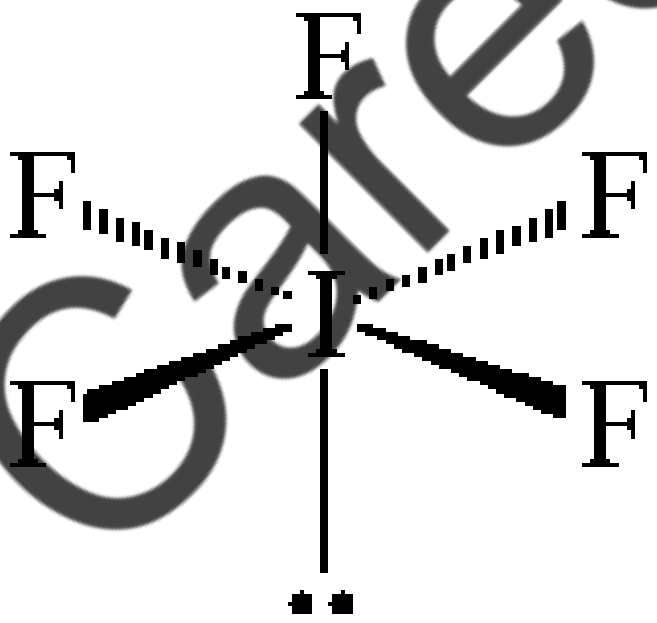
Chlorine Monofluoride - No hybridization, polar

ClF is linear in shape

Chlorine Trifluoride and iodine Trifluoride - sp^3d hybridized and T-shaped.

Iodine Pentafluoride and Bromine pentafluoride -

sp^3d^2 hybridized and square pyramidal



Iodine heptafluoride - sp^3d^3 hybridised and pentagonal bipyramidal

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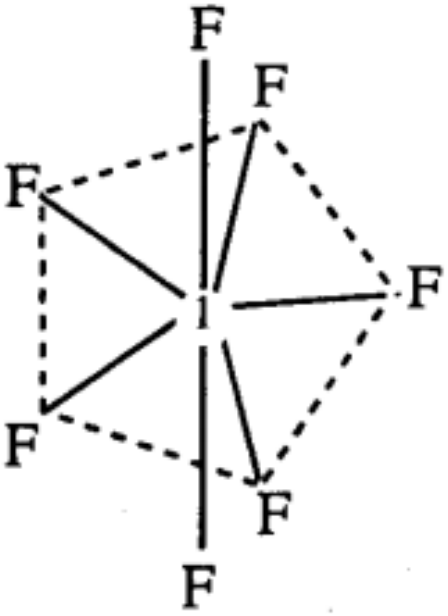
IF_7 exhibits Pentagonal bipyramidal shape



	Compound	Structure
1.	XX'	linear
2.	XX_3'	T-shaped
3.	XX_5'	Square pyramidal
4.	XX_7'	Pentagonal bipyramidal

The correct answer is option 1

Q. 44 Which one of the following pairs of species have the same bond order?

Option 1:
CO, NO

Option 2:
O_2, NO^+

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Option 3:
CN^-, CO

Option 4:
N_2, O_2^-

Correct Answer:
CN^-, CO

Solution:
Bond Order -

Bond order is defined as one-half the difference between the number of electrons present in the bonding and the antibonding orbitals.

- wherein

$$Bond\,Order = \frac{N_b - N_a}{2}$$

$$\text{Bond order} = \frac{1}{2}(N_b - N_a)$$

O_2 : $N_b = 10$, $N_a = 6$, Bond order = 2

NO^+ : $N_b = 10$, $N_a = 4$, Bond order = 3

CN^- : $N_b = 10$, Na = 4, Bond order = 3

CO : $N_b = 10$, Na = 4, Bond order = 3

N_2 : $N_b = 10$, Na = 4, Bond order = 3

O^-_2 : $N_b = 10$, Na = 7, Bond order = 1.5

NO : $N_b = 10$, Na = 5, Bond order = 2.5

Hence, the answer is the option (3).

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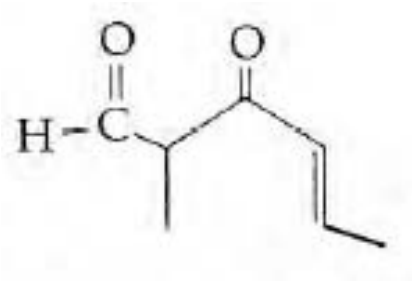
Q. 45 The IUPAC name of the compound



is _____.

Option 1:
3-keto-2-methylhex-4-enal

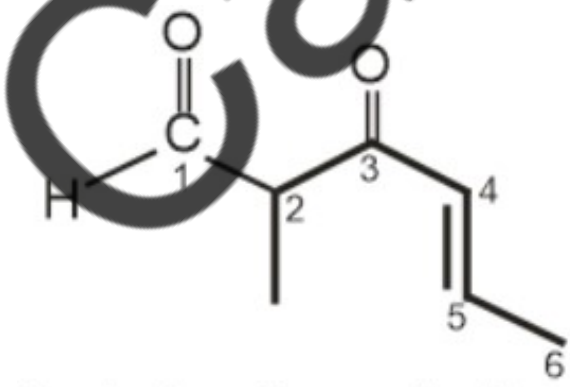
Option 2:
5-formylhex-2-en-3-one

Option 3:
5-methyl-4-oxohex-2-en-5-al

Option 4:
3-keto-2-methylhex-5-enal

Correct Answer:
3-keto-2-methylhex-4-enal

Solution:



3 - keto - 2 - methylhex - 4 enal

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Biology

Q. 1 The genotypes of a Husband and Wife are I^AI^B and I^Ai.

Among the blood types of their children, how many different genotypes and phenotypes are possible?

Option 1:
3 genotypes ; 3 phenotypes

Option 2:
3 genotypes ; 4 phenotypes

Option 3:
4 genotypes ; 3 phenotypes

Option 4:
4 genotypes ; 4 phenotypes

Correct Answer:
4 genotypes ; 4 phenotypes

Solution:
The possible genotypes of their children can be determined by combining the alleles from both parents:

Each child has a 50% chance of inheriting the IA allele from the husband and a 50% chance of inheriting the IA allele from the wife. Therefore, the IA genotype is possible.
Each child also has a 50% chance of inheriting the IB allele from the husband and a 50% chance of inheriting the i allele from the wife. Therefore, the IBi genotype is possible.

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Finally, each child has a 50% chance of inheriting the i allele from both parents. Therefore, the ii genotype is also possible.
In summary, the possible genotypes of their children are IAIA, IAIB, IAi, and ii, resulting in four genotypes.

Now let's consider the phenotypes:

IAIA and IAi genotypes both result in blood type A phenotype.
IBIB and IBi genotypes both result in blood type B phenotype.
IAIB genotype results in blood type AB phenotype.
The ii genotype results in blood type O phenotype.
Therefore, the possible phenotypes of their children are A, B, AB, and O, resulting in four phenotypes.

Based on this analysis, the correct option is Option 4) 4 genotypes; 4 phenotypes.

Q. 2 Adult human RBCs are enucleate. Which of the following statement(s) is/are **most appropriate** explanation for this feature?

(a) They do not need to reproduce

(b) They are somatic cells

(c) They do not metabolize

(d) All their internal space is available for oxygen transport

Options :

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Option 1:
Only (d)

Option 2:
Only (a)

Option 3:
(a), (c) and (d)

Option 4:
(b) and (c)

Correct Answer:
Only (d)

Solution:
In human RBC, the nucleus is lost during maturation. This provides lot of space for oxygen transport due to presence of haemoglobin. RBC lacks other organelles like mitochondria as well.

Hence, the statement ' All their internal space is available for oxygen transport' is correct.

Q. 3 An important characteristic that Hemichordates share with Chordates is:

Option 1:
Absence of notochord

Option 2:
Ventral tubular nerve cord

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Option 3:
Pharynx with gill slits

Option 4:
Pharynx without gill slits

Correct Answer:
Pharynx with gill slits

Solution:
As we learnt in

Sub - phylum Cephalochordata -

The notochord is present throughout the life and extents from head to tail. More number of pharyngeal gill slits and better developed.

- wherein

Eg. Branchiostoma, Amphioxus.

Pharyngeal gill slits are present in hemichordates as well as in chordates. Notochord is present in chordates only. Ventral tubular nerve cord is characteristic feature of non-chordates.

Q. 4 Good vision depends on adequate intake of carotene rich food. Select the best option from the following statements.

a. Vitamin A derivatives are formed from carotene.
b. The photopigments are embedded in the membrane discs of the inner segment.
c. Retinal is a derivative of vitamin A.
d. Retinal is a light absorbing part of all the visual photopigments.

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Option 1:
(a) and (b)

Option 2:
(a), (c) and (d)

Option 3:
(a) and (c)

Option 4:
(b), (c) and (d)

Correct Answer:
(b), (c) and (d)

Solution:

As we have learnt in the mechanism of Vision -

-

- The retina of the eyes has proteins called opsin which binds to the retinal.
- Retinal is the light-sensitive pigments.
- Vitamin A is necessary for the production of retinal, a component of rhodopsin.
- Rhodopsin is required for sensing dim light.
- Since deficiency of vitamin A leads to inadequate production of retinal it results in poor vision.
- Carotenes are the pigments present in red-orange fruits and vegetables.
- Carotenes synthesize vitamin A derivatives.

Hence, the correct answer is option d.

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Q. 5 Zygotic meiosis is characteristic of :

Option 1:
Marchantia

Option 2:
Fucus

Option 3:
Funaria

Option 4:
Chlamydomonas

Correct Answer:
Chlamydomonas

Solution:
As we learnt in

Types of life cycle in chlorophyceae -

1. Haplontic → Eg. Ulothrix, Spirogyra
2. Diplontic → Eg. Caulerpa
3. Diplohaplontic → Eg. Ulva, Cladophora.

-

Chlamydomonas has haplontic life cycle hence showing zygotic meiosis or initial meiosis.

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Q. 6 A decrease in blood pressure/volume will not cause the release of:

Option 1:
Renin

Option 2:
Atrial Natriuretic Factor

Option 3:
Aldosterone

Option 4:
ADH

Correct Answer:
Atrial Natriuretic Factor

Solution:

- ANF is atrial natriuretic factor causes vasodilation and leads to decrease of blood pressure.
- The increase in the blood flow to the atria of the heart causes the release of ANF.
- It is mainly released by the atria of the heart in response to high blood volume.
- It reduces the water, sodium, and adipose concentration in the circulatory system and thus reduces blood pressure.

Hence, the correct answer is 'Atrial Natriuretic Factor'

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Q. 7 Lungs are made up of air-filled sacs, the alveoli. They do not collapse even after forceful expiration, because of :

Option 1:
Residual Volume

Option 2:
Inspiratory Reserve Volume

Option 3:
Tidal Volume

Option 4:
Expiratory Reserve Volume

Correct Answer:
Residual Volume

Solution:

As we have learnt in

Residual Volume (RV): Volume of air remaining in the lungs even after a forcible expiration. This averages 1100 mL to 1200 mL. It prevents lungs from collapsing.

Hence,t he correct answer is 'Residual Volume'

-

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Q. 8 Which one of the following statements is **correct,** with reference to enzymes?

Option 1:
Apoenzyme = Holoenzyme + Coenzyme

Option 2:
Holoenzyme = Apoenzyme + Coenzyme

Option 3:
Coenzyme = Apoenzyme + Holoenzyme

Option 4:
Holoenzyme = Coenzyme + Cofactor

Correct Answer:
Holoenzyme = Apoenzyme + Coenzyme

Solution:
As we learnt in

Co-enzymes as cofactor -

A cofactor is an organic compound and their association with the apoenzyme is only transient, usually occurs during the course of catalysis.

- wherein

Eg: NAD and NADP.

Coenzymes are transiently associated with apoenzyme, forming a holoenzyme

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Q. 9 Mycorrhizae are the example of :

Option 1:
Fungistasis

Option 2:
Amensalism

Option 3:
Antibiosis

Option 4:
Mutualism

Correct Answer:
Mutualism

Solution:
As we learnt in

Mutualism -

This interaction confers benefits on both the interacting species.

- wherein

Lichens represent an intimate mutualistic relationship between the fungus and photosynthesising algae or cyanobacteria. Similarly, the mycorrhizae are associations between fungi and the roots of the higher plants. The fungi help the plants in the absorption of essential nutrients from the soil while the plant, in turn, provides the fungi with energy-yielding carbohydrates.

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Q. 10 Which of the following are not polymeric?

Option 1:
Nucleic acids

Option 2:
Proteins

Option 3:
Polysaccharides

Option 4:
Lipids

Correct Answer:
Lipids

Solution:
As we learnt in

Lipids -

Lipids are water insoluble. They could be simple fatty acids, that has a carboxyl group attached to an R-group. The R-group could be a methyl (-CH_3) or ehtyl (-C_2H_5)

-

Lipids are fatty acids.

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Q. 11 Which among the following are the smallest living cells, known without a definite cell wall, pathogenic to plants as well as animals and can survive without oxygen?

Option 1:
Pseudomonas

Option 2:
Mycoplasma

Option 3:
Nostoc

Option 4:
Bacillus

Correct Answer:
Mycoplasma

Solution:
As we learnt in

Mycoplasm / PPLO (Pleuropneumonia like organism) -

- Smallest monerans
- Unicellular, non – motile
- Lack cell wall
- Resistance to antibiotics that affect cell wall synthesis like penicillin.
- Inhibited by antibiotics like tetracycline which acts on metabolic pathways.
- DNA is linear, double stranded extending almost throughout the cell.

- wherein

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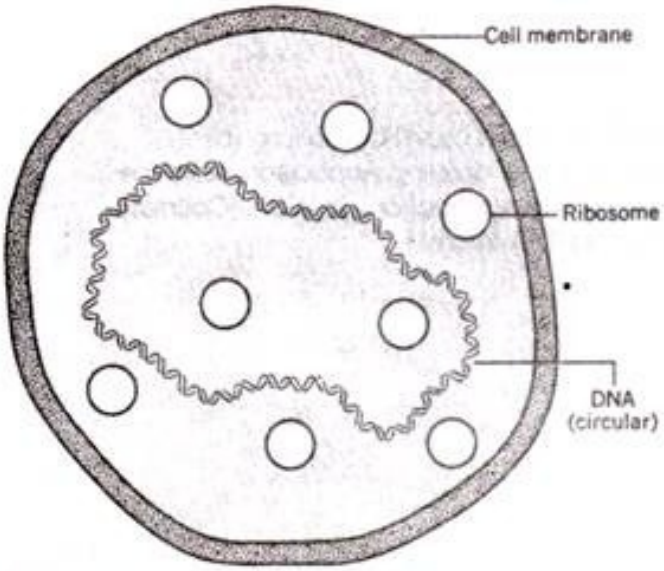

Q. 12 Asymptote in a logistic growth curve is obtained when :

Option 1:
The value of 'r' approaches zero

Option 2:
K = N

Option 3:
K > N

Option 4:
K < N

Correct Answer:
K = N

Solution:

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- Asymptote is a straight line associated with a curve.
- A population that survives in a habitat with limited resources shows a lag phase during the initial stage as the organism needs to adjust during the initial stage and the population is less than the carrying capacity.
- After, the initial stage it shows accelerated growth as the population increases and utilizes the resources.
- The resources get utilized at the later stage and thus the growth deaccelerates and further, it turns to an asymptote when the population density becomes equivalent to carrying capacity. So, in the case of an asymptote, K=N.

Q. 13 Plants which produce characterstic pneumatophores and show vivipary belong to :

Option 1:
Mesophytes

Option 2:
Halophytes

Option 3:
Psammophytes

Option 4:
Hydrophytes

Correct Answer:
Halophytes

Solution:
Halophytes have modified roots(pneumatophores) and seed germination happens even attached to the parent plant (vivipary).so the correct option is b.

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Q. 14 Identify the wrong statement in context of heartwood:

Option 1:
Organic compounds are deposited in it

Option 2:
It is highly durable

Option 3:
It conducts water and minerals efficiently

Option 4:
It comprises dead elements with highly lignified walls

Correct Answer:
It conducts water and minerals efficiently

Solution:
As we learnt in

Heart wood -

New secondary xylem tissues degenerated into the old secondary xylem. Waste materials like lignin, suberin, tannin and resin gums are formed due to degeneration. All waste materials are filled in lumen of tracheids and vessels of secondary xylem.

- wherein

Wood in center becomes dark coloured which is called heart wood or duramen.

and

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Sap wood -

The peripheral or outer wood which looks light in color is known as Sap wood or Alburnum.

- wherein

The active work of sap wood is water conduction. Heart wood provides the mechanical strength.

Heartwood doesn't conduct water and minerals efficiently but provide mechanical strength.

Q. 15 With reference to factors affecting the rate of photosynthesis, which of the following statements is **not** correct?

Option 1:
Light saturation for CO_2 fixation occurs at 10% of full sunlight

Option 2:
Increasing atmospheric CO_2 concentration upto 0.05% can enhance CO_2 fixation rate

Option 3:
C_3 plants responds to higher temperatures with enhanced photosynthesis while C_4 plants have much lower temperature optimum

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Option 4:
Tomato is a greenhouse crop which can be grown in CO_2 - enriched atmosphere for higher yield

Correct Answer:
C_3 plants responds to higher temperatures with enhanced photosynthesis while C_4 plants have much lower temperature optimum

Solution:

- C4 plants do not perform photorespiration while C3 plants do. Photosynthetic activity is mainly affected by enzymes rather than temperatures. High temperature leads to the destruction of photosynthetic pigments. Hence, the correct answer is option c.

Q. 16 Artificial selection to obtain cows yielding higher milk output represents:

Option 1:
Stabilizing selection as it stabilizes this character in the population.

Option 2:
Directional as it pushes the mean of the character in one direction.

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Option 3:
Disruptive as it splits the population into two, one yielding higher output and the other lower output.

Option 4:
Stabilizing followed by disruptive as it stabilizes the population to produce higher yielding cows.

Correct Answer:
Directional as it pushes the mean of the character in one direction.

Solution:
There are three types of selections:

- **Stabilizing**
- **Directional**
- **Disruptive**

The extremity of the trait distribution that is being selected against is known as directional selection. So we use directional selection for high yield of milk form cows.

Q. 17 Which of the following represents order of 'Horse'?

Option 1:
Equidae

Option 2:
Perissodactyla

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Option 3:
Caballus

Option 4:
Ferus

Correct Answer:
Perissodactyla

Solution:
The horse belongs to Orders - Perissodactyla Family - Equidae Genus - Equus Species - ferus Subspecies - caballus

---

Q. 18 An example of colonial alga is :

Option 1:
Chlorella

Option 2:
Volvox

Option 3:
Ulothrix

Option 4:
Spirogyra

Correct Answer:
Volvox

Solution:
As we learnt in

Thallophyta / Algae -

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Chlorophyll - bearing, thalloid, avascular plants and are usually aquatic.

- wherein

Eg. Chlamydomonas, Volvox.

Q. 19 The DNA fragments separated on an agarose gel can be visualised after staining with:

Option 1:
Bromophenol blue

Option 2:
Acetocarmine

Option 3:
Aniline blue

Option 4:
Ethidium Bromide

Correct Answer:
Ethidium Bromide

Solution:
As we learnt in

Ethidium Bromide -

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The separated DNA fragments can be seen only after staining with a compound known as Ethidium Bromide followed by exposure to U.V. radiation as bright orange coloured bands. Ethidium bromide is a fluorescent dye, which intercalate with DNA molecules and flourish when exposed to UV light (302 nm). EtBr is used to stain DNA fragments.

Q. 20 The hepatic portal vein drains blood to liver from

Option 1:
Heart

Option 2:
Stomach

Option 3:
Kidneys

Option 4:
Intestine

Correct Answer:
Intestine

Solution:
Hepatic portal veins carry the deoxygenated blood from the gastrointestinal tract, gallbladder, pancreas and spleen to the liver. It carries the nutrient rich blood to the liver which is further processed there and returned back to the heart via inferior vena cava.

Hence the correct answer is 'intestine'

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Q. 21 MALT constitutes about_____ percent of the lymphoid tissue in human body.

Option 1:
50%

Option 2:
20%

Option 3:
70%

Option 4:
10%

Correct Answer:
50%

Solution:
As we learnt in

Mucosaassociated lymphoid tissue (MALT) -

There is a lymphoid tissue also located within the lining of the major tracts (respiratory, digestive and urogenital tracts) called mucosaassociated lymphoid tissue (MALT).

- wherein

It constitutes about 50 percent of the lymphoid tissue in the human body.

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Q. 22 Which of the following is **correctly** matched for the product produced by them?

Option 1:
Acetobacter aceti : Antibiotics

Option 2:
Methanobacterium : Lactic acid

Option 3:
Penicillium notatum : Acetic acid

Option 4:
Saccharomyces cerevisiae : Ethanol

Correct Answer:
Saccharomyces cerevisiae : Ethanol

Solution:
As we learnt in

Principles -

Biotechnology deals in two principles -

1- Maintenance of microbial contamination free (sterile) surrounding in chemical engineering (old biotechnology).

Ex. production of ethanol by fermentation

2 - Genetic Engineering - It is also known as Recombinant DNA Technology or RDI.

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- wherein

Only desired organism grow under sterile condition to produce large amount of desired product.

It is the asexual method of gene transfer across taxonomy boundary.

Ethanol is produced by fermentation of carbohydrates by Saccharomyces cerevisiae (Brewer's yeast).

Q. 23 Select the **correct** route for the passage of sperms in male frogs :

Option 1:
Testes → Bidder's canal → Kidney → Vasa efferentia → Urinogenital duct → Cloaca

Option 2:
Testes → Vasa efferentia → Kidney → Seminal Vesicle → Urinogenital duct → Cloaca

Option 3:
Testes → Vasa efferentia → Bidder's canal → Ureter → Cloaca

Option 4:
Testes → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca

Correct Answer:
Testes → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca

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Solution:
The correct route for the passage of sperm in male frog is as follows:

Testes→ Vasa efferentia→kidney→bidder's canal→urinogenital duct→cloaca.

Thus, the correct answer is option d.

Q. 24 A temporary endocrine gland in the human body is :

Option 1:
Pineal gland

Option 2:
Corpus cardiacum

Option 3:
Corpus luteum

Option 4:
Corpus allatum

Correct Answer:
Corpus luteum

Solution:
The human corpus luteum is a temporary endocrine gland because it secretes progesterone for maintenance of early pregnancy. It develops after ovulation from the ruptured follicle during the luteal phase.

Hence, the correct answer is 'Corpus luteum'

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Q. 25 Homozygous purelines in cattle can be obtained by:

Option 1:
mating of related individuals of same breed

Option 2:
mating of unrelated individuals of same breed

Option 3:
mating of individuals of different breed

Option 4:
mating of individuals of different species

Correct Answer:
mating of related individuals of same breed

Solution:
Homozygous pure lines can be obtained by mating related individuals of the same breed.

Inbreeding is defined as the mating of related individuals of the same breed for 4-6 generations. hence correct option is (1).

Q. 26 Which one from those given below is the period for Mendel's hybridization experiments?

Option 1:
1856 - 1863

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Option 2:
1840 - 1850

Option 3:
1857 - 1869

Option 4:
1870 - 1877

Correct Answer:
1856 - 1863

Solution:
As we learnt in

Mendel's law of inheritance -

Mendel conducted hybridisation experiments on garden peas for seven years (1856 - 1863) and proposed the law of inheritance in living organism

- wherein

It is for the first time that statistical analysis and mathematical logic were applied to problem in biology.

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Q. 27 Which of the following cell organelles is responsible for extracting energy from carbohydrates to form ATP?

Option 1:
Lysosome

Option 2:
Ribosome

Option 3:
Chloroplast

Option 4:
Mitochondrion

Correct Answer:
Mitochondrion

Solution:
Mitochondrion cell is the “Power house of Cell” it oxidizes the carbohydrates to derive ATP.

Q. 28 The final proof for DNA as the genetic material came from the experiments of :

Option 1:
Griffith

Option 2:
Hershey and Chase

Option 3:
Avery, Mcleod and McCarty

Option 4:
Hargobind Khorana

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Correct Answer:
Hershey and Chase

Solution:
As we learnt in

Conclusion Made by Hershey and Chase -

DNA in the genetic material that is passed from virus to bacteria.

-

Q. 29 Select the **mismatch**

Option 1:
Cycas -Dioecious

Option 2:
Salvinia-Heterosporous

Option 3:
Equisetum-Homosporous

Option 4:
Pinus - Dioecious

Correct Answer:
Pinus - Dioecious

Solution:
As we learnt in

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Gymnosperms -

Evergreen, perennial woody trees or shrubs. They are non-flowering plants producing naked seeds.

- wherein

Eg Cycas, Pinus, Cedrus

Reproduction in Gymnosperms -

Pollen grains are produced from microspores within microsporangia. Naked ovules are borne on megasporophyllus and has nucellus which develops into megaspore mother cell.

- wherein

Pollen grains are transfer by wind to ovules where fertilization takes place by releasing male gamates by pollen tube inside archegonia (female sex organ).

Pinus belong to gymnosperms in which both sex organs are present on plant, i.e, monoecious

Q. 30 Transplantation of tissues/organs fails often due to non-acceptance by the patient's body. Which type of immune-response is responsible for such rejections?

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Option 1:
Autoimmune response

Option 2:
Cell-mediated immune response

Option 3:
Hormonal immune response

Option 4:
Physiological immune response

Correct Answer:
Cell-mediated immune response

Solution:
As we learnt in

Transplantation -

Very often, when some human organs like heart, eye, liver, kidney fail to function satisfactorily, transplantation is the only remedy to enable the patient to live a normal life

- wherein

For this a suitable donor is required. Grafts from just any source – an animal, another primate, or any human beings cannot be made since the grafts would be rejected sooner or later. Tissue matching, blood group matching are essential before undertaking any graft/transplant and even after this the patient has to take immuno–suppresants all his/her life. The body is able to differentiate 'self ' and 'nonself' and the cell-mediated immune response is responsible for the graft rejection

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Transplantation may be rejected in a patient due to the cell-mediated immune response that is mediated by T-lymphocytes.

Q. 31 Which statement is wrong for Krebs' cycle?

Option 1:
There are three points in the cycle where NAD^+ is reduced to $NADH+H^+$

Option 2:
There is one point in the cycle where FAD^+ is reduced to $FADH_2$

Option 3:
During conversion of succinyl CoA to succinic acid, a molecule of GTP is synthesised

Option 4:
The cycle starts with condensation of acetyl group (acetyl CoA) with pyruvic acid to yield citric acid

Correct Answer:
The cycle starts with condensation of acetyl group (acetyl CoA) with pyruvic acid to yield citric acid

Solution:

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As we have learnt in:

Steps of Krebs Cycle -

Step 1: Formation of Citrate
This is a condensation step, combining the two-carbon acetyl group with a four-carbon oxaloacetate molecule to form a six-carbon molecule of citrate.

Step 2: Formation of Isocitrate
The citrate is rearranged to form an isomeric form, isocitrate by an enzyme aconitase. This reaction is inhibited by fluoroacetate

Step 3: Oxidation of Isocitrate to α-Ketoglutarate
In this step, isocitrate dehydrogenase catalyzes the oxidative decarboxylation of isocitrate to form α-ketoglutarate. It generates NADH from NAD

Step 4: Oxidation of α-Ketoglutarate to Succinyl-CoA
α-Ketoglutarate is oxidized, carbon dioxide is removed, and coenzyme A is added to form the 4-carbon compound succinyl-CoA. During this oxidation, NAD+ is reduced to NADH + H+. The enzyme that catalyzes this reaction is α-Ketoglutarate dehydrogenase. This reaction is inhibited by arsenite dehydrogenase

Step 5: Conversion of Succinyl-CoA to Succinate
CoA is removed from succinyl-CoA to produce succinate. The energy released is used to make guanosine triphosphate (GTP) from guanosine diphosphate (GDP) and Pi by substrate-level phosphorylation. GTP can then be used to make ATP. The enzyme succinyl-CoA synthase catalyzes this reaction of the citric acid cycle.

-

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Krebs cycle is the reactions that help in converting pyruvic acid to carbon dioxide and water in mitochondria. In the first step, acetyl-CoA undergoes condensation with the oxaloacetic acid to yield citric acid. This reaction is catalyzed by the enzyme citric acid synthetase. Since citric acid is the first intermediate formed in Krebs cycle it is also called citric acid cycle.

Q. 32 Which of the following statements is correct?

Option 1:
The ascending limb of loop of Henle is impermeable to water

Option 2:
The descending limb of loop of Henle is impermeable to water

Option 3:
The ascending limb of loop of Henle is permeable to water

Option 4:
The descending limb of loop of Henle is permeable to electrolytes

Correct Answer:
The ascending limb of loop of Henle is impermeable to water

Solution:
As we learnt in

MECHANISM OF CONCENTRATION OF THE FILTRATE -

The flow of blood through the two limbs of vasa recta is also in a counter current pattern. The proximity between the Henle's loop and vasa recta, as well as the counter current in them help in maintaining an increasing

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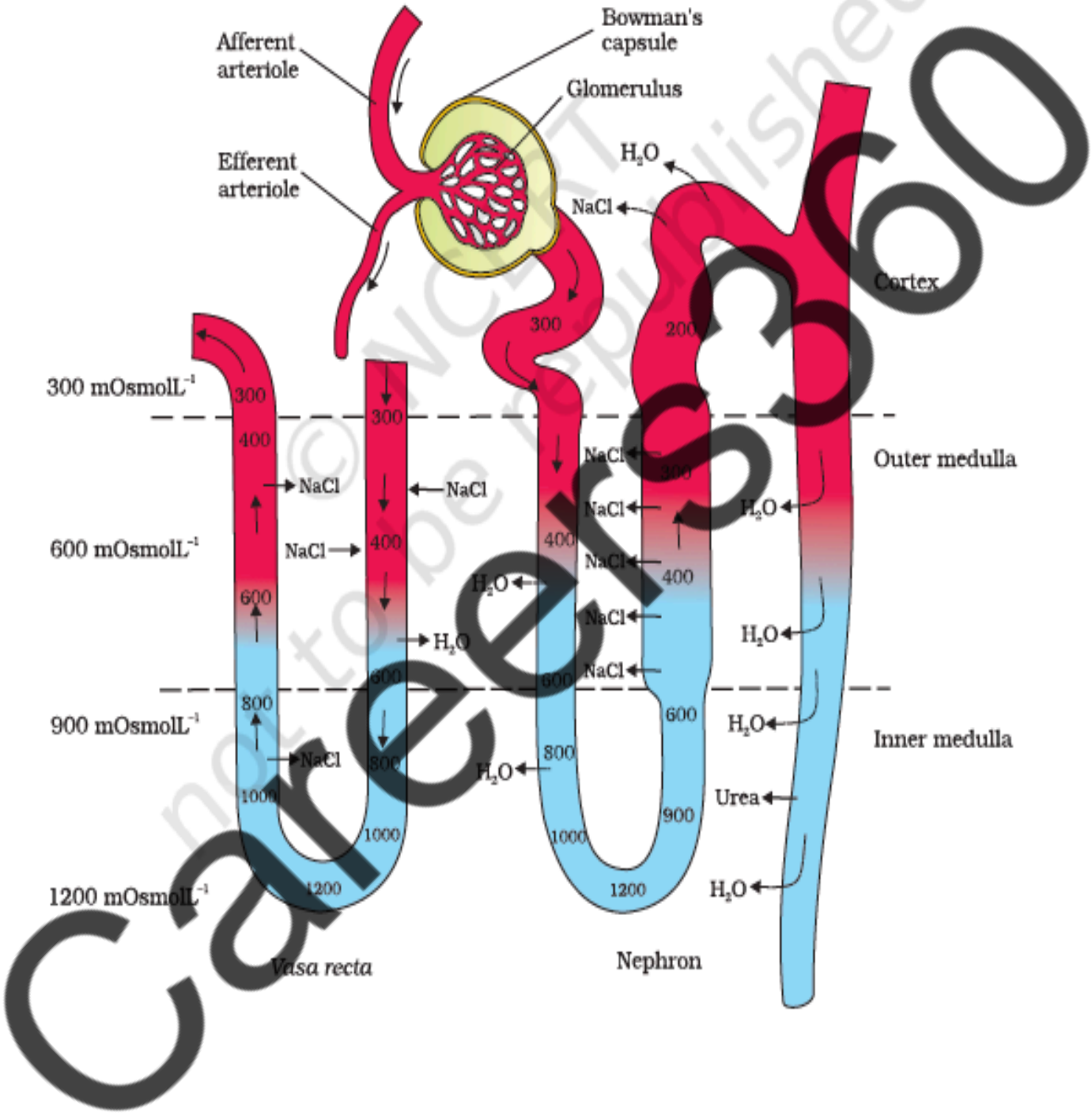
osmolarity towards the inner medullary interstitium, i.e., from 300 mOsmol L^{-1} in the cortex to about 1200 mOsmol L^{-1} in the inner medulla.

- wherein



Only salt goes outside in ascending loop, not water.

Q. 33 In case of poriferans the spongocoel is lined with flagellated cells called :

Option 1:
ostia

Option 2:
oscula

Option 3:
choanocytes

Option 4:
mesenchymal cells

Correct Answer:
choanocytes

Solution:
As we learnt in

Choanoderm -

Gastral epithelium layer consists of flagellated cell or choanocytes.

- wherein

These are characteristics cell of phylum porifera responsible for ingestion of food and differentiation of sex cell.

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Choanocytes (collar cells) form lining of spongocoel in poriferans (sponges). Flagella in collar cells provide circulation to water in water canal system

---

Q. 34 Select the mismatch :

Option 1:
Frankia - *Alnus*

Option 2:
Rhodospirillum - Mycorrhiza

Option 3:
Anabaena - Nitrogen fixer

Option 4:
Rhizobium - Alfalfa

Correct Answer:
Rhodospirillum - Mycorrhiza

Solution:

- **Mycorrhiza** represents the **symbiotic or mutualistic relationship** between fungi with roots of higher vascular plants.
- For example, Pinus and Orchid plants
- While ***Rhodospirillum*** is a **facultative anaerobe and free-living nitrogen fixer which does not form a symbiosis.**

Hence, the correct option is **2).**

Q. 35 Which cells of 'Crypts of Lieberkuhn' secrete antibacterial lysozyme?

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Option 1:
Argentaffin cells

Option 2:
Paneth cells

Option 3:
Zymogen cells

Option 4:
Kupffer cells

Correct Answer:
Paneth cells

Solution:
Crypts of Lieberkuhn are tubular glands occur throughout the intestine between of the wild. Paneth cells are present in the mucous membrane living of the small intestine.

Hence, the correct answer is 'Paneth cells'

Q. 36 Viroids differ from viruses in having:

Option 1:
DNA molecules without protein coat

Option 2:
RNA molecules with protein coat

Option 3:
RNA molecules without protein coat

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Option 4:
DNA molecules with protein coat

Correct Answer:
RNA molecules without protein coat

Solution:
Virus is the microorganism that made up of protein and has a protein coat as outer covering. Virus is of two types on the basis of genetic material either DNA containing or RNA containing. Whereas viroid is the smallest particle which contain RNA as a genetic material but do not contain protein as outer covering.

Q. 37 Out of 'X' pairs of ribs in humans only 'Y' pairs are true ribs. Select the option that correctly represents values of X and Y and provides their explanation :

Option 1:
X = 12, Y = 7 True ribs are attached dorsally to vertebral column and ventrally to the sternum

Option 2:
X = 12, Y = 5 True ribs are attached dorsally to vertebral column and sternum on the two ends

Option 3:
X = 24, Y = 7 True ribs are dorsally attached to vertebral column but are free on ventral side

Option 4:
X = 24, Y = 12 True ribs are dorsally attached to vertebral column but are free on ventral side

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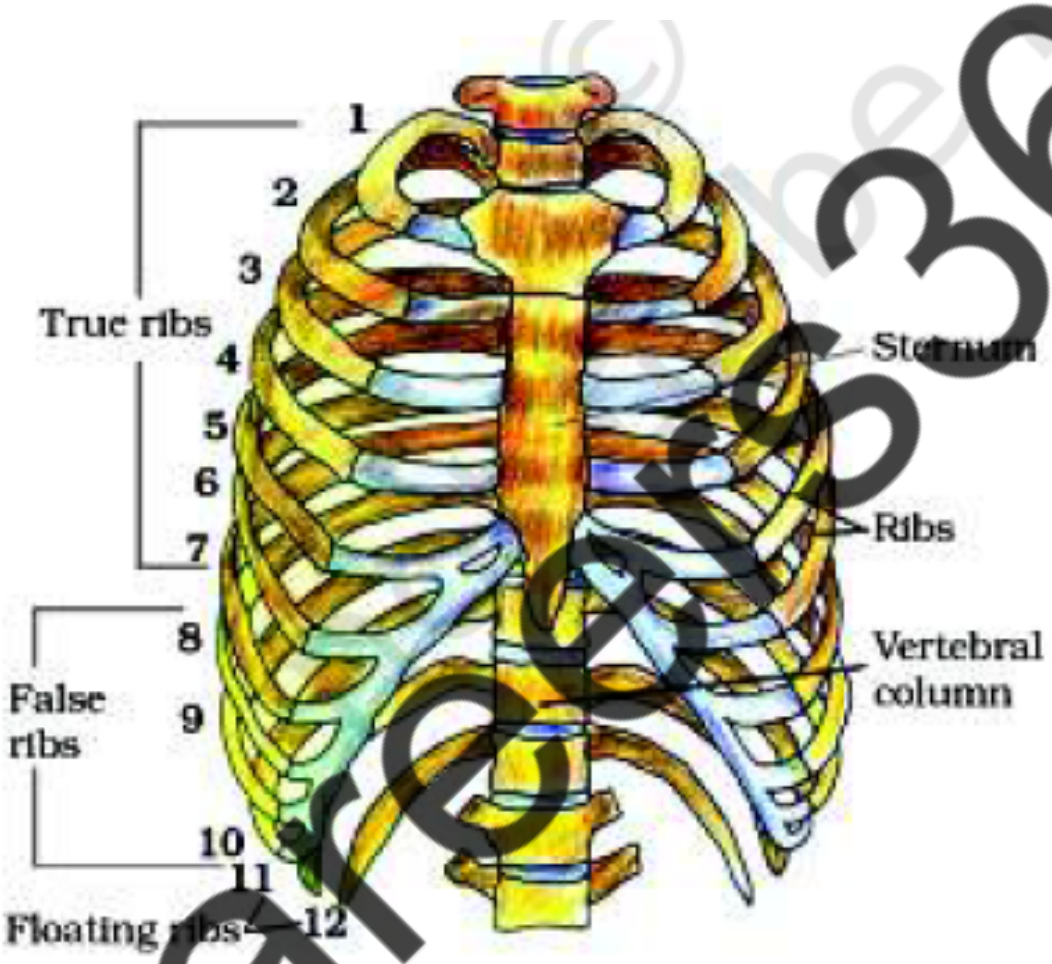
Correct Answer:

X = 12, Y = 7 True ribs are attached dorsally to vertebral column and ventrally to the sternum

Solution:

As we learnt in

True ribs -



The human rib cage is made up of 24 ribs, that is 12 pairs, of which 7 pairs are true as they are attached to the sternum, while other 5 ribs are false ribs attached to the sternum by coastal cartilages of the ribs above them. So, the correct answer is 'X=12, Y=5 True ribs are attached dorsally to vertebral column and sternum on the two ends'

Q. 38 The region of Biosphere Reserve which is legally protected and where no human activity is allowed is known as:

Option 1:
Core zone

Option 2:
Buffer zone

Option 3:
Transition zone

Option 4:
Restoration zone

Correct Answer:
Core zone

Solution:
The region of the biosphere reserve which is legally protected and where no human activity is allowed is known as the core zone. The correct option is a.

Q. 39 Which of the following is made up of dead cells?

Option 1:
Xylem parenchyma

Option 2:
Collenchyma

Option 3:
Phellem

Option 4:
Phloem

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Correct Answer:
Phellem

Solution:
As we learnt in

Cork cambium -

Cork cambium or extrastelar cambium or phellogen arises from the hypodermis or from the outer layer of cortex. Cork cambium divides periclinically(divides both outer and inner side). The outer side cells become dead and known as cork or phellem. The inner side cells are differentiated into parenolyma and contain chloroplasts. These cells are called secondary cortex or phelloderm.

-

Q. 40 The morphological nature of the edible part of coconut is:

Option 1:
Perisperm

Option 2:
Cotyledon

Option 3:
Endosperm

Option 4:
Pericarp

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Correct Answer:
Endosperm

Solution:
As learnt in

free-nuclear endosperm -

During endosperm development, the PEN undergoes successive nuclear divisions to give rise to free nuclei. This stage of endosperm development is called free-nuclear endosperm.

e.g., The coconut water from tender coconut that is a free-nuclear endosperm

Hence, the correct answer is option c, i.e., Endosperm.

Q. 41 What is the criterion for DNA fragments movement on agarose gel during gel electrophoresis?

Option 1:
The larger the fragment size, the farther it moves

Option 2:
The smaller the fragment size, the farther it moves

Option 3:
Positively charged fragments move to farther end

Option 4:
Negatively charged fragments do not move

Correct Answer:
The smaller the fragment size, the farther it moves

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Solution:
As we learnt in

Gel Electrophoresis -

This technique is used to separate DNA fragments on the basis of charge and size. Agarose is a polysaccharide which is extracted from red algae (gelidium). Gel electrophoresis is used to separate DNA fragments. Smaller the fragments, further it will move.

Q. 42 Presence of plants arranged into well defined vertical layers depending on their height can be seen best in :

Option 1:
Tropical Savannah

Option 2:
Tropical Rain Forest

Option 3:
Grassland

Option 4:
Temperate Forest

Correct Answer:
Tropical Rain Forest

Solution:
As we learnt in

Stratification -

Vertical distribution of different species occupying different levels is called stratification.

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- wherein

For example, trees occupy top vertical strata or layer of a forest, shrubs the second and herbs and grasses occupy the bottom layers.

Tropical rain forest shows stratification. It can be defined as the grouping of plants into two or more well defined layers depending upon their height. These layers are called strata or storeys.

Q. 43 A baby boy aged two years is admitted to play school and passes through a dental check-up. The dentist observed that the boy had twenty teeth. Which teeth were absent?

Option 1:
Incisors

Option 2:
Canines

Option 3:
Pre-molars

Option 4:
Molars

Correct Answer:
Pre-molars

Solution:

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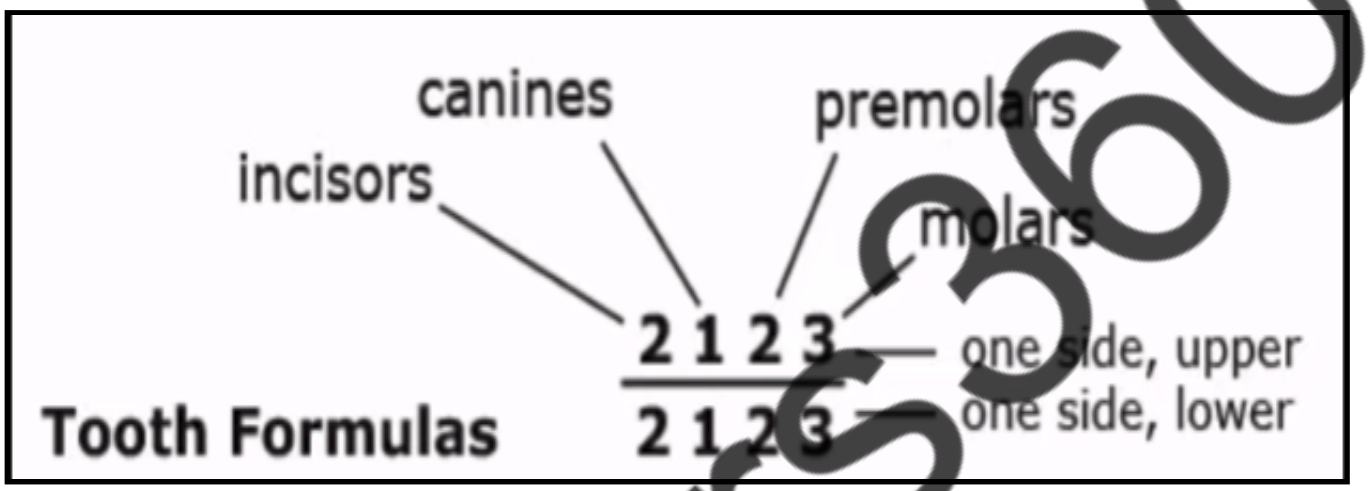
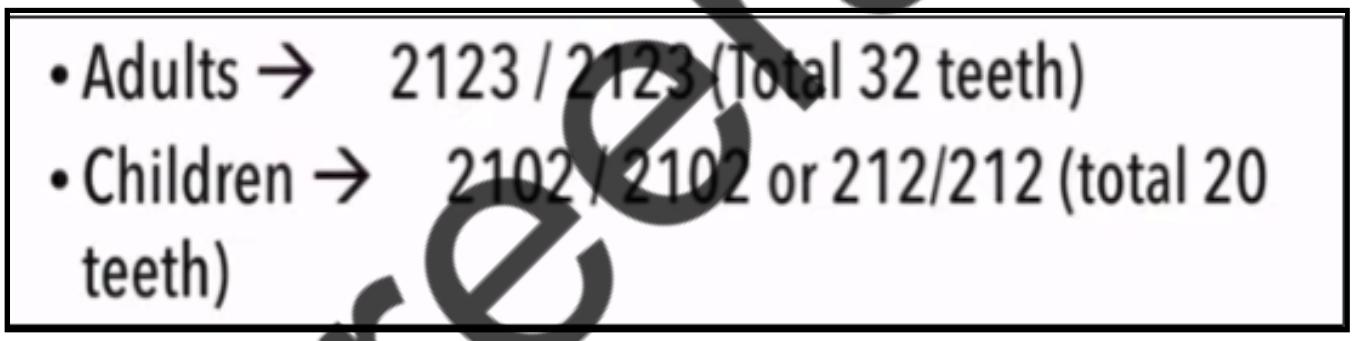
As we have learnt in:

Dental Formula:

- It shows the arrangement of teeth in each half of the upper and lower laws in order of incisors, canines, premolars and molars.





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Premolars are located between the canine and molar teeth. It is also known as transitional teeth. They help to tear and crush the food. It is four in number and is absent in infants which is actually replaced by the molars in the later stages.

Hence, the correct answer is 'Pre-molars'

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Q. 44 Which of the following components provides sticky character to the bacterial cell?

Option 1:
Cell wall

Option 2:
Nuclear membrane

Option 3:
Plasma membrane

Option 4:
Glycocalyx

Correct Answer:
Glycocalyx

Solution:
Sticky character of the bacterial wall is due to glycocalyx or slime layer. This layer is rich in glycoproteins.

Q. 45 Double fertilization is exhibited by:

Option 1:
Gymnosperms

Option 2:
Algae

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Option 3:
Fungi

Option 4:
Angiosperms

Correct Answer:
Angiosperms

Solution:
As we learnt in

Primary Endosperm Cell (PEC) -

Since two types of fusions, syngamy and triple fusion take place in an embryo sac the phenomenon is termed double fertilisation, an event unique to flowering plants. The central cell after triple fusion becomes the primary endosperm cell (PEC) and develops into the endosperm while the zygote develops into an embryo.

Double fertilization is unique to flowering plants, angiosperms.

Q. 46 In Bougainvillea thorns are the modifications of :

Option 1:
Stipules

Option 2:
Adventitious root

Option 3:
Stem

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Option 4:
Leaf

Correct Answer:
Stem

Solution:
As we learnt in

Types of special stem modifications -

Phyllacodes - Flat, fleshy, green leaf like structure which has a succession of nodes and internodes at long and short intervals. Leaves are modified into spine or scale to check transpiration. Photosynthesis takes place in phyllacodes. Ex: Opuntia, cereus, Euphorbia

Cladode - Phylloclade with one internode is called cladode. Ex: Aspargus & Ruscus

Thorn - Modification of axillary bud or terminal bud of the stem. It may bear leaves, flowers & Endogenous in origin. Ex: Catissa (Karonda), pomegranate, citrus, Bougainvillae

Tendrils - Helps the plant to climb, When it comes in contact with the neighbouring object it coils round it. It is modification of axillary bud and exogenous in origin. Ex: Grapes, cucumber, pumkins, watermelon etc

-

Thorns are modified stems.

Q. 47 Which of the following in sewage treatment removes suspended solids?

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Option 1:
Tertiary treatment

Option 2:
Secondary treatment

Option 3:
Primary treatment

Option 4:
Sludge treatment

Correct Answer:
Primary treatment

Solution:
As we learnt in

Primary treatment of sewage -

These treatment steps basically involve physical removal of particles – large and small – from the sewage through filtration and sedimentation.

These are removed in stages; initially, floating debris is removed by sequential filtration. Then the grit (soil and small pebbles) are removed by sedimentation. All solids that settle form the primary sludge and the supernatant forms the effluent. The effluent from the primary settling tank is taken for secondary treatment.

Primart treatment steps basically involve physical removal of particles – large and small – from the sewage through filtration and sedimentation.

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Q. 48 Receptor sites for neurotransmitters are present on:

Option 1:
Membranes of synaptic vesicles

Option 2:
Pre-synaptic membrane

Option 3:
Tips of axons

Option 4:
Post-synaptic membrane

Correct Answer:
Post-synaptic membrane

Solution:

-

Neurotransmitters are the chemical which helps to transfer the nerve signal from one neuron to the other during chemical synapse. These are stored in the synaptic vesicles at the axon terminal. These are released across the synaptic cleft, where they bind to specific receptors located on the postsynaptic side of the synapse.-

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Q. 49 A dioecious flowering plant prevents both:

Option 1:
Autogamy and xenogamy

Option 2:
Autogamy and geitonogamy

Option 3:
Geitonogamy and xenogamy

Option 4:
Cleistogamy and xenogamy

Correct Answer:
Autogamy and geitonogamy

Solution:

Mechanisms to control inbreeding depression -

1. pollen release and stigma receptivity are not synchronised
2. the anther and stigma are placed at different positions so that the pollen cannot come in contact with the stigma of the same flower
3. self-incompatibility
4. production of unisexual flowers

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- wherein

If both male and female flowers are present on the same plant such as castor and maize (monoecious), it prevents autogamy but not geitonogamy. In several species such as papaya, male and female flowers are present on different plants, that is each plant is either male or female (dioecy). This condition prevents both autogamy and geitonogamy.

Dioecious plants prevent both autogamy and geitonogamy.

Q. 50 The pivot joint between atlas and axis is a type of :

Option 1:
Fibrous joint

Option 2:
Cartilaginous joint

Option 3:
Synovial joint

Option 4:
Saddle joint

Correct Answer:
Synovial joint

Solution:

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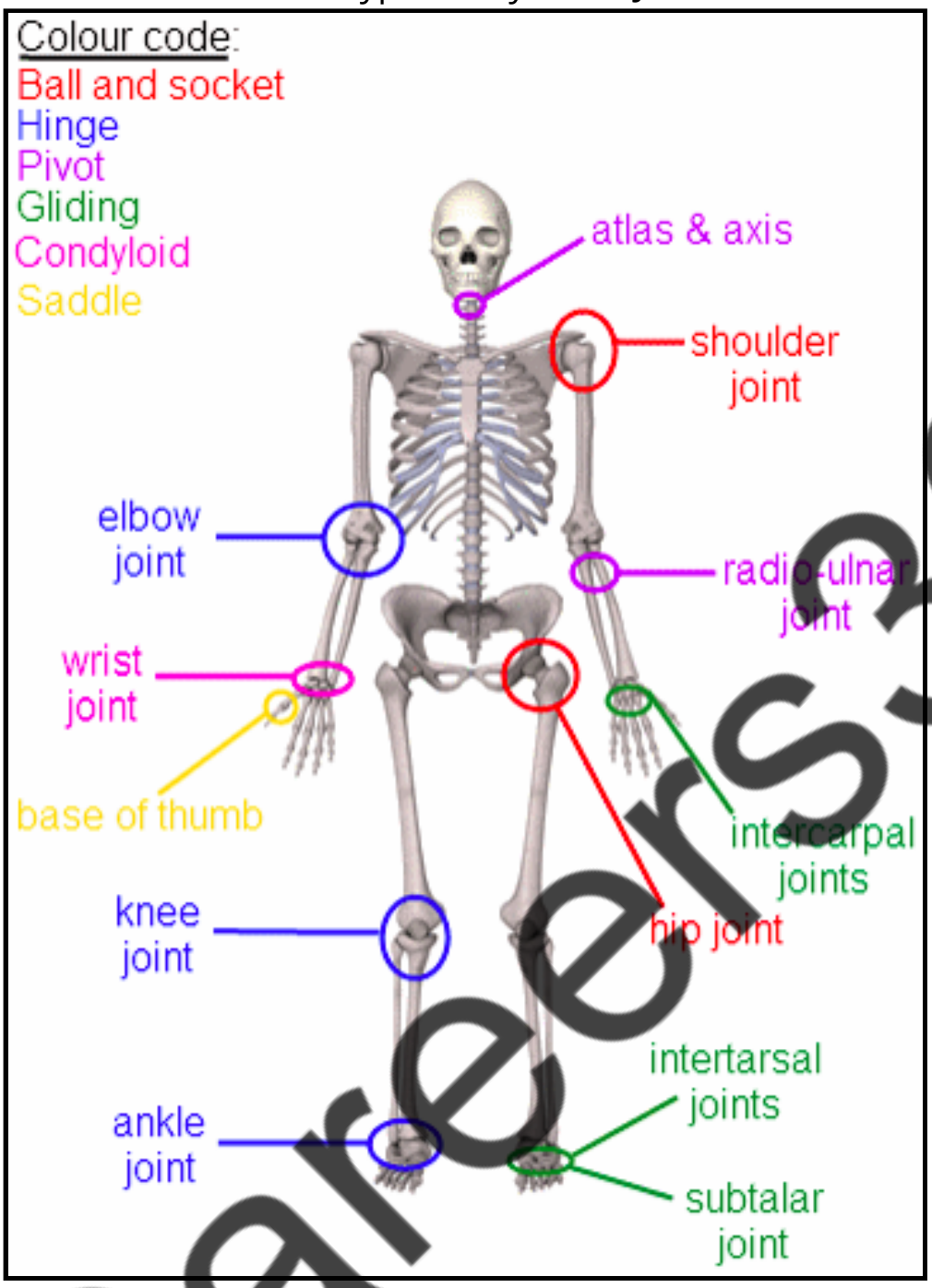
As we have learnt in types of Synovial Joints -



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The six types of synovial joints are the pivot, hinge, saddle, plane, condyloid and ball and socket joints. Pivot joints are found in neck region between atlas and axis.

Hence, the correct answer is 'Synovial joint'

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Q. 51 The water potential of pure water is:

Option 1:
Zero

Option 2:
Less than zero

Option 3:
More than zero but less than one

Option 4:
More than one

Correct Answer:
Zero

Solution:

As we have learnt in:

Water Potential:

- The concept of water potential was given by Slatyer and Taylor in 1960.
- The potential energy of water is called water potential. e.g., water is stored behind a dam.
- The term 'water potential' is used to describe the tendency of the water molecules to move from one place to another.
- Water moves from the region of high water potential to low water potential.
- Water potential of pure water is maximum which is taken as zero.
- Water potential of solutions is always less than zero, that is, they have negative water potential.

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- Water potential is denoted by ψ_w
- It is measured in Mega Pascal (MPa).
-

-Hence, the correct answer is option a.

Q. 52 Which ecosystem has the maximum biomass?

Option 1:
Forest ecosystem

Option 2:
Grassland ecosystem

Option 3:
Pond ecosystem

Option 4:
Lake ecosystem

Correct Answer:
Forest ecosystem

Solution:

As we have already discussed in Pyramid of Biomass & Pyramid of Energy
-

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Biomass refers to the total weight of living matter per unit area. In an ecosystem, the biomass decreases from the producer level to the consumer level. Forest ecosystem have the maximum biomass because it includes organism of all trophic level.

Hence, the correct option is (a).

Q. 53 GnRH, a hypothalamic hormone, needed in reproduction, acts on:

Option 1:
Anterior pituitary gland and stimulates secretion of LH and oxytocin

Option 2:
Anterior pituitary gland and stimulates secretion of LH and FSH

Option 3:
Posterior pituitary gland and stimulates secretion of oxytocin and FSH

Option 4:
Posterior pituitary gland and stimulates secretion of LH and relaxin

Correct Answer:
Anterior pituitary gland and stimulates secretion of LH and FSH

Solution:
GnRH is a gonadotrophin hormone which is released by the hypothalamus. It acts on the anterior pituitary and stimulates the secretion of the LH and FSH which further helps in the development of gonads.

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Hence, the correct answer is 'Anterior pituitary gland and stimulates secretion of LH and FSH'

Q. 54 Alexander Von Humbolt described for the first time:

Option 1:
Ecological Biodiversity

Option 2:
Laws of limiting factor

Option 3:
Species area relationships

Option 4:
Population Growth equation

Correct Answer:
Species area relationships

Solution:
As we learnt in

Species- Area relationships -

Every area has different species of plants and animals. It is important to note that the relation between species richness with area of wide variety of taxa (birds, plants, freshwater fishes and bats) turns out to be rectangular hyperbola.

- wherein

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log S = log C + Z log A
where
S= Species richness A= Area
Z = slope of the line (regression coefficient)
C=Y-intercept

Q. 55 Fruit and leaf drop at early stages can be prevented by the application of

Option 1:
Cytokinins

Option 2:
Ethylene

Option 3:
Auxins

Option 4:
Gibberellic acid

Correct Answer:
Auxins

Solution:
As we learnt in

Auxin -

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Tryptophan is the precursor of Indole acetic acid (IAA, natural auxin). Important biassay of auxins are Avena curvature test and cross root inhibition test.

- wherein

Discovered by Went.

NAA and 2,4-D are auxins which prevent fruit drop in tomato and citrus respectively.

Q. 56 Which of the following facilitates opening of stomatal aperture?

Option 1:
Contraction of outer wall of guard cells

Option 2:
Decrease in turgidity of guard cells

Option 3:
Radial orientation of cellulose microfibrils in the cell wall of guard cells

Option 4:
Longitudinal orientation of cellulose microfibrils in the cell wall of guard cells

Correct Answer:
Radial orientation of cellulose microfibrils in the cell wall of guard cells

Solution:

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As we learnt in

Opening and closing of stomata -

This is governed by the change in osmotic pressure or turgidity of guard cells.

- wherein

When guard cells are turgid, stomata open and when guard cells are flaccid, stomata close.

Increase in the turgidity of guard cells, which is due to radial orientation of cellulose microfibrils in the cell wall.

Q. 57 DNA replication in bacteria occurs:

Option 1:
During S-phase

Option 2:
Within nucleolus

Option 3:
Prior to fission

Option 4:
Just before transcription

Correct Answer:
Prior to fission

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Solution:
DNA replication is the process of creating new DNA from existing DNA. DNA replication occurs in microbes before fission. Mesosomes help in its occurrence.

Q. 58 Phosphonol pyruvate (PEP) is the primary CO_2 acceptor in :

Option 1:
C_3 plants

Option 2:
C_4 plants

Option 3:
C_2 plants

Option 4:
C_3 and C_4 plants

Correct Answer:
C_4 plants

Solution:
The C4 Pathway -

- The primary CO2 acceptor is a 3-carbon molecule phosphoenolpyruvate (PEP) and is present in the mesophyll cells.
- The CO2 is received in the form of bicarbonate ions.
- The carbonic anhydrase hydrates CO2 and produce bicarbonate
- The enzyme responsible for this fixation is PEP carboxylase or PEPcase.
- It is important to register that the mesophyll cells lack RuBisCO enzyme.
- The C4 acid OAA is formed in the mesophyll cells.
- It then forms other 4-carbon compounds like malic acid or aspartic acid in the mesophyll cells itself, which are transported to the bundle sheath cells.
- In the bundle sheath cells these C4 acids are broken down to release CO2 and a 3-carbon molecule.

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- The 3-carbon molecule is transported back to the mesophyll where it is converted to PEP again, thus, completing the cycle
- The CO_2 released in the bundle sheath cells enters the C3 or the Calvin pathway, a pathway common to all plants.
- The bundle sheath cells are rich in an enzyme Ribulose bisphosphate carboxylase-oxygenase (RuBisCO), but lack PEPcase.
- Thus, the basic pathway that results in the formation of the sugars, the Calvin pathway, is common to the C3 and C4 plants.

-Hence, the correct answer is option (b).

Q. 59 Which of the following options best represents the enzyme composition of pancreatic juice?

Option 1:
Amylase, peptidase, trypsinogen, rennin

Option 2:
Amylase, pepsin, trypsinogen, maltase

Option 3:
Peptidase, amylase, pepsin, rennin

Option 4:
Lipase, amylase, trypsinogen, procarboxypeptidase

Correct Answer:
Lipase, amylase, trypsinogen, procarboxypeptidase

Solution:

As we have learnt in pancreas:

- The pancreatic juice contains:
- Sodium bicarbonate

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- Proenzymes: trypsinogen, chymotrypsinogen and procarboxypeptidase
- Enzymes: pancreatic amylase, DNAse, RNAse, pancreatic lipase

- The pancreas plays an important role in food digestion.
- Its exocrine portion secretes a fluid called pancreatic juice that contains enzymes such as lipase, amylase, trypsinogen, procarboxypeptidase, chymotrypsinogen, nucleases and elastase.
- The pancreatic duct unites with the common bile duct and both these ducts open into the duodenum.
- In this manner, the pancreatic juices reaches the duodenum to perform its digestive activity.

Q. 60 Among the following characters, which one was not considered by Mendel in his experiments on pea?

Option 1:
Stem - Tall or Dwarf

Option 2:
Trichomes - Glandular or non-glandular

Option 3:
Seed - Green or Yellow

Option 4:
Pod - Inflated or Constricted

Correct Answer:
Trichomes - Glandular or non-glandular

Solution:
As we learnt in

Mendel's law of inheritance -

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Mendel conducted hybridisation experiments on garden peas for seven years (1856 - 1863) and proposed the law of inheritance in living organism

- wherein

It is for the first time that statistical analysis and mathematical logic were applied to problem in biology.

Mendel experimented on

1. Seed colour

2. Seed shape

3. Flower colour

4. Pod colour

5. Pod shape

6. Flower position

7. Plant height

Q. 61 A gene whose expression helps to identify transformed cell is known as :

Option 1:
Selectable marker

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Option 2:
Vector

Option 3:
Plasmid

Option 4:
Structural gene

Correct Answer:
Selectable marker

Solution:
In recombinant DNA technology, selectable markers are the specific genes that are used to identify the transformants from the non-transformants after the process of recombination. These genes are used to detect whether the incorporation of a nucleic acid sequence has been successful into an organism's DNA. These marker genes are resistant to various antibiotics and are used in cloning vectors.

Hence, the correct answer is 'selectable markers'

Q. 62 In case of a couple where the male is having a very low sperm count, which technique will be suitable for fertilisation?

Option 1:
Intrauterine transfer

Option 2:
Gamete intracytoplasmic fallopian transfer

Option 3:
Artificial Insemination

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Option 4:
Intracytoplasmic sperm injection

Correct Answer:
Artificial Insemination

Solution:
As we learnt in

Artificial insemination (AI) -

Infertility cases either due to inability of the male partner to inseminate the female or due to very low sperm counts in the ejaculates, could be corrected by artificial insemination (AI) technique

- wherein

In this technique, the semen collected either from the husband or a healthy donor is artificially introduced either into the vagina or into the uterus (IUI-intra-uterine insemination) of the female

AI is carried out when there is very low sperm count.

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Q. 63 Match the following sexually transmitted diseases (column-I) with their causative agent (column-II) and select the correct option:

Column-I	Column-II
A) Gonorrhea	i) HIV
B) Syphilis	ii) Neisseria
C) Genital Warts	iii) Treponema
D) AIDS	iv) Human Papilloma-Virus

Option 1:

A	B	C	D
iii	iv	i	ii

Option 2:

A	B	C	D
iv	ii	iii	i

Option 3:

A	B	C	D
iv	iii	ii	i

Option 4:

A	B	C	D
ii	iii	iv	i

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Correct Answer:

A	B	C	D
ii	iii	iv	i

Solution:
As we learnt in

Cause of AIDS -

AIDS is caused by the Human Immuno Deficiency Virus (HIV), a member of a group of viruses called retrovirus.

- wherein

Retrovirus has an envelope enclosing the RNA genome.

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Q. 64 Which among these is the correct combination of aquatic mammals?

Option 1:
Seals, Dolphins, Sharks

Option 2:
Dolphins, Seals, *Trygon*

Option 3:
Whales, Dolphins, Seals

Option 4:
Trygon, Whales, Seals

Correct Answer:
Whales, Dolphins, Seals

Solution:
As we learnt in

Class Mammalia -

Unique characteristic is presence of mammary glands. Two pairs of limbs and external ears or pinnae present.

-

Sharks and Trygon (sting ray) are the members of chondrichthyes (cartilaginous fish) while whale, Dolphin and Seals are aquatic mammals belong to class mammalia.

Q. 65 Flowers which have single ovule in the ovary and are packed into inflorescence are usually pollinated by:

Option 1:
Water

Option 2:
Bee

Option 3:
Wind

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Option 4:
Bat

Correct Answer:
Wind

Solution:
As we learnt in

Wind as an Agent of Pollination -

Wind pollination also requires that the pollen grains are light and non-sticky so that they can be transported in wind currents. They often possess well-exposed stamens so that the pollens are easily dispersed into wind currents. Stigma is large and often-feathery to easily trap air-borne pollen grains. They have a single ovule in each ovary and numerous flowers packed into an inflorescence.

Hence, the correct answer is option c.

Q. 66 Life cycle of *Ectocarpus* and *Fucus* respectively are :

Option 1:
Haplontic, Diplontic

Option 2:
Diplontic, Haplodiplontic

Option 3:
Haplodiplontic, Diplontic

Option 4:
Haplodiplontic, Haplontic

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Correct Answer:
Haplodiplontic, Diplontic

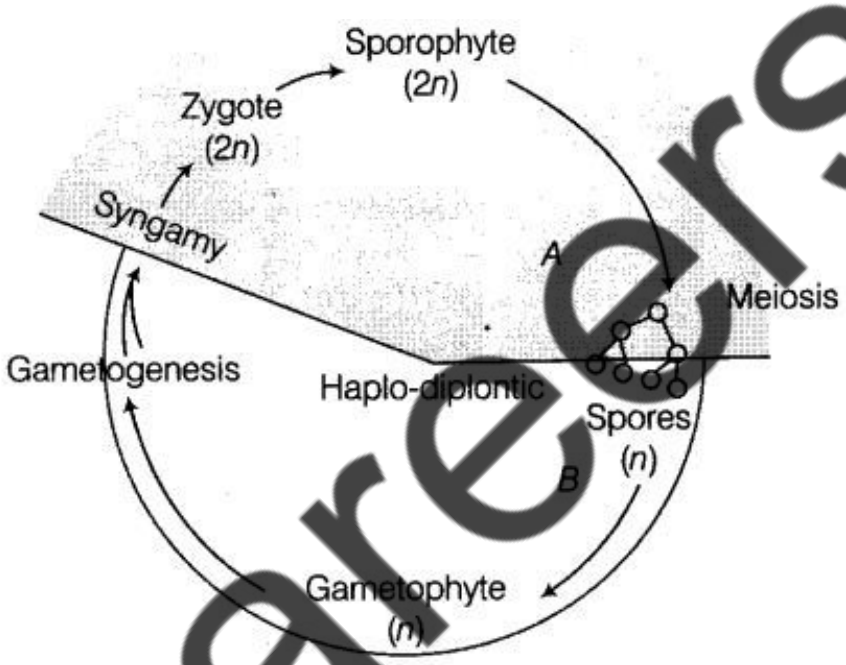
Solution:
As we learnt in

Haplo-diplontic life cycle -

Sporophyte and gametophyte both are photosynthetic and free living.

- wherein

Eg All bryophytes, pteridophytes, some brown algae (Laminaria), Ectocarpus and some green algae (ulva and cladophora)



Ectocarpus has haplodiplontic life cycle and Fucus has diplontic life cycle.

Q. 67 The association of histone H1 with a nucleosome indicates:

Option 1:
Transcription is occurring.

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Option 2:
DNA replication is occurring.

Option 3:
The DNA is condensed into a Chromatin Fibre.

Option 4:
The DNA double helix is exposed.

Correct Answer:
The DNA is condensed into a Chromatin Fibre.

Solution:
As we learnt in

Histones -

This is a set of positively charged basic proteins.

- wherein

Histones are rich in the basic amino acid residues lysines and arginines. They are organised to form a unit of eight molecules called histone octamer.

The DNA is condensed into a Chromatin Fibre.

Q. 68 During DNA replication, Okazaki fragments are used to elongate:

Option 1:
The leading strand towards replication fork.

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Option 2:
The lagging strand towards replication fork.

Option 3:
The leading strand away from replication fork.

Option 4:
The lagging strand away from the replication fork.

Correct Answer:
The lagging strand away from the replication fork.

Solution:
As discussed in

Replication Fork -

For long DNA molecules since the two strands, od DNA cannot be separated, the reparation occur within a small opening of DNA helix, referred to as replication fork.

- wherein

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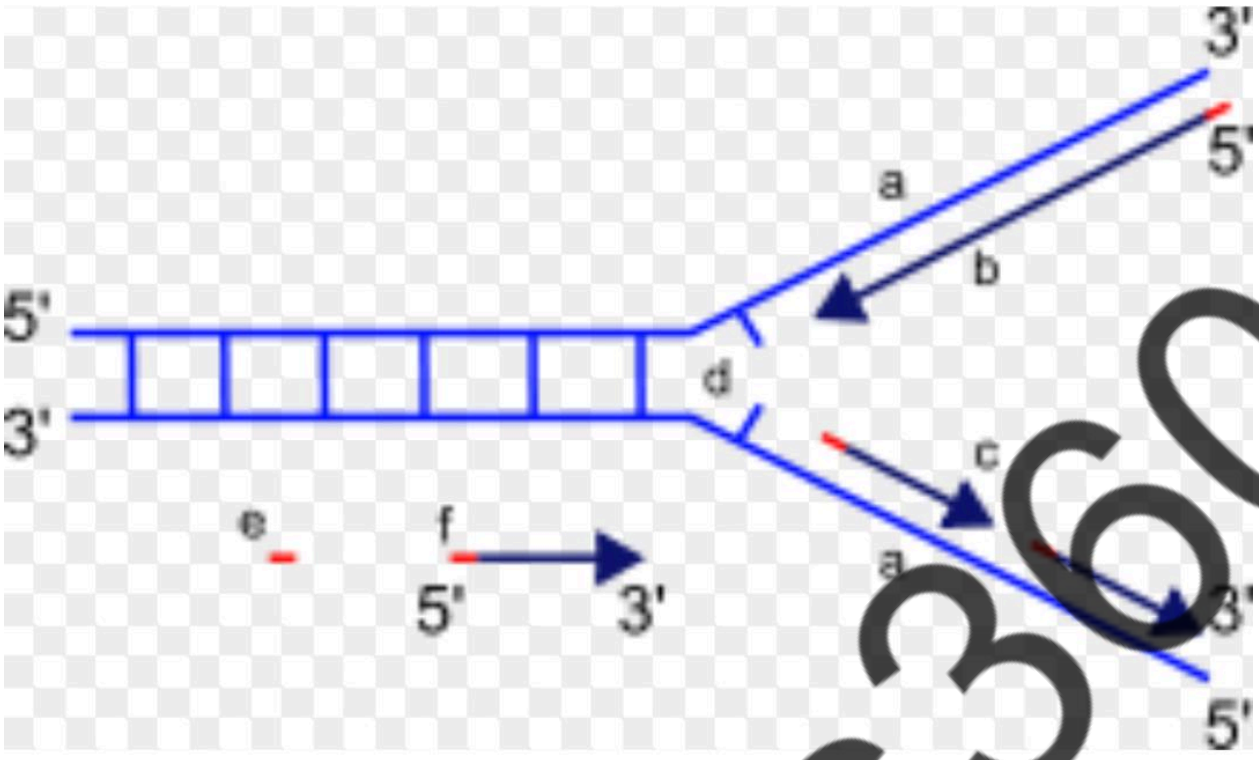

Two DNA polymerase molecules work simultaneous at the DNA fork, one on the leading strand and the other on the lagging strand.
Each Okazaki fragment is synthesized by DNA polymerase at lagging strand in $5' \rightarrow 3'$ direction.
New Okazaki fragments appear as the replication fork opens further.
As the first Okazaki fragment appears away from the replication fork, the direction of elongation would be away from replication fork.

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Q. 69 Thalassemia and sickle cell anemia are caused due to a problem in globin molecule synthesis. Select the correct statement.

Option 1:
Both are due to a qualitative defect in globin chain synthesis.

Option 2:
Both are due to a quantitative defect in globin chain synthesis.

Option 3:
Thalassemia is due to less synthesis of globin molecules.

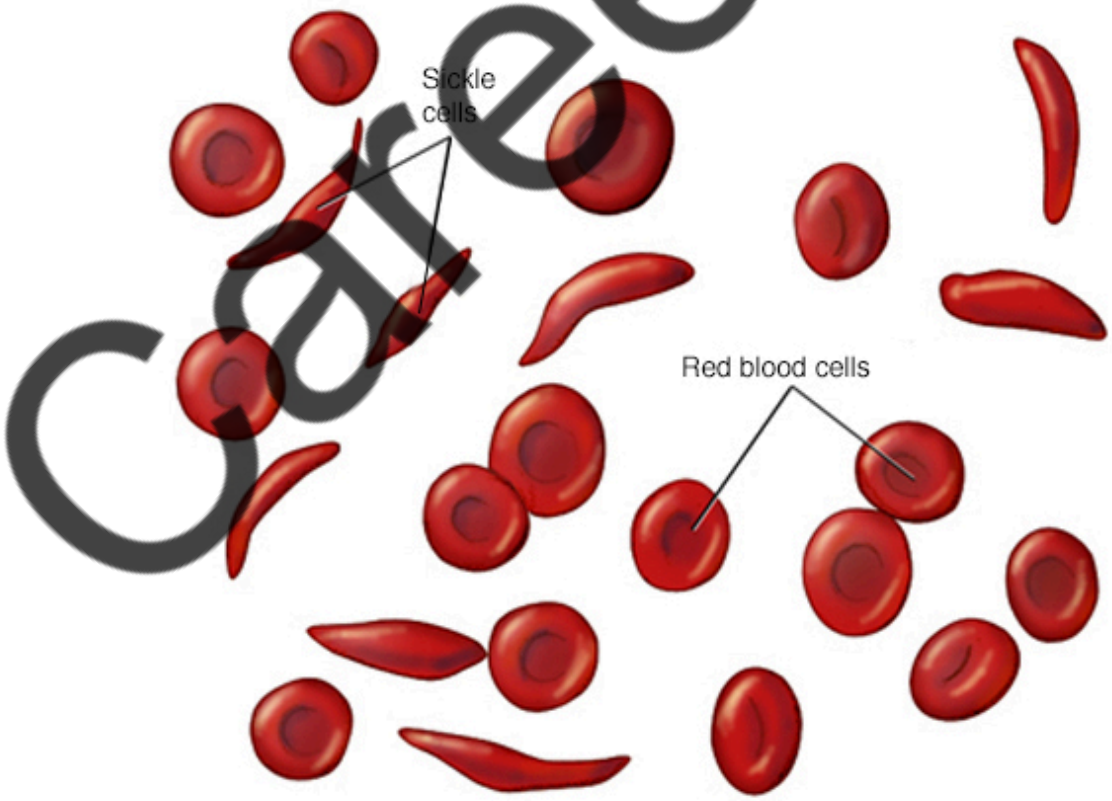
Option 4:
Sickle cell anemia is due to a quantitative problem of globin molecules.

Correct Answer:
Thalassemia is due to less synthesis of globin molecules.

Solution:
As we learnt in

Cause of Sickle-cell anaemia -

It is caused by the substitution of glutamic acid (Glu) by Valine (Val) at the 6th position of beta globin chain of the haemoglobin molecule.



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- wherein

The mutant haemoglobic molecule undergoes polymerisation under low oxygen tension causing the change in shape of RBC from bioconcave disc to elongated sickle line structure.

Q. 70 Coconut fruit is a :

Option 1:
Drupe

Option 2:
Berry

Option 3:
Nut

Option 4:
Capsule

Correct Answer:
Drupe

Solution:

As we have seen **Table showing important fruits and their edible parts:**

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Simple Fruits	**Scientific Name**	**Types**	Edible Part
Wheat	*Triticum aestivum*	Caryopsis	Endosperm & Embryo
Maize	*Zea mays*	Caryopsis	Endosperm & Embryo
Rice	*Oryza sativa*	Caryopsis	Endosperm & Embryo
Tomato	*Solanum lycopersicum*	Berry	Pericarp & Placenta
Litchi	*Litchi chinensis*	Nut	Fresh aril
Banana	*Musa paradisiaca*	Berry	Mesocarp & Endocarp
Brinjal	*Solanum melongena*	Berry	Pericarp & Placenta
Coconut	*Cocos nucifera*	Drupe	Endosperm, testa, cotyledon & embryo
Almond	*Prunus amygdalus*	Drupe	Seeds
Aggregate Fruits			
Strawberry	*Fragaria vesco*	Etaerio of achenes	Fleshy thalamus & seeds
Lotus	*Nelumbo nucifera*	Etaerio of achenes	Fleshy thalamus & seeds
Composite Fruits			
Jackfruit	*Artocarpus heterophyllus*	Sorosis	Fleshy axis, bracts, perianth & leaves
Fig	*Ficus carica*	Syconus	Peduncle & seeds
Mulberry	*Morus alba*	Sorosis	Fleshy axis and succulent perianth

-

Hence, coconut is a drupe.

Q. 71 Attractants and rewards are required for :

Option 1:
Anemophily

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Option 2:
Entomophily

Option 3:
Hydrophily

Option 4:
Cleistogamy

Correct Answer:
Entomophily

Solution:
As we learnt in

Pollination through insects -

To sustain animal visits, the flowers have to provide rewards to the animals. Nectar and pollen grains are the usual floral rewards. For harvesting the rewards from the flower, the animal visitor comes in contact with the anthers and the stigma. The body of the animal gets a coating of pollen grains, which are generally sticky in animal-pollinated flowers. When the animal carrying pollen on its body comes in contact with the stigma, it brings about pollination.

- wherein

In some species, floral rewards are in providing safe places to lay eggs; an example is that of the tallest flower of Amorphophallus.

One biotic agent for pollination is insects which transfer pollen grains to a stigma of flower. this is known as entomophily and plants are characterised by attractive and brightly coloured flowers.

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Q. 72 Spliceosomes are **not** found in cells of :

Option 1:
Plants

Option 2:
Fungi

Option 3:
Animals

Option 4:
Bacteria

Correct Answer:
Bacteria

Solution:
A spliceosome is a large and complex molecular machine found primarily within the splicing speckles of the cell nucleus of eukaryotic cells.
The spliceosome is assembled from snRNAs and protein complexes.

Q. 73 Hypersecretion of Growth Hormone in adults does not cause further increase in height, because:

Option 1:
Growth Hormone becomes inactive in adults.

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Option 2:
Epiphyseal plates close after adolescence.

Option 3:
Bones loose their sensitivity to Growth Hormone in adults.

Option 4:
Muscle fibres do not grow in size after birth .

Correct Answer:
Epiphyseal plates close after adolescence.

Solution:
As learnt in

Epiphyseal plates or receptors of growth hormones close after adulthood or adolescence.

Q. 74 Which one of the following statements is not valid for aerosols?

Option 1:
They are harmful to human health

Option 2:
They alter rainfall and monsoon patterns

Option 3:
They cause increased agricultural productivity

Option 4:
They have negative impact on agricultural land

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Correct Answer:
They cause increased agricultural productivity

Solution:
As we learnt in

Causes of air pollution -

Aerosols: Aerosols are small particles of all sorts of solid or liquid substances suspended in the air. They block the stomata of plants and prevent the gaseous exchanges between plants and atmosphere. In the absence of photosynthesis, agricultural productivity gets decreased. -

---

Q. 75 The vascular cambium normally gives rise to

Option 1:
Phelloderm

Option 2:
Primary phloem

Option 3:
Secondary xylem

Option 4:
Periderm

Correct Answer:
Secondary xylem

Solution:

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Secondary Growth in Intrastelar Region of Dicot Stem -

Formation of Secondary Vascular Tissues

- The vascular cambium produces secondary xylem towards inside, that is, towards the pith.
- It produces secondary phloem towards the outside.
- The continuous formation of secondary xylem crushes the primary phloem. The primary xylem remains intact at some places.
- The vascular cambium also produces radial parenchymatous cells that passes through the secondary xylem and secondary phloem. These are called secondary medullary rays.
- The secondary xylem has axial and radial (horizontal) system while radial system is absent in primary xylem.
- Primary xylem has long vessels and tracheids as compared to secondary xylem.

-

Hence, the correct answer is option c.

Q. 76 If there are 999 bases in RNA that codes for a protein with 333 amino acids, and the base at position 901 is deleted such that the length of the RNA becomes 998 bases, how many codons will be altered?

Option 1:
1

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Option 2:
11

Option 3:
33

Option 4:
333

Correct Answer:
33

Solution:
999 bases in RNA codes for a protein with 333 amino acids correct that is 900 + 99 bases or you can say **900 + 901 + 902** and so on..... and **300 + 33** codons..
If base at **901** position is deleted then first 900 bases will be normal while **bases after 901th base will be altered** that is **99 bases will be altered.**
3 bases stand for 1 codon, that is **99 bases are coded by 33** codons will be altered due to the deletion of one base at 901 position.

Hence, the correct answer is (c).

Q. 77 Which of the following are found in extreme saline conditions?

Option 1:
Archaebacteria

Option 2:
Eubacteria

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Option 3:
Cyanobacteria

Option 4:
Mycobacteria

Correct Answer:
Archaebacteria

Solution:
As we learnt in

Archaebacteria -

Live in harsh habitats.

Cell walls lack muramic acid and peptidoglycan.

Branched lipids having Phytanyl side groups decreases membrane fluidity.

- wherein

They are of three major type

- Methanogens =produces metane
- Halophiles = salt loving i.e. grow in salty area
- Thermoacidophiles = live in high temperature & low pH.

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Q. 78 The process of separation and purification of expressed protein before marketing is called :

Option 1:
Upstream processing

Option 2:
Downstream processing

Option 3:
Bioprocessing

Option 4:
Postproduction processing

Correct Answer:
Downstream processing

Solution:

As we learnt in
Downstream processing is used for separation and purification of the desired product before marketing. It is mainly used during large scale production of metabolites. Hence, the correct answer is 'Downstream processing'

Q. 79 Capacitation occurs in :

Option 1:
Rete testis

Option 2:
Epididymis

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Option 3:
Vas deferens

Option 4:
Female Reproductive tract

Correct Answer:
Female Reproductive tract

Solution:
As we learnt in

Events during Fertilisation -

Fertilisation occurs when sperm comes in contact with ovum inside the female reproductive tract. Capacitation in the female tract enhances sperm function which then comes in contact with zona pellucida.

Hence, the correct answer is option d.

Q. 80 Functional megaspore in an angiosperm develops into:

Option 1:
Ovule

Option 2:
Endosperm

Option 3:
Embryo sac

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Option 4:
Embryo

Correct Answer:
Embryo sac

Solution:
As we learnt in

Structure of Ovule -

Each ovule has one or two protective envelopes called integuments. Integuments encircle the nucellus except at the tip where a small opening called the micropyle is organised. Opposite the micropylar end, is the chalaza, representing the basal part of the ovule. Enclosed within the integuments is a mass of cells called the nucellus. Located in the nucellus is the embryo sac or female gametophyte.

Embryo sac develops from the functional megaspore. Hence, the correct answer is option c.

Q. 81 Anaphase promoting complex (APC) is a protein degradation machinery necessary for proper mitosis of animal cells. If APC is defective in a human cell, which of the following is expected to occur?

Option 1:
Chromosomes will not condense

Option 2:
Chromosomes will be fragmented

Option 3:
Chromosomes will not segregate

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Option 4:
Recombination of chromosome arms will occur

Correct Answer:
Chromosomes will not segregate

Solution:
Anaphase-promoting complex or cyclosome is a necessary protein degradation machinery for the proper mitosis of animal cells. It separates chromatids to form chromosomes during anaphase. In case of any defect in APC leads to the failure of the segregation of chromosomes.

Q. 82 Myelin sheath is produced by:

Option 1:
Schwann Cells and Oligodendrocytes

Option 2:
Astrocytes and Schwann Cells

Option 3:
Oligodendrocytes and Osteoclasts

Option 4:
Osteoclasts and Astrocytes

Correct Answer:
Schwann Cells and Oligodendrocytes

Solution:

As we have learnt in neuron:

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- There are two types of axons, namely, myelinated and unmyelinated.
- The myelinated nerve fibres are enveloped with Schwann cells, which form a myelin sheath around the axon.
- The gaps between two adjacent myelin sheaths are called nodes of Ranvier.
- Myelinated nerve fibres are found in spinal and cranial nerves.
- Unmyelinated nerve fibre is enclosed by a Schwann cell that does not form a myelin sheath around the axon and is commonly found in autonomous and the somatic neural systems.
- Myelin sheath is also known as medullary sheath. It is an insulating layer which covers the axon and helps to maintain a constant membrane potential. It is produced by Schwann cell in the peripheral nervous system and from oligodendroglia cells.

-

Hence, the correct answer is 'Schwann cells and oligodendrocytes'

Q. 83 Which of the following options gives the correct sequence of events during mitosis?

Option 1:
Condensation → nuclear membrane disassembly → crossing over → segregation → telophase

Option 2:
Condensation → nuclear membrane disassembly → arrangement at equator → centromere division → segregation → telophase

Option 3:
Condensation → crossing over → nuclear membrane disassembly → segregation → telophase

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Option 4:

Condensation → arrangement at equator → centromere division → segregation → telophase

Correct Answer:

Condensation → nuclear membrane disassembly → arrangement at equator → centromere division → segregation → telophase

Solution:

As we learnt in

Prophase -

Prophase is first stage of mitosis, marked by the initiation of condensation of chromosomal material, to form compact mitotic chromosomes. Mitotic spindle appears at the end of this phase , do not show golgi body, ER, nucleus.

- wherein

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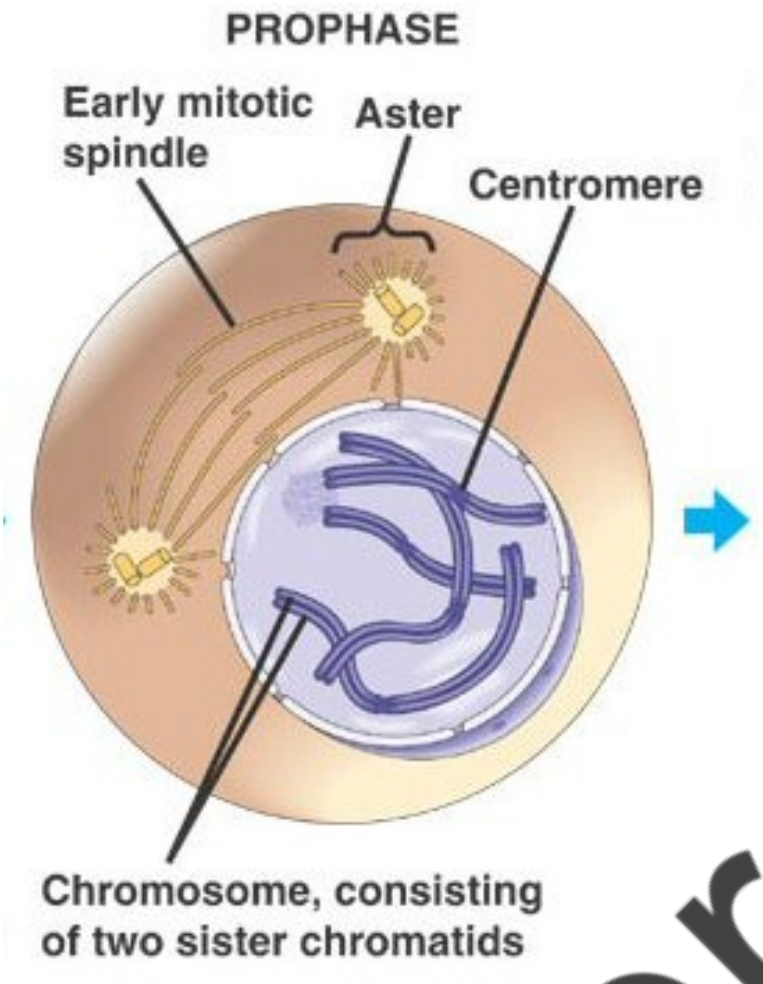

Condensation (chromosomes become visible) ⟶ Nuclear membrane disassembly(Late prophase) ⟶ Arrangement of equator(Metaphase) ⟶ Centromere division (Anaphase) ⟶ Segregation ⟶ Telophase

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Q. 84 A disease caused by an autosomal primary non-disjunction is :

Option 1:
Down's syndrome

Option 2:
Klinefelter's syndrome

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Option 3:
Turner's syndrome

Option 4:
Sickle cell anemia

Correct Answer:
Down's syndrome

Solution:
As we learnt in

Down's Syndrome -

It is a chromosomal disorder caused due toa gain of extra copy of chronosome.

- wherein

The affected indivisual is short, of small round head, burrowed tounge and partially open mouth.

It is a chromosomal disorder caused due to a gain of extra copy of chronosome.

Q. 85 Which one of the following is related to Ex-situ conservation of threatened animals and plants?

Option 1:
Wildlife Safari parks

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Option 2:
Biodiversity hot spots

Option 3:
Amazon rainforest

Option 4:
Himalayan region

Correct Answer:
Wildlife Safari parks

Solution:
Wildlife Safari parks conserve threatened species from their natural habitat. The correct option is a.

Q. 86 The function of copper ions in copper releasing IUD's is :

Option 1:
They suppress sperm motility and fertilising capacity of sperms

Option 2:
They inhibit gametogenesis

Option 3:
They make uterus unsuitable for implantation

Option 4:
They inhibit ovulation

Correct Answer:
They suppress sperm motility and fertilising capacity of sperms

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Q. 87 Which of the following RNAs should be most abundant in animal cell?

Option 1:
r-RNA

Option 2:
t-RNA

Option 3:
m-RNA

Option 4:
mi-RNA

Correct Answer:
r-RNA

Solution:
As we learnt in

Functions of types of RNA -

The mRNA provides the templete tRNA brings aminoacids and reads genetic code and rRNAs plays structural and catalytic during translation.

-

rRNA is most abundant in animal cell. It constitutes 80% of total RNA of the cell.

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Q. 88 Root hairs develop from the region of :

Option 1:
Maturation

Option 2:
Elongation

Option 3:
Root cap

Option 4:
Meristematic activity

Correct Answer:
Maturation

Solution:
Root hairs grow from the root's maturation region. The root tip is divided into several zones, including the zone of cell division (meristematic activity), the zone of elongation, and the zone of maturation. The zone of maturation refers to where freshly generated cells differentiate and mature into specialized root tissues, where root hairs emerge. Root hairs are elongated, finger-like projections that enhance the surface area of the root for improved water and nutrient absorption from the soil. Hence, the correct answer is Option 1.

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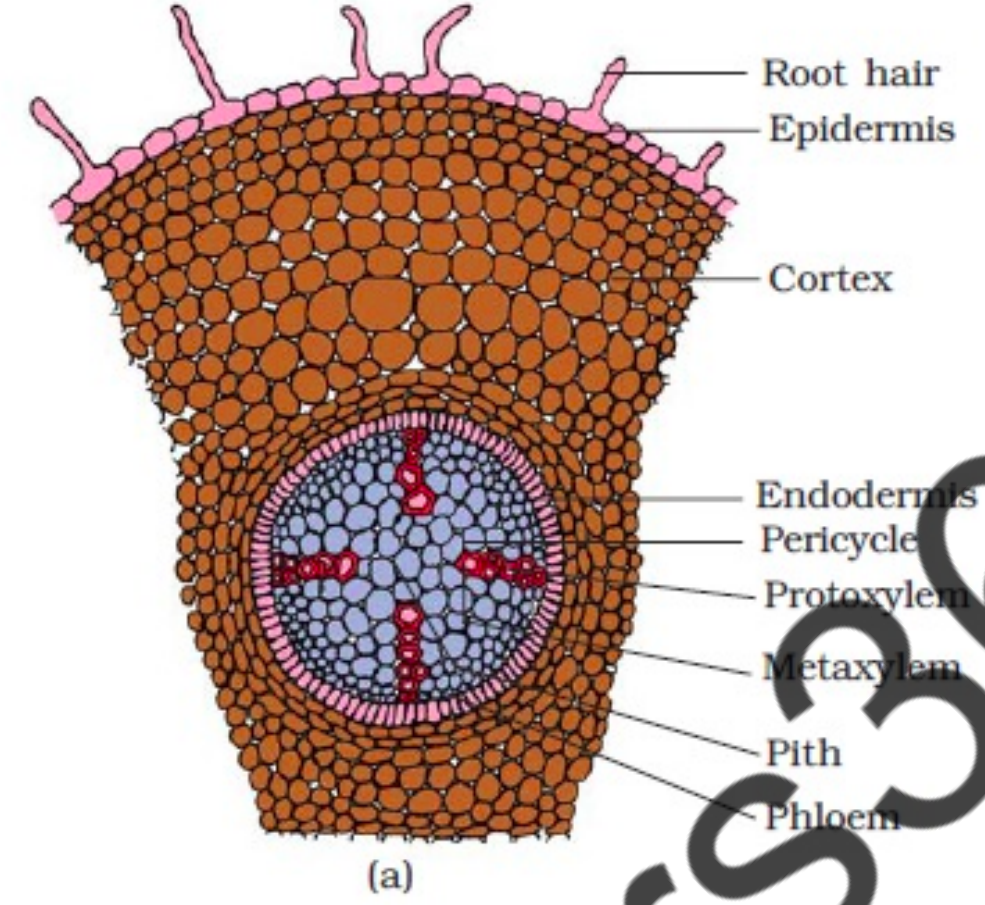

Q. 89 Frog's heart when taken out of the body continues to beat for some time.

Select the best option from the following statements.

(a) Frog is a poikilotherm.

(b) Frog does not have any coronary circulation.

(c) Heart is "myogenic" in nature.

(d) Heart is autoexcitable Options.

Options :

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Option 1:
Only (c)

Option 2:
Only (d)

Option 3:
(a) and (b)

Option 4:
(c) and (d)

Correct Answer:
(c) and (d)

Solution:
Frog's heart when taken out of the body continues to beat for some time because the heart is myogenic in nature and is autoexcitable. Thus, the correct answer is option d.

Q. 90 DNA fragments are :

Option 1:
Positively charged

Option 2:
Negatively charged

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Option 3:
Neutral

Option 4:
Either positively or negatively charged depending on their size

Correct Answer:
Negatively charged

Solution:
The DNA molecule is negatively charged due to phosphate group in their sugar-phosphate backbone, so they get attracted to the positive pole.

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